

Anna University Regional Campus - Tirunelveli

Tirunelveli - 627007

BONAFIDE CERTIFICATE

NAME :

REGISTER NUMBER :

PROGRAMME :

COURSE CODE & TITLE :

DEPARTMENT :

SEMESTER :

Certified that this is a Bonafide Record of Practical Work Done
by Mr. /Ms. _____ in the
_____ Laboratory during the Period of
_____ 2024-2025.

DATE:

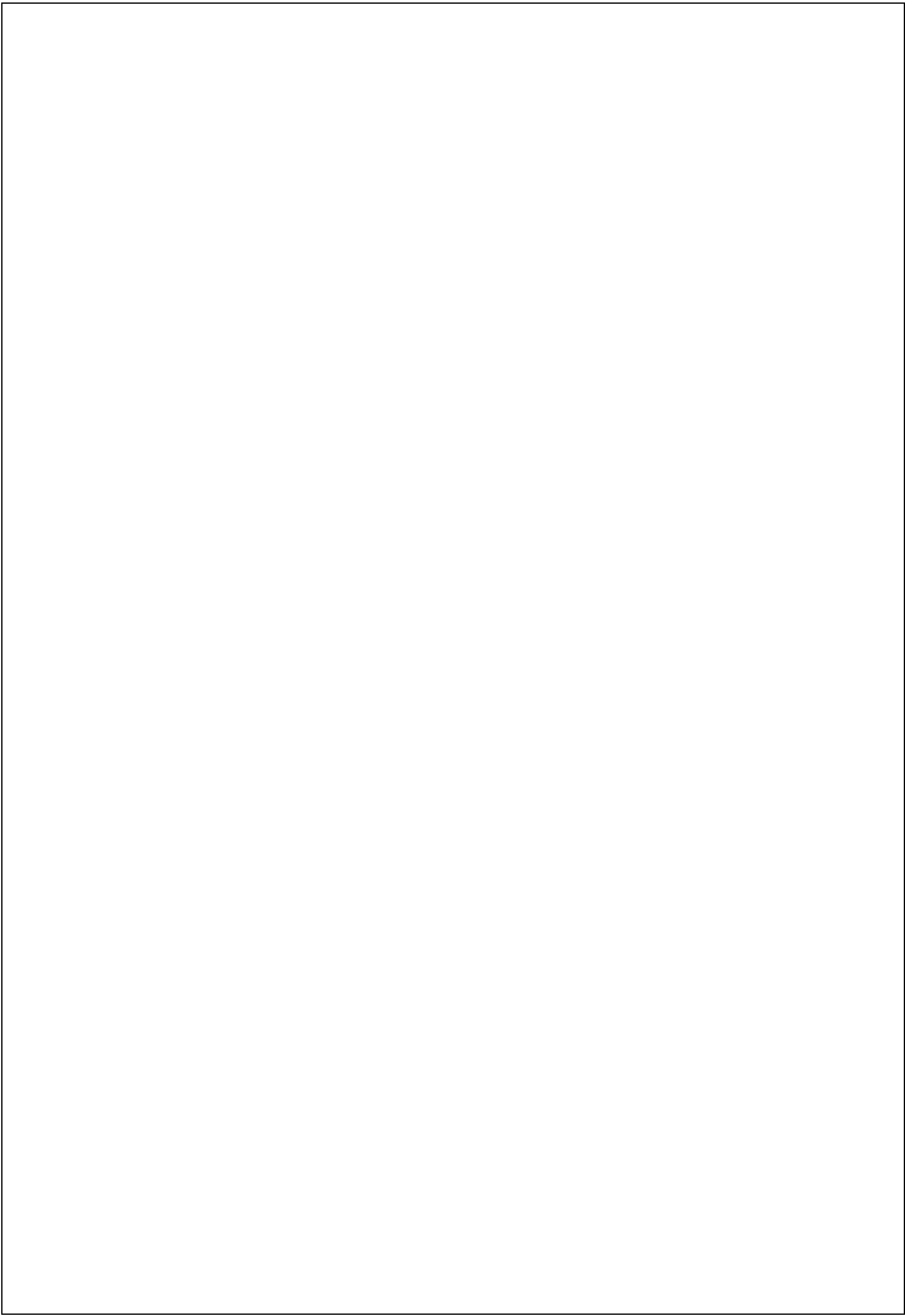
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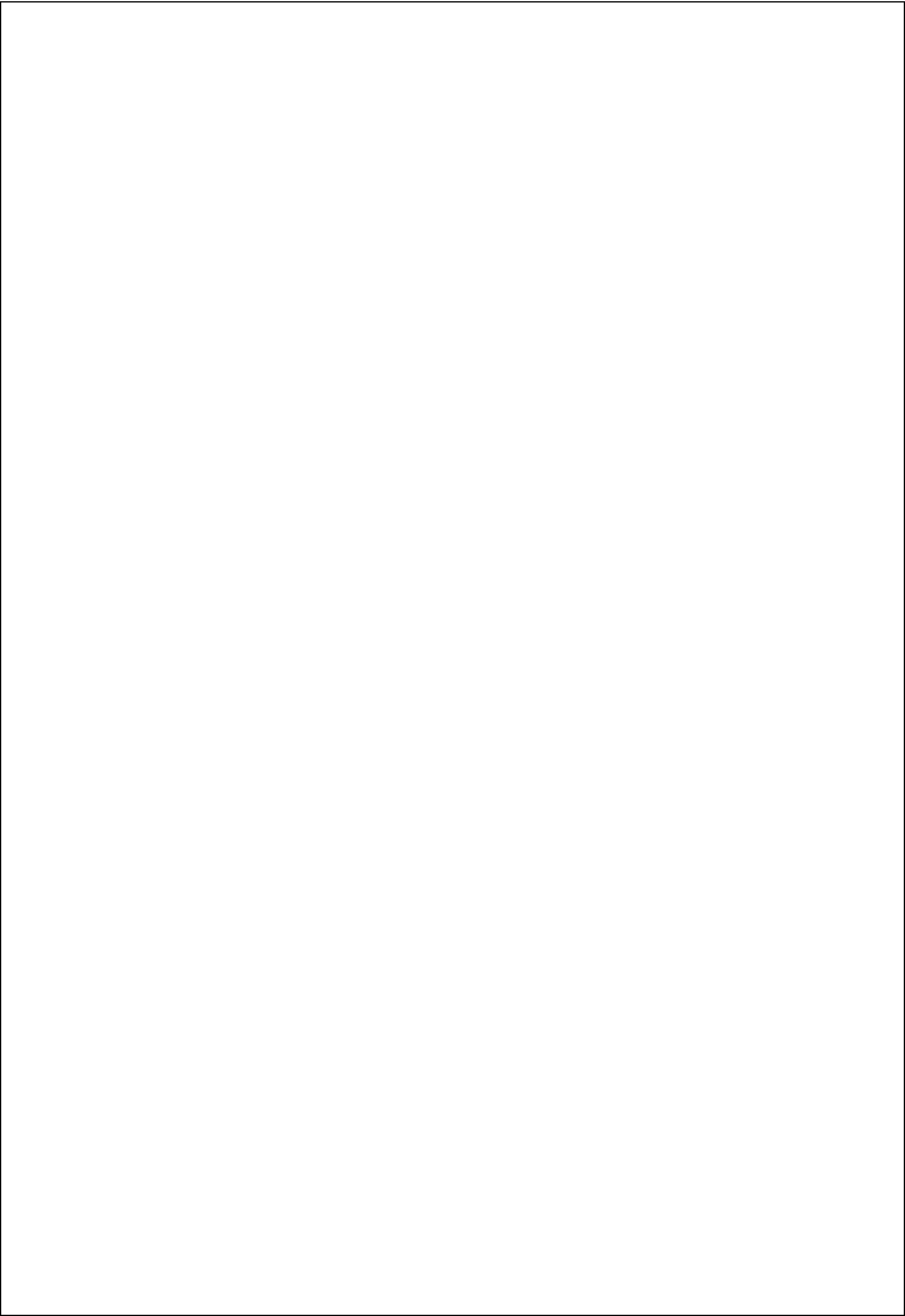
INTERNAL EXAMINER

EXTERNAL EXAMINER



Index

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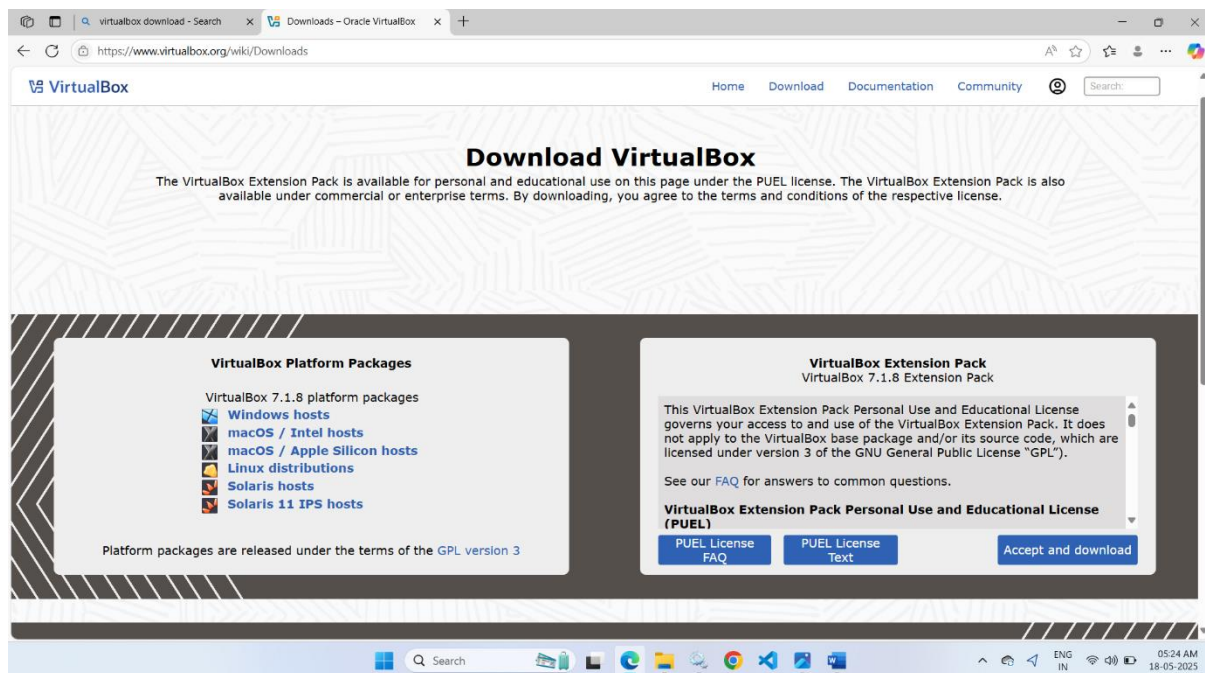


CREATE TYPE 2 VIRTUALIZATION IN VIRTUALBOX

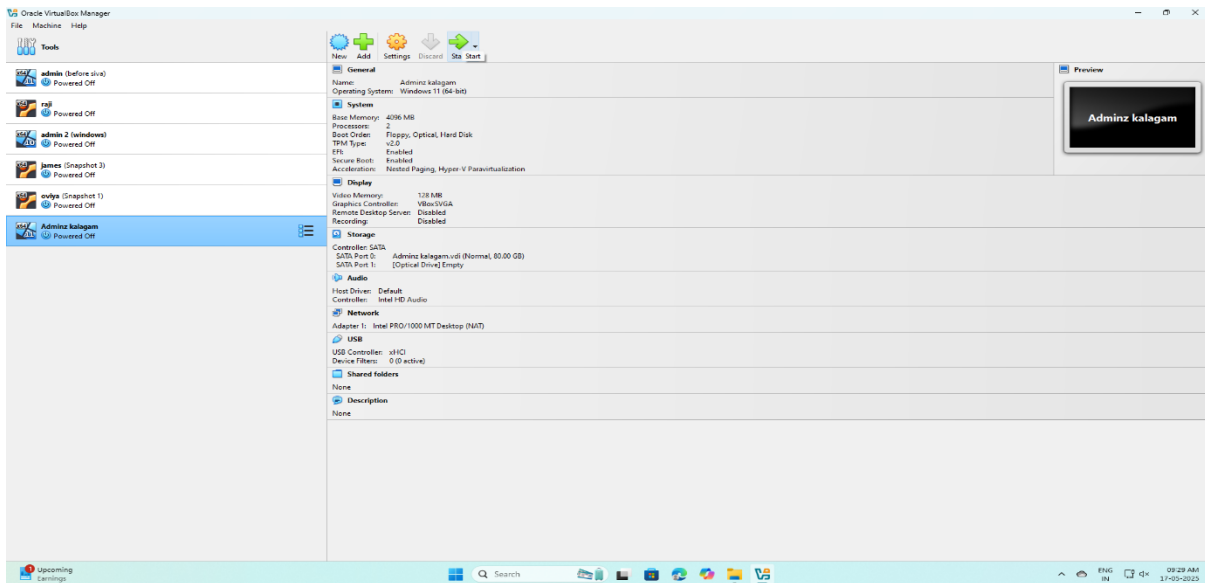
AIM

PROCEDURE

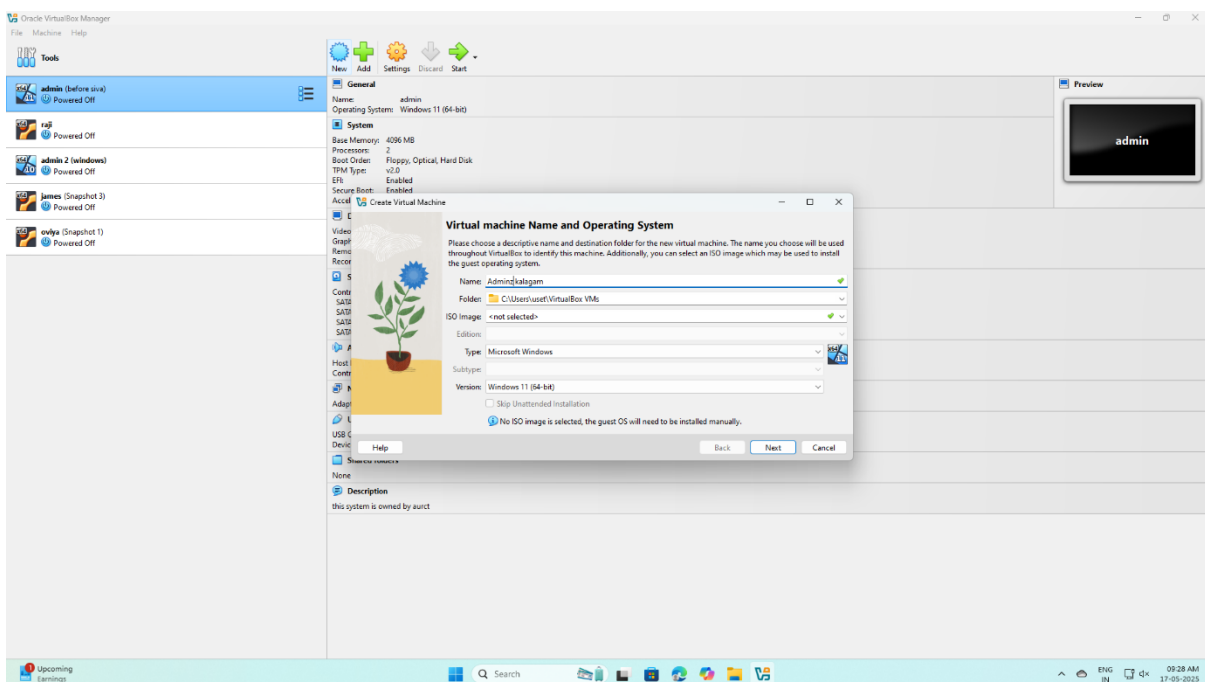
Step 1 : Download and Install VirtualBox and Ubuntu ISO from their official websites to get the latest versions needed for virtualization and OS installation.



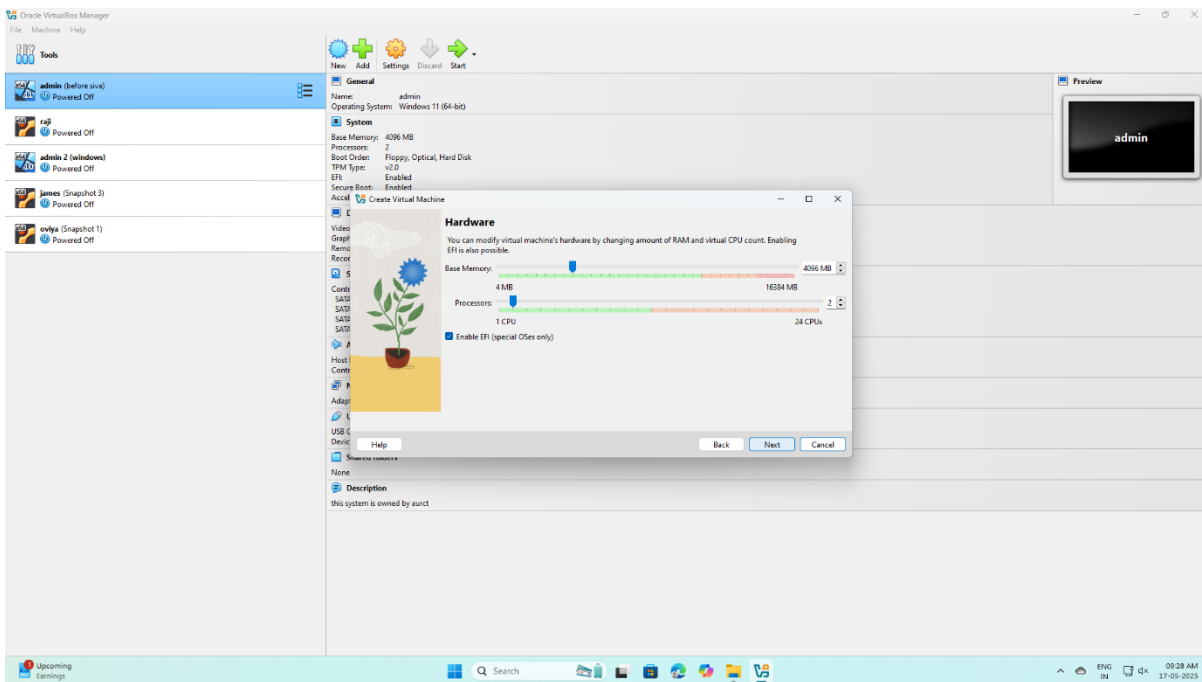
Step 2 : Open VirtualBox and click “New” to begin creating a new virtual machine for Ubuntu.



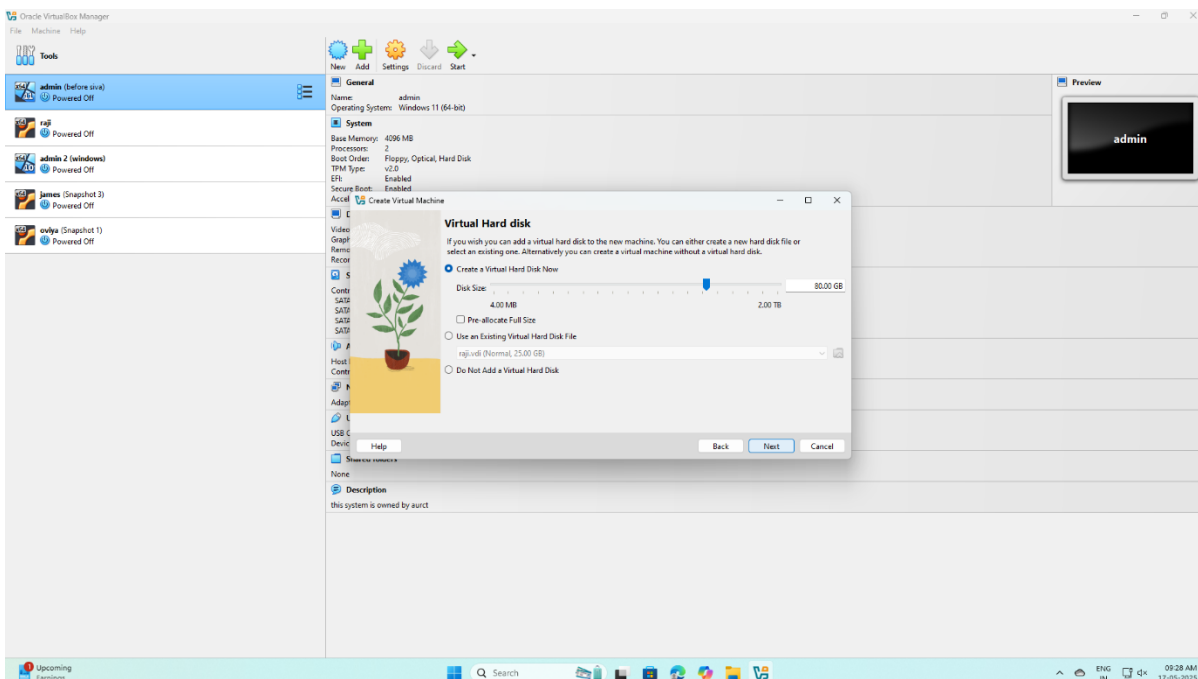
Step 3 : Enter VM name and OS type, like “Ubuntu 22.04”, set Type to “Linux” and Version to “Ubuntu (64-bit)”, and check “Skip Unattended Installation”.



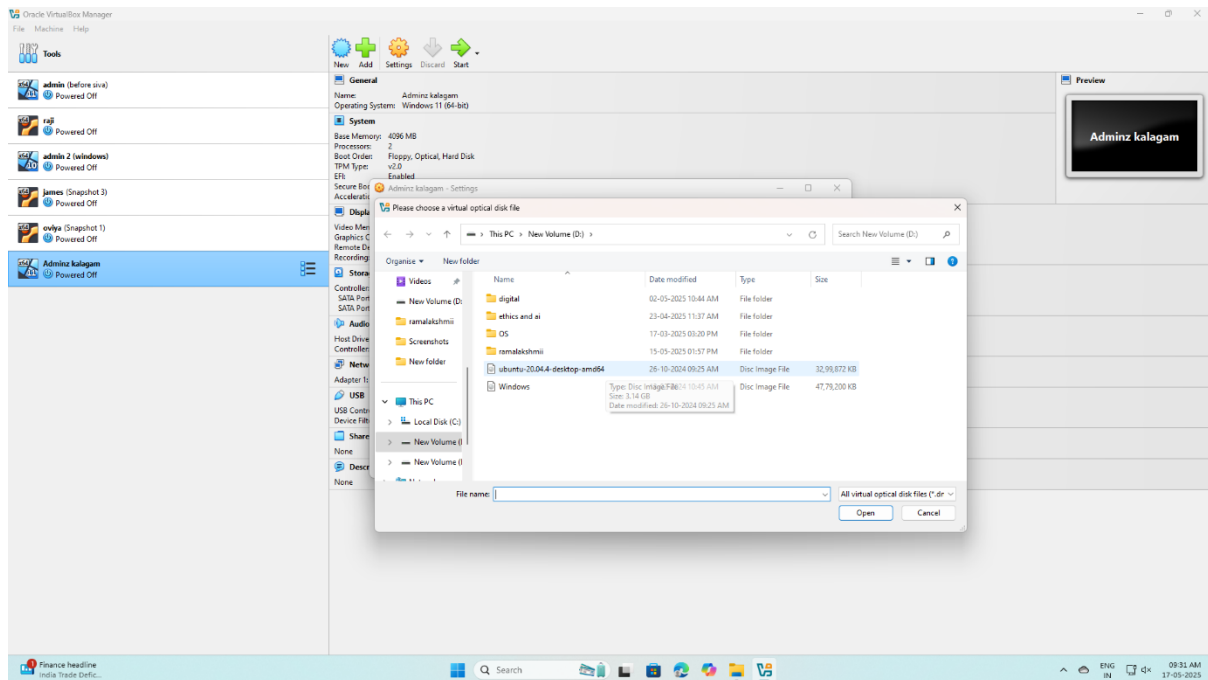
Step 4 : Allocate memory (RAM) by choosing at least 2048 MB (2 GB) to ensure Ubuntu runs smoothly.



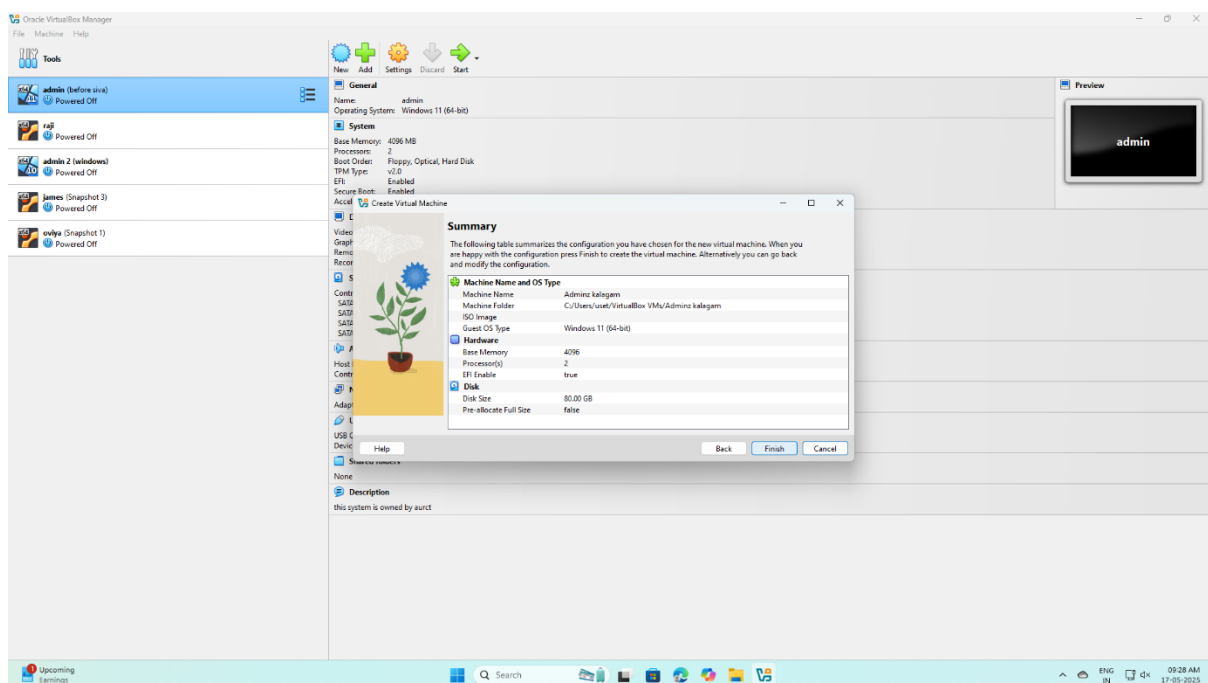
Step 5 : Create a virtual hard disk, choose VDI, select “Dynamically allocated”, and set disk size to at least 25 GB for system and app files.



Step 6 : Upload Ubuntu ISO before finishing VM creation by clicking the folder icon next to “ISO Image”, selecting your downloaded Ubuntu ISO, and clicking “Choose”.

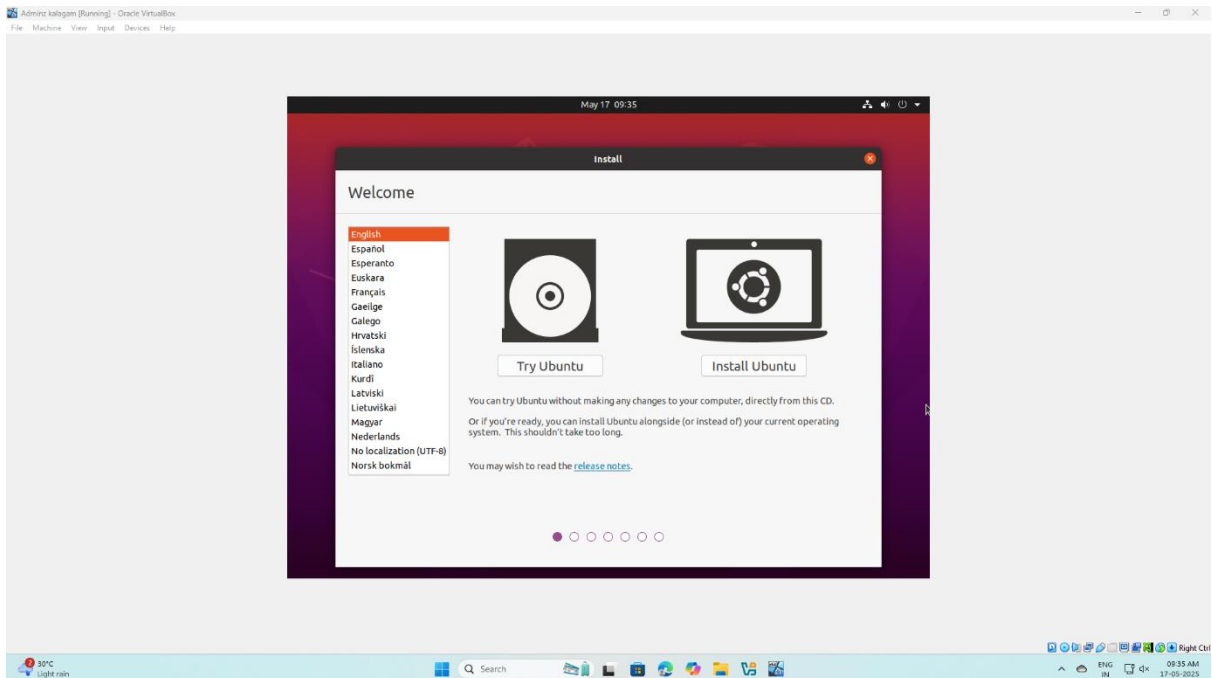


Step 7 : Click Finish to create the VM, which now has the Ubuntu ISO mounted and ready to boot.

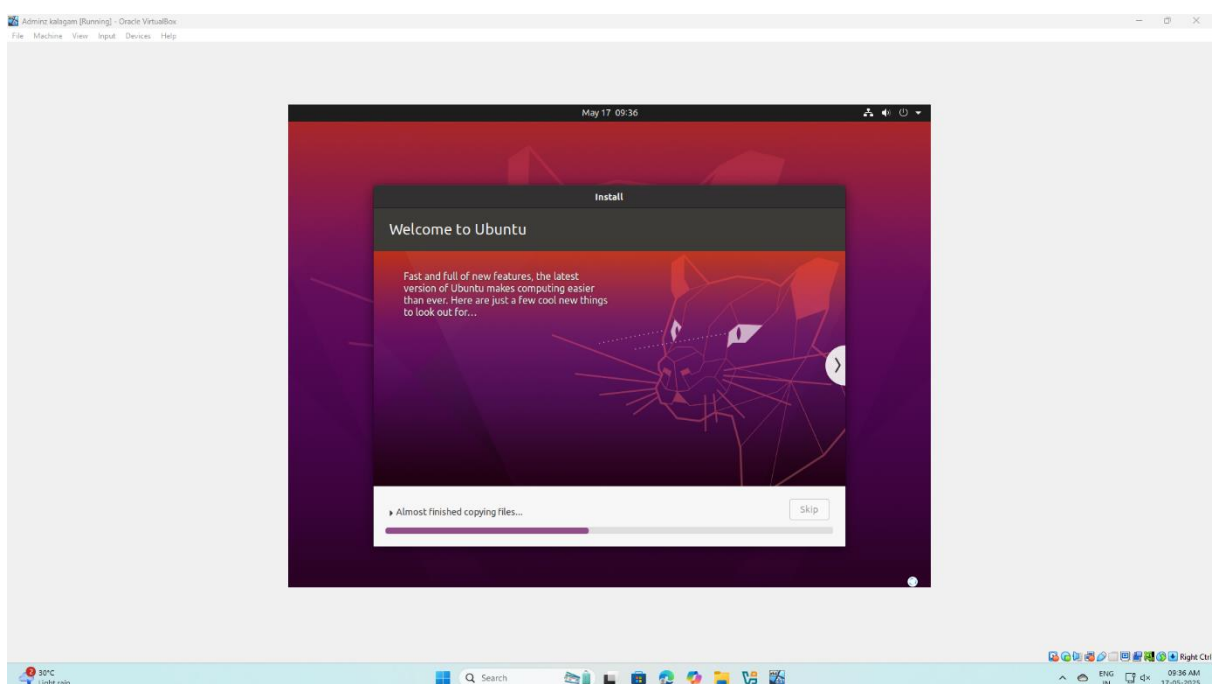


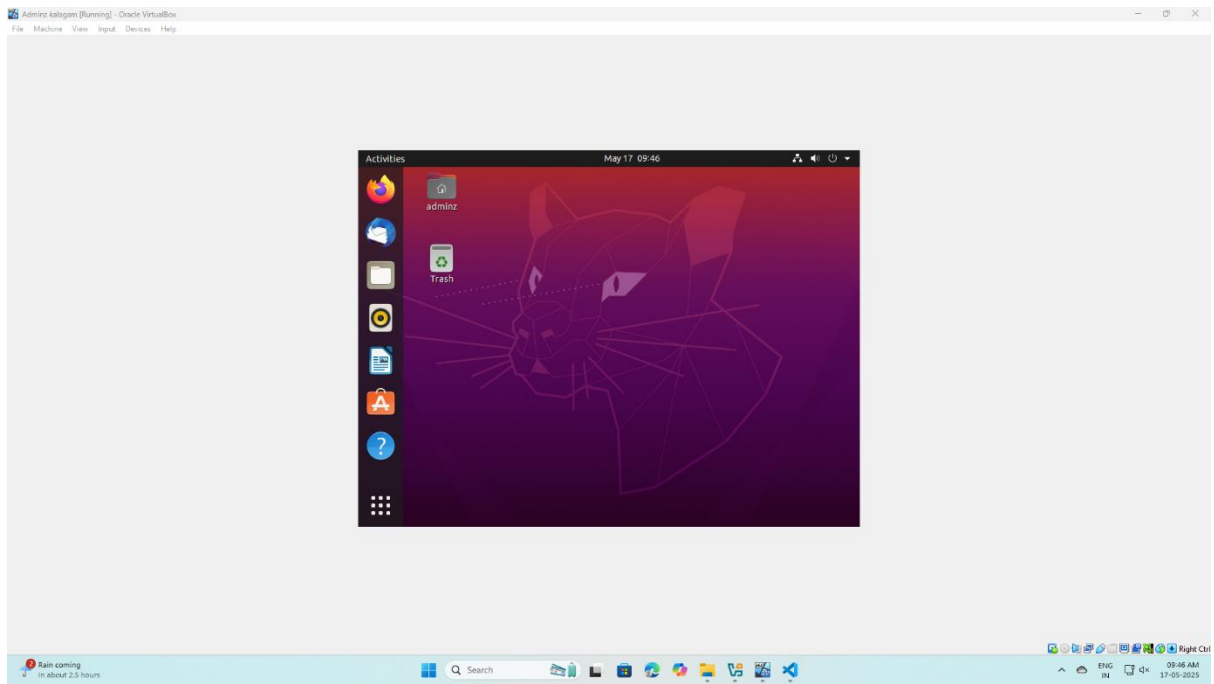
Step 8 : Start the VM, which will automatically boot into the Ubuntu installer from the ISO.

Step 9 : Choose “Install Ubuntu” and proceed with the normal installation options like keyboard layout and software selection.



Step 10 : Start using Ubuntu on VirtualBox, where you can now install software, browse the internet, write code, run terminal commands, and fully manage the guest operating system just like on a physical machine.





RESULT

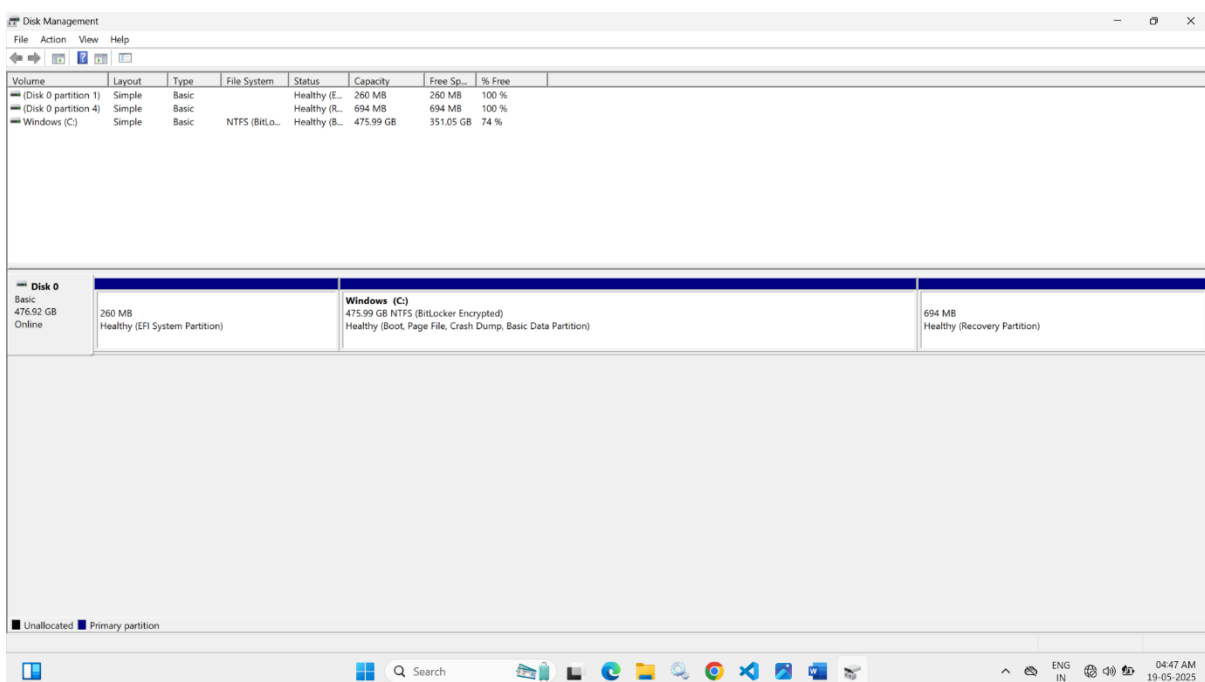
SHRINK AND EXTEND VIRTUAL DISK

AIM

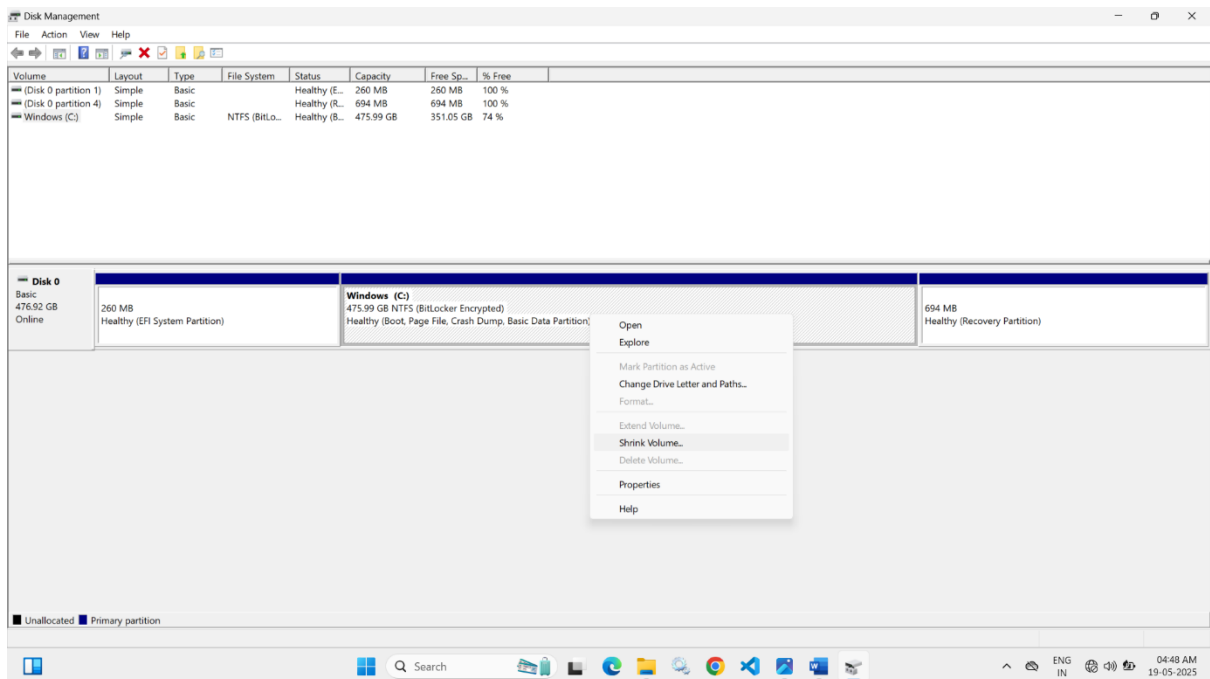
PROCEDURE

SHRINK VIRTUAL DISK

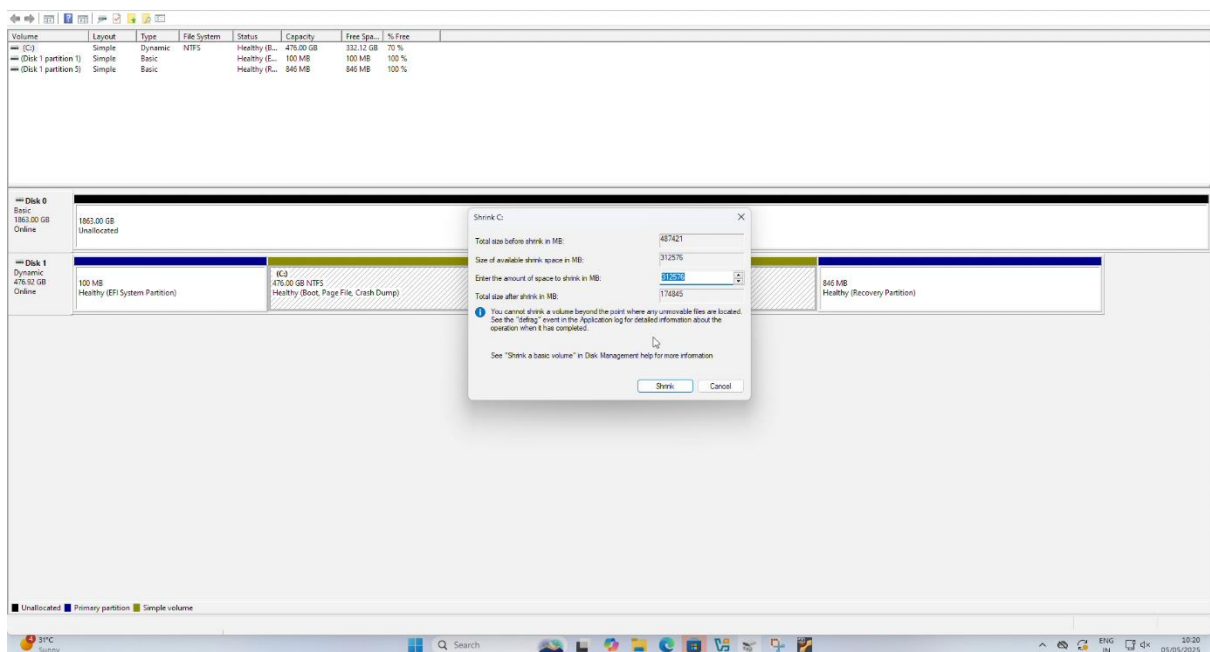
Step 1 : Press Windows + X, then select Disk Management from the menu to open the disk management console, which allows you to manage all your disk partitions and volumes.

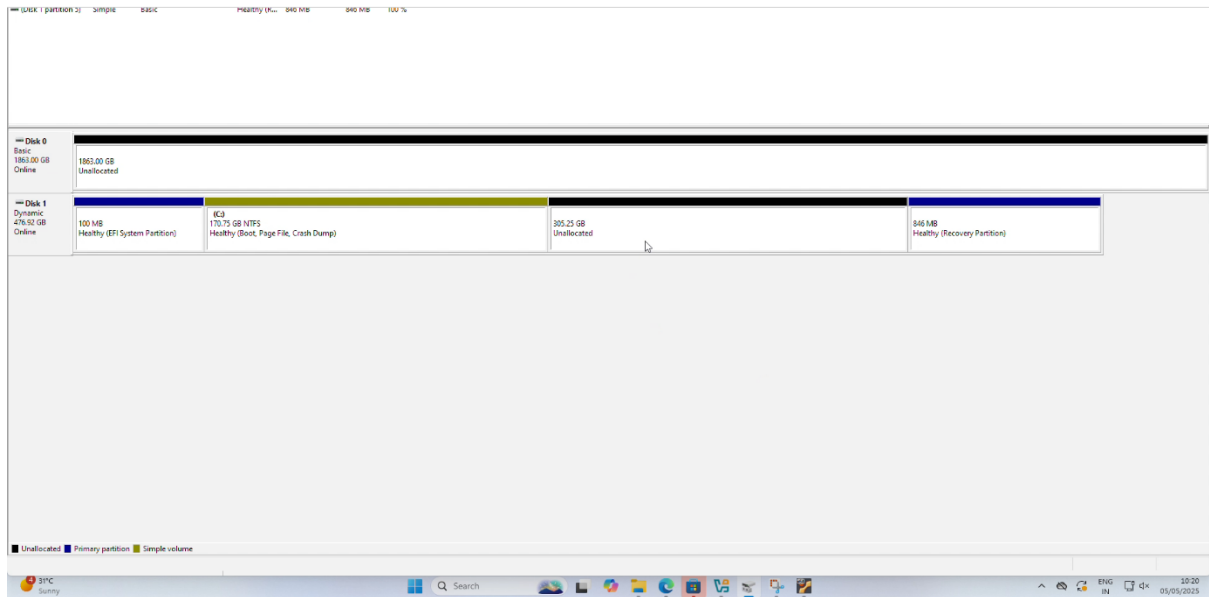


Step 2 : In the Disk Management window, locate the partition or volume you want to shrink, right-click on it, and choose Shrink Volume to begin the shrinking process.



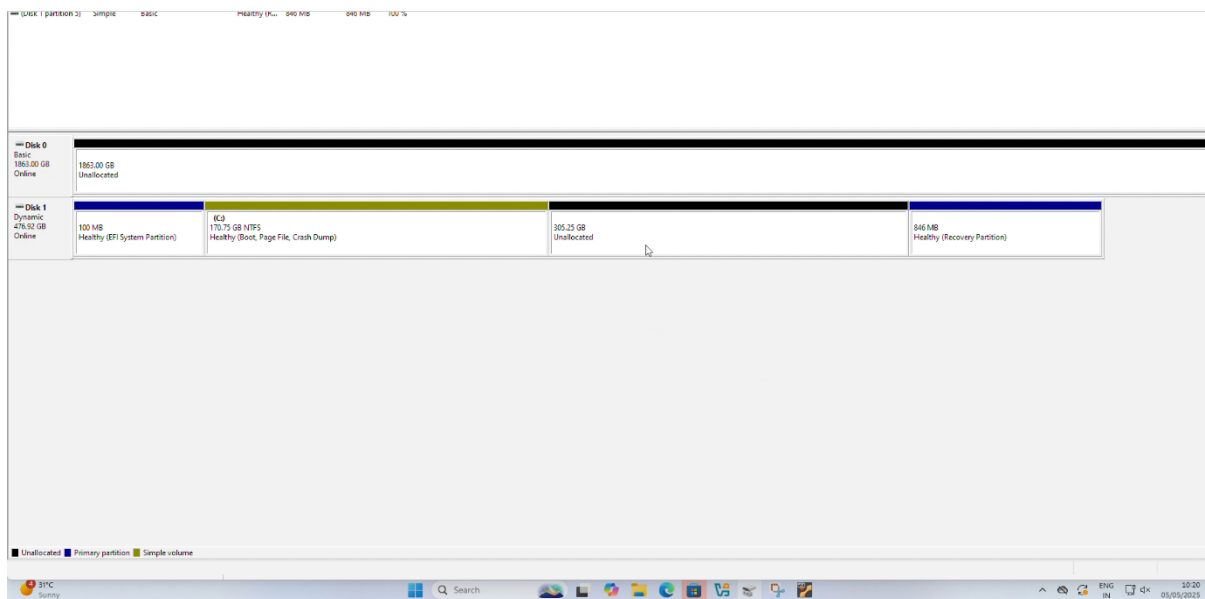
Step 3 : A prompt will appear asking how much space you want to shrink the volume by enter the amount in megabytes (MB) and click Shrink to reduce the volume size and free up unallocated space.



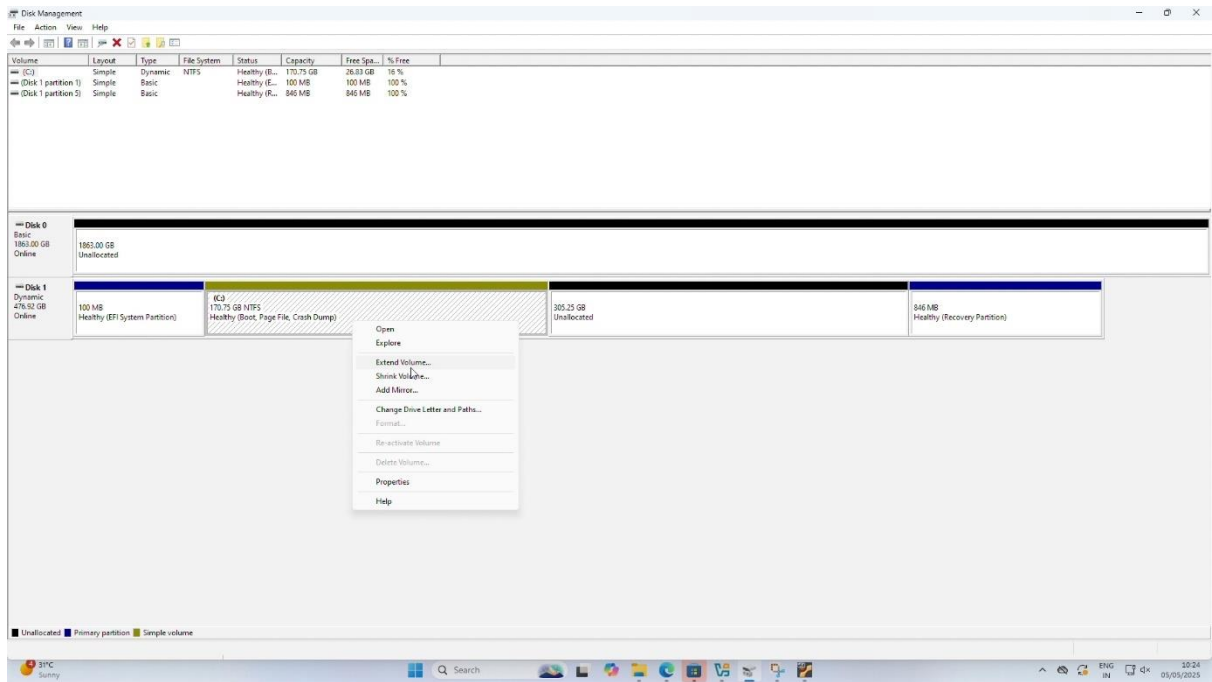


EXTEND VIRTUAL DISK:

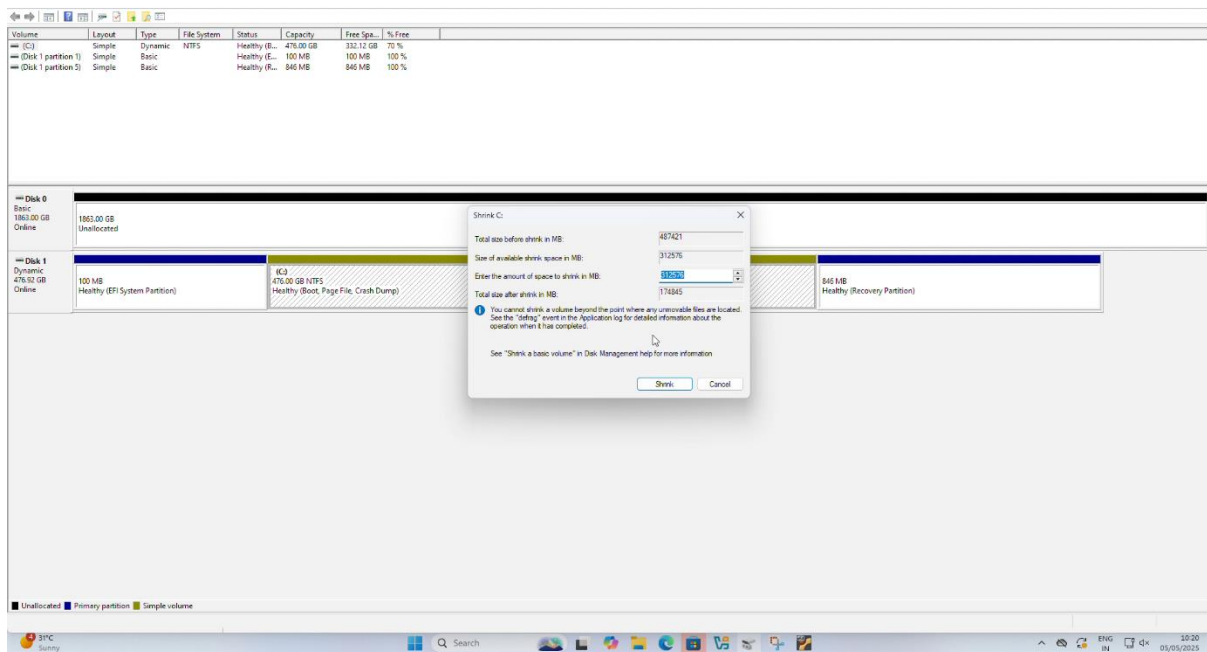
Step 1 : Press Windows + X, then select Disk Management to open the partition management interface, which provides tools to modify your disk volumes.



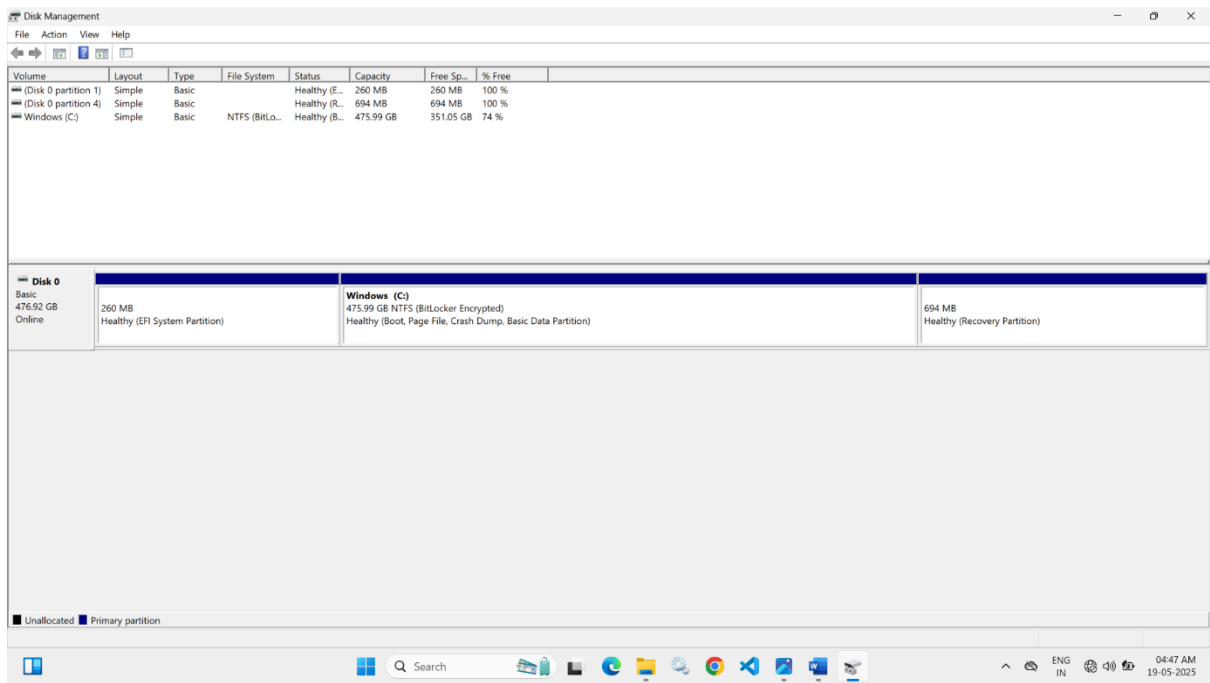
Step 2 : In the Disk Management window, locate the volume you wish to extend, right-click on it, and select Extend Volume to begin the extension process.



Step 3 : A wizard will pop up, allowing you to specify how much unallocated space to add to the selected volume; choose the size in megabytes (MB) and click Next to proceed.



Step 4 : After reviewing the changes, click Finish to apply the extension, and the selected volume will be expanded to include the newly allocated space.



RESULT

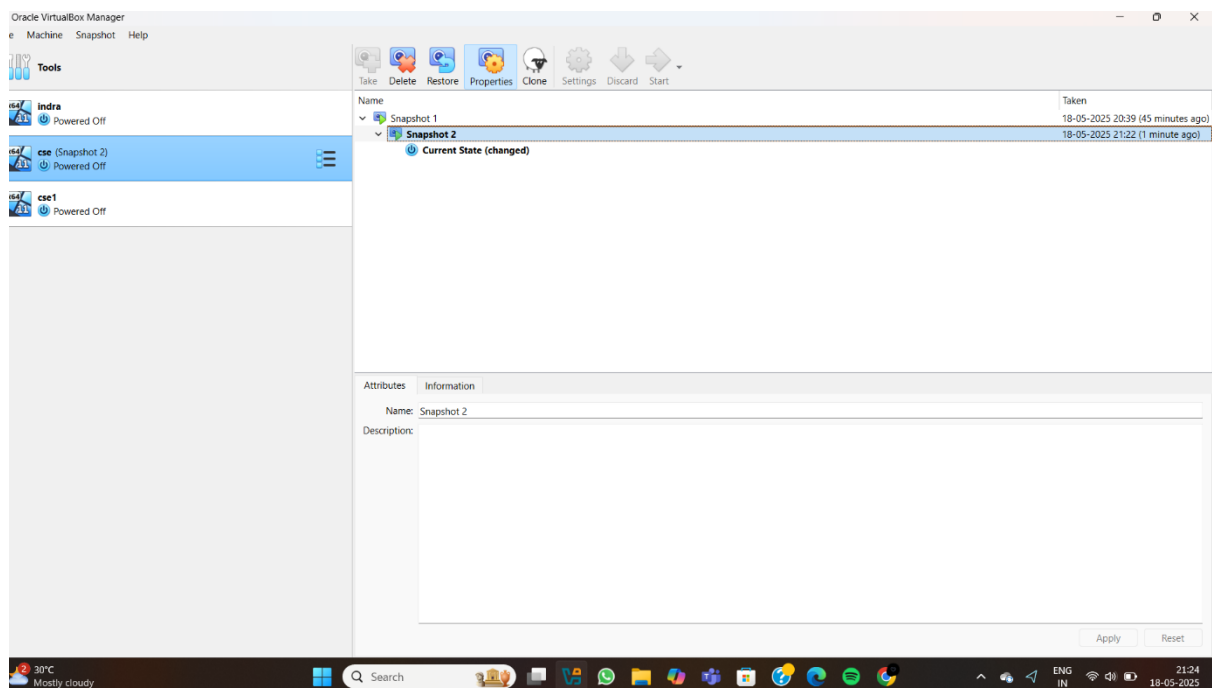
CREATE, MANAGE, CONFIGURE AND SCHEDULE SNAPSHOTS

AIM

PROCEDURE

CREATION OF SNAPSHOT

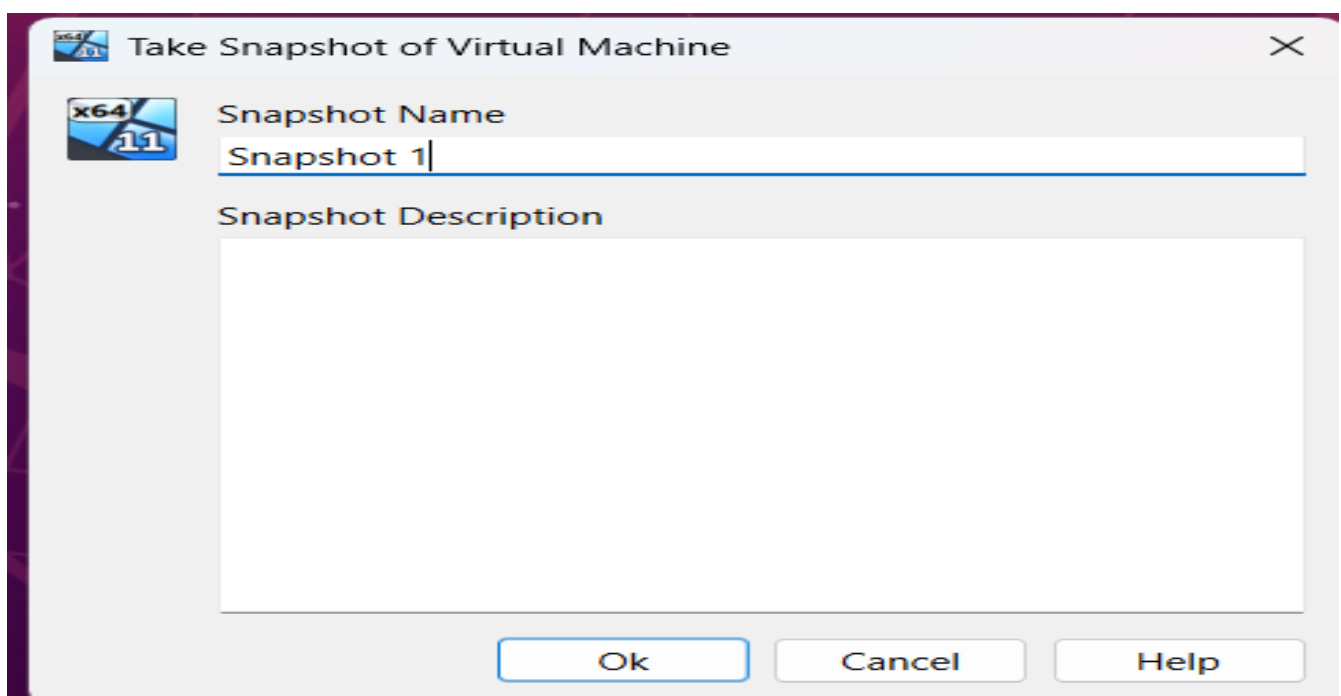
Step 1 : Launch VirtualBox on your host machine and select the running Ubuntu guest virtual machine.



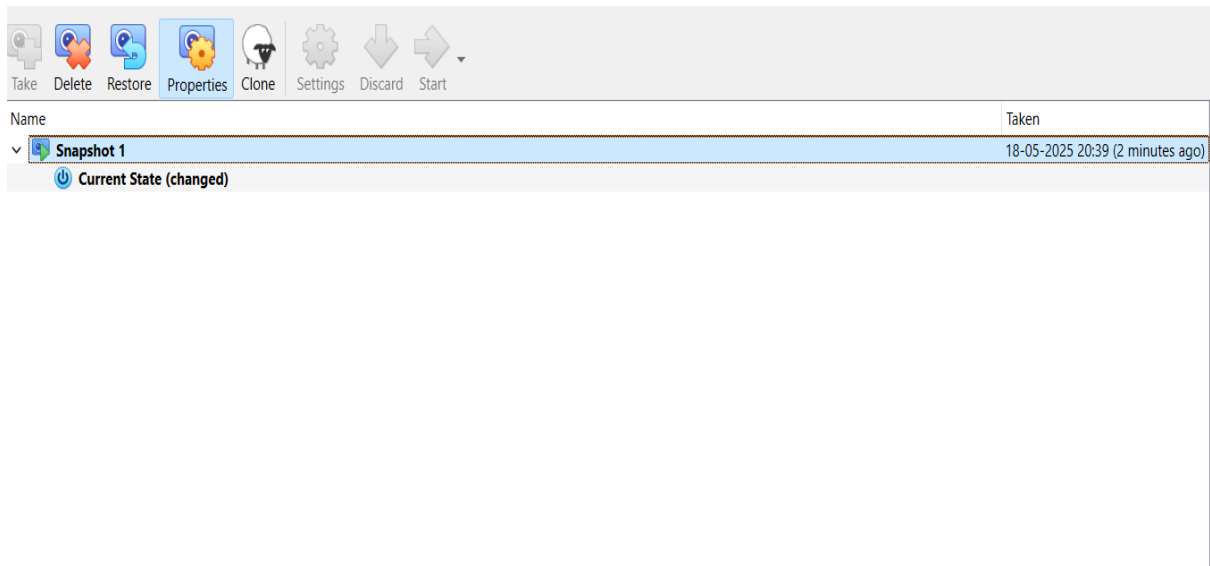
Step 2 : With the Ubuntu VM selected, click on Machine in the top menu, then choose Take Snapshot to open the snapshot dialog.



Step 3 : Provide a meaningful name and description for the snapshot to make it easy to identify, and click OK to create the snapshot.

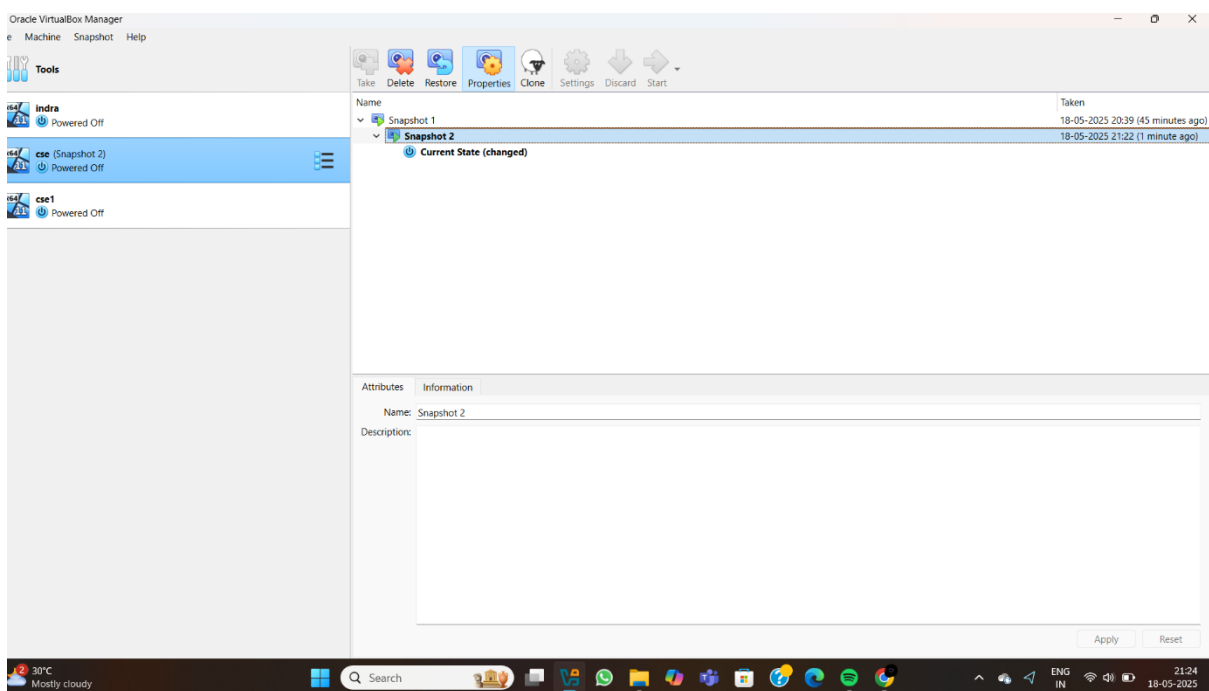


Step 4 : The snapshot will now be stored, and you can restore the machine to this point at any time by selecting the snapshot from the Snapshots tab in VirtualBox.

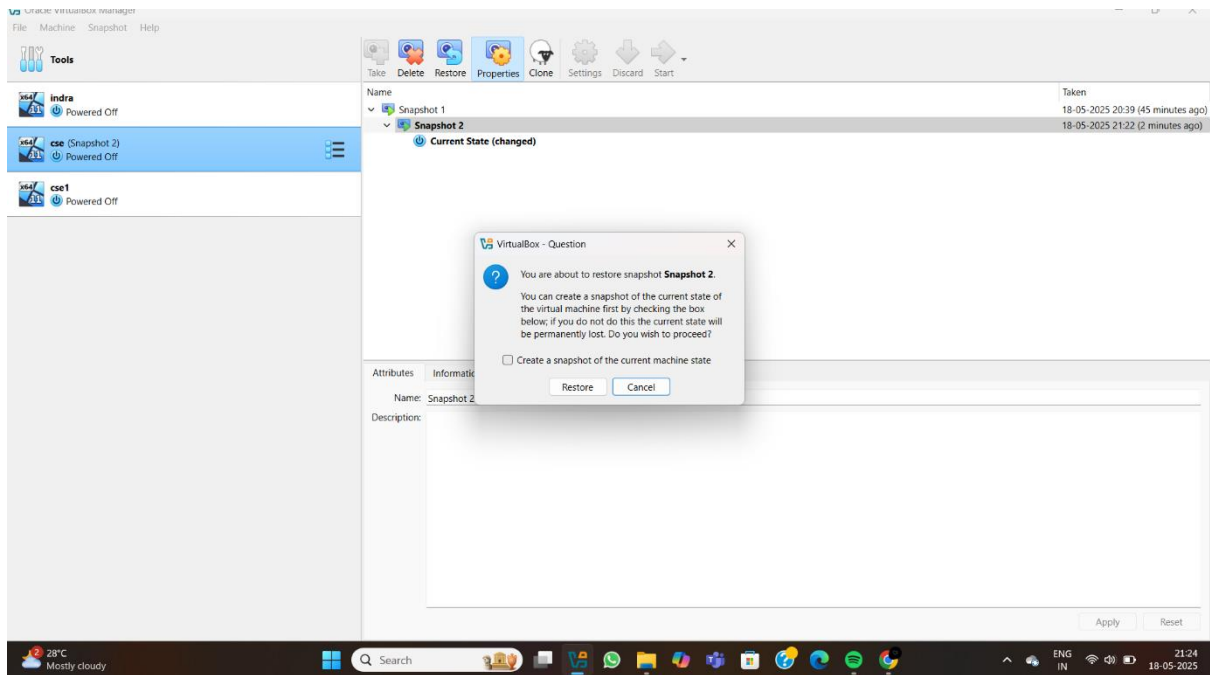


MANAGING SNAPSHOTS

Step 1 : With the Ubuntu VM selected in VirtualBox, click on the Snapshots tab in the top right section of the VirtualBox window to view your existing snapshots.



Step 2 : To restore a snapshot, right-click on a snapshot in the Snapshots tab and select Restore to roll back to the selected point in time.



Step 3 : To remove a snapshot, right-click on it and choose Delete Snapshot to remove it from the system.

Step 4 : If needed, right-click the snapshot and select Rename to change its name and make it more descriptive.

RESULT

CREATE SPANNED, MIRRORED AND STRIPED VOLUME

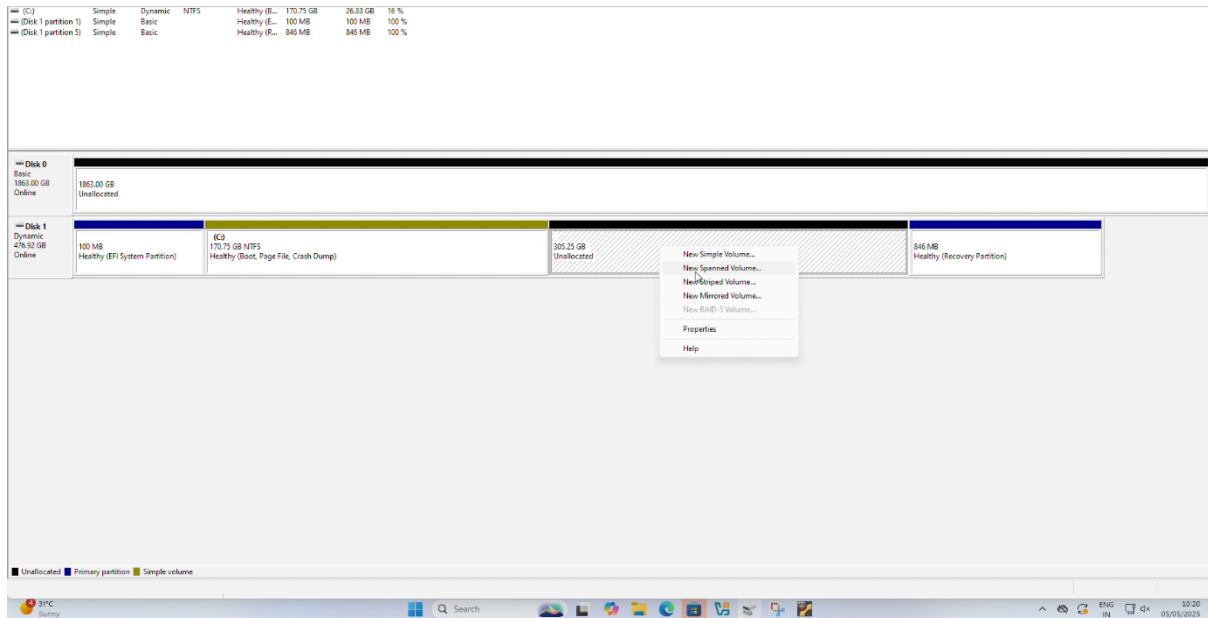
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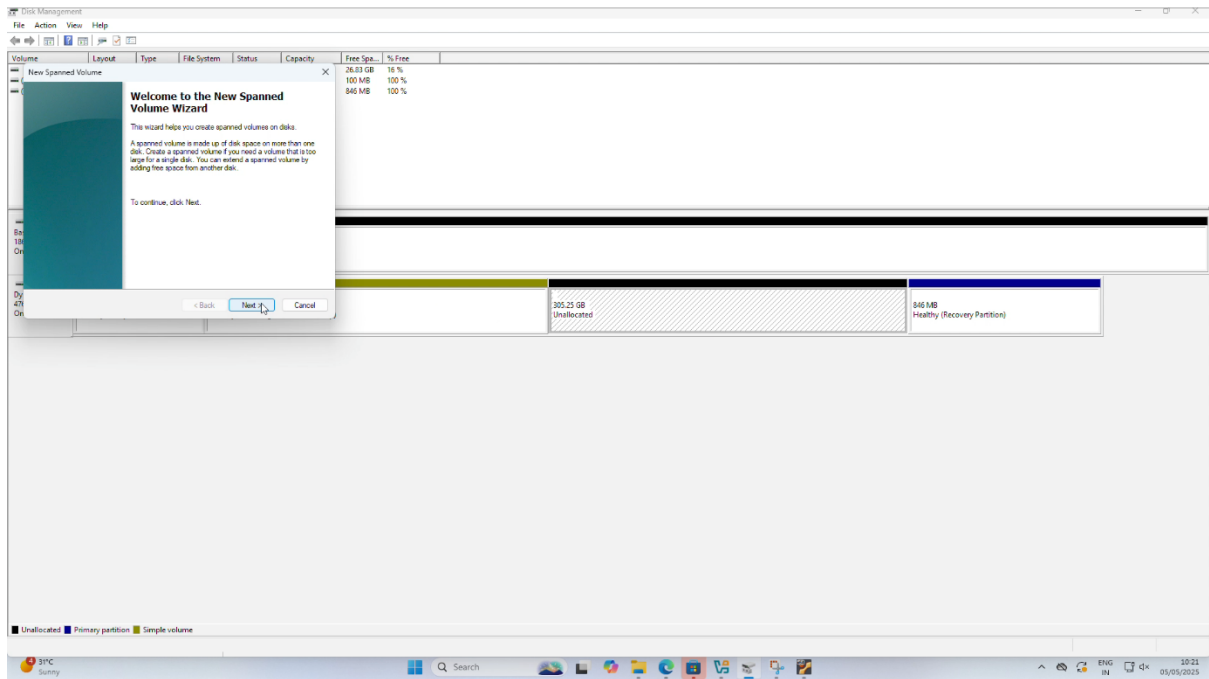
PROCEDURE

CREATE A SPANNED VOLUME

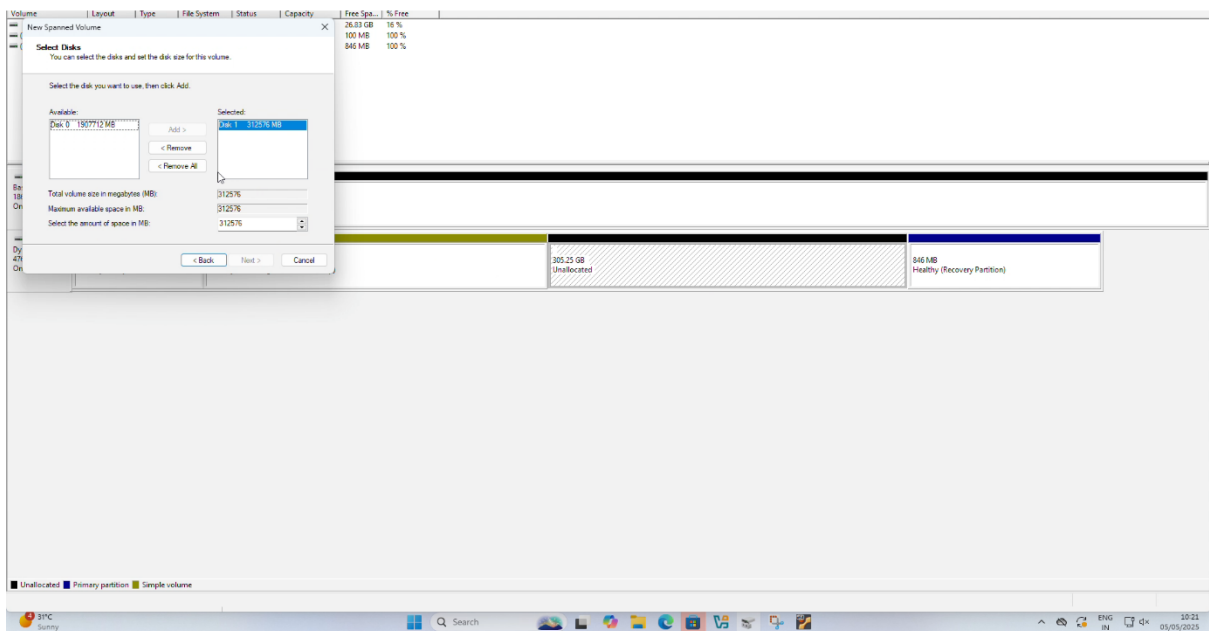
Step 1 : Press Windows + X and select Disk Management to open the disk management console.

Step 2 : Right-click on unallocated space on a disk and choose **New Spanned Volume** to start the volume creation wizard.

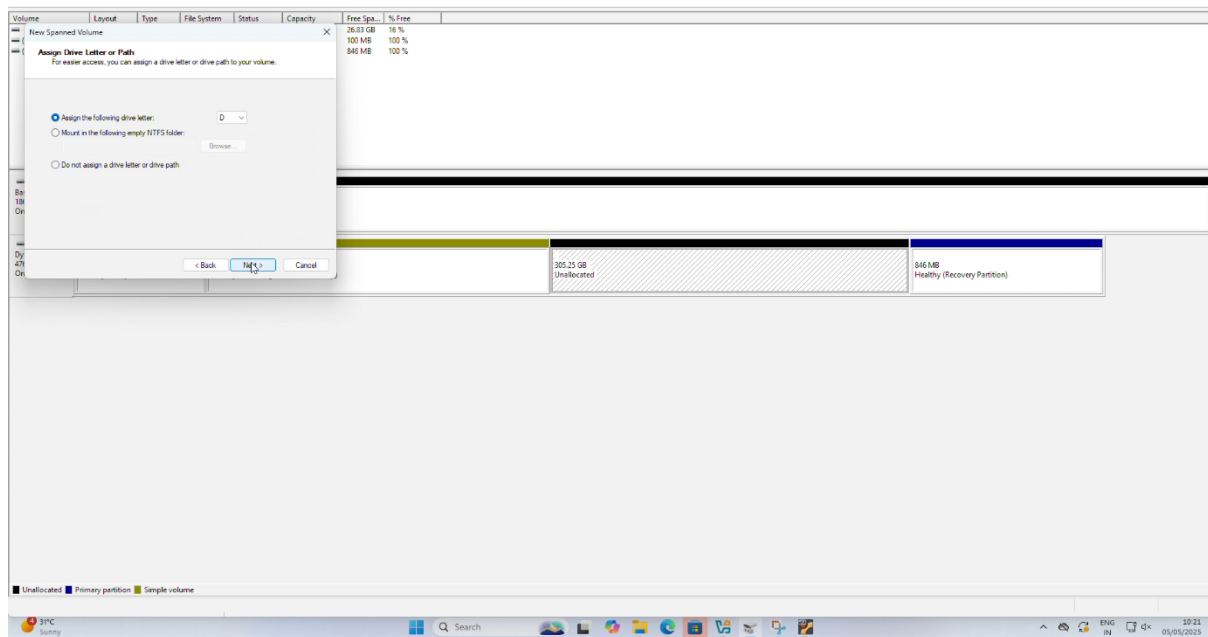




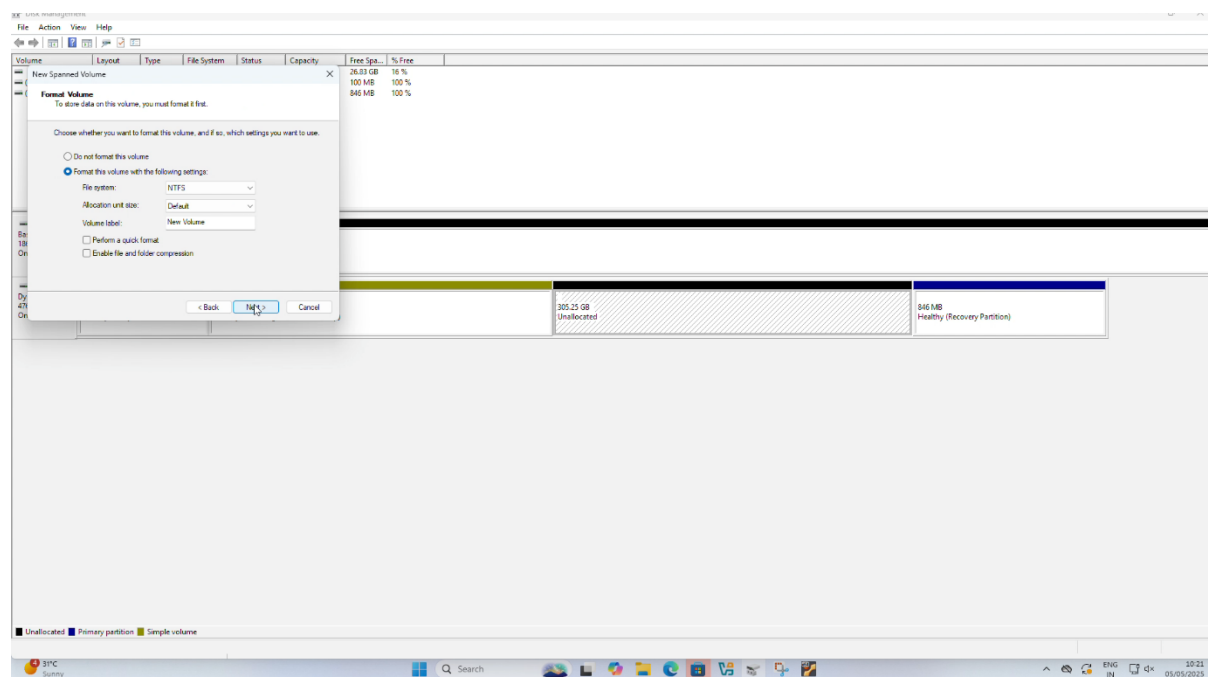
Step 3 : In the wizard, select the disks that you want to include in the spanned volume and click **Next**.

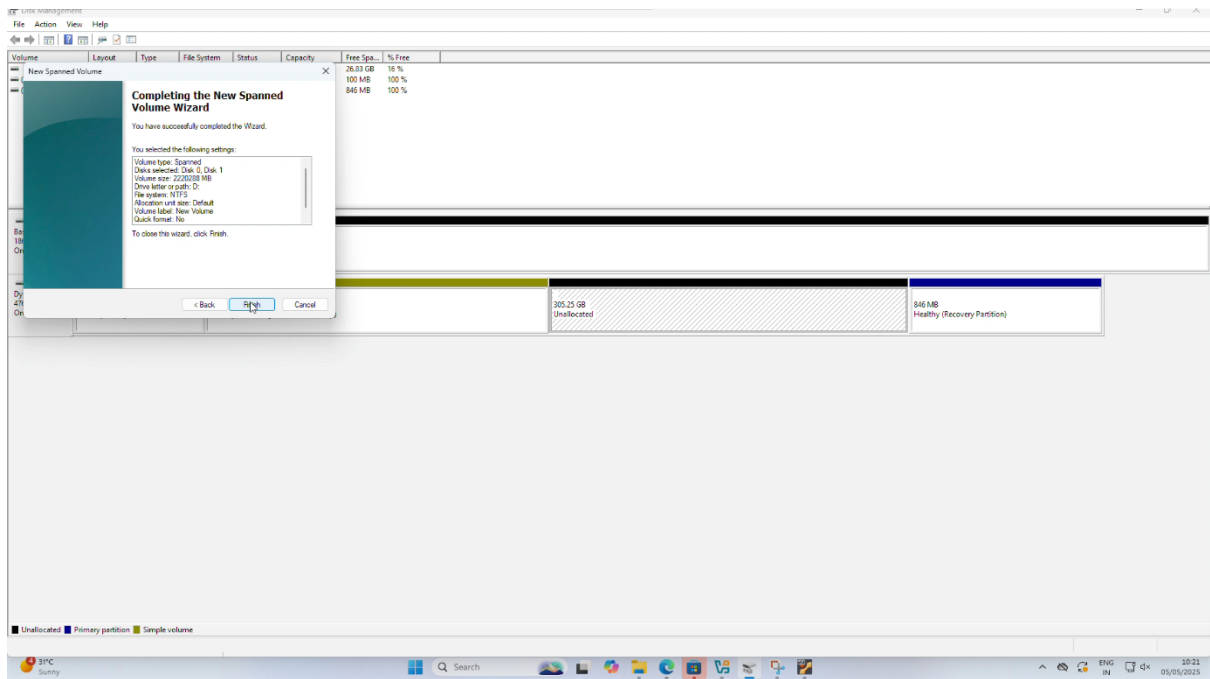


Step 4 : Assign a drive letter or mount path to the new spanned volume and click **Next**.



Step 5 : Choose the file system (e.g., NTFS) and perform a quick format, then click **Next** and **Finish** to create the spanned volume.

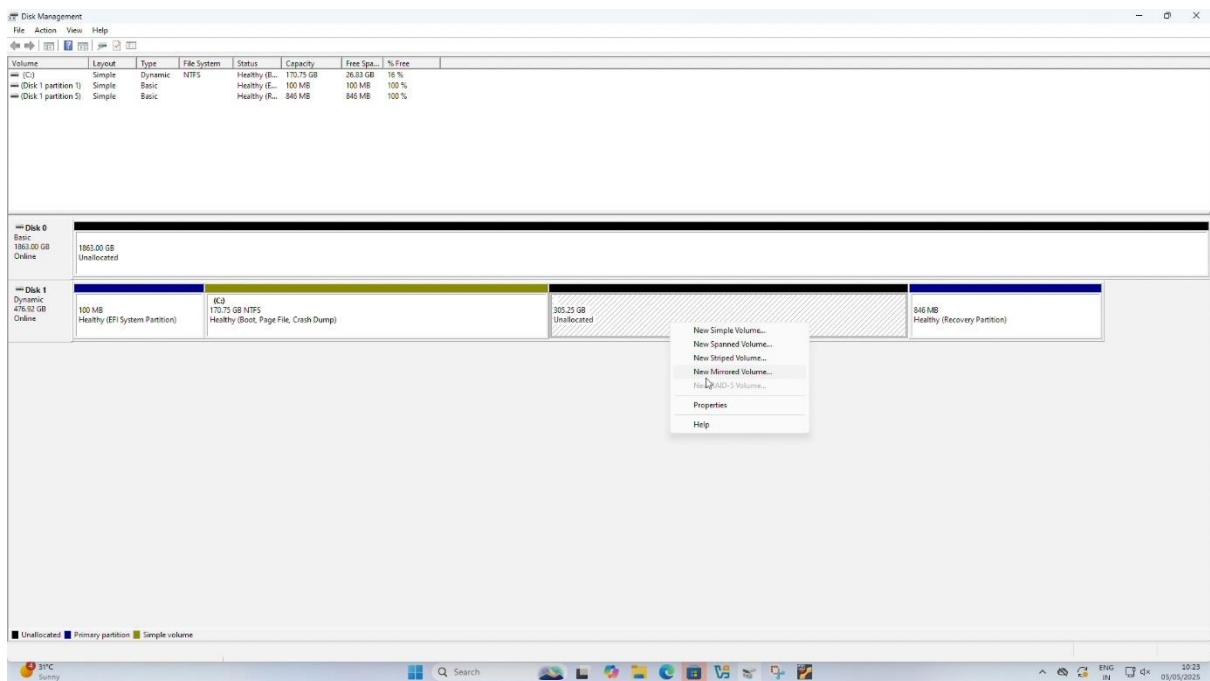




CREATE A MIRRORED VOLUME

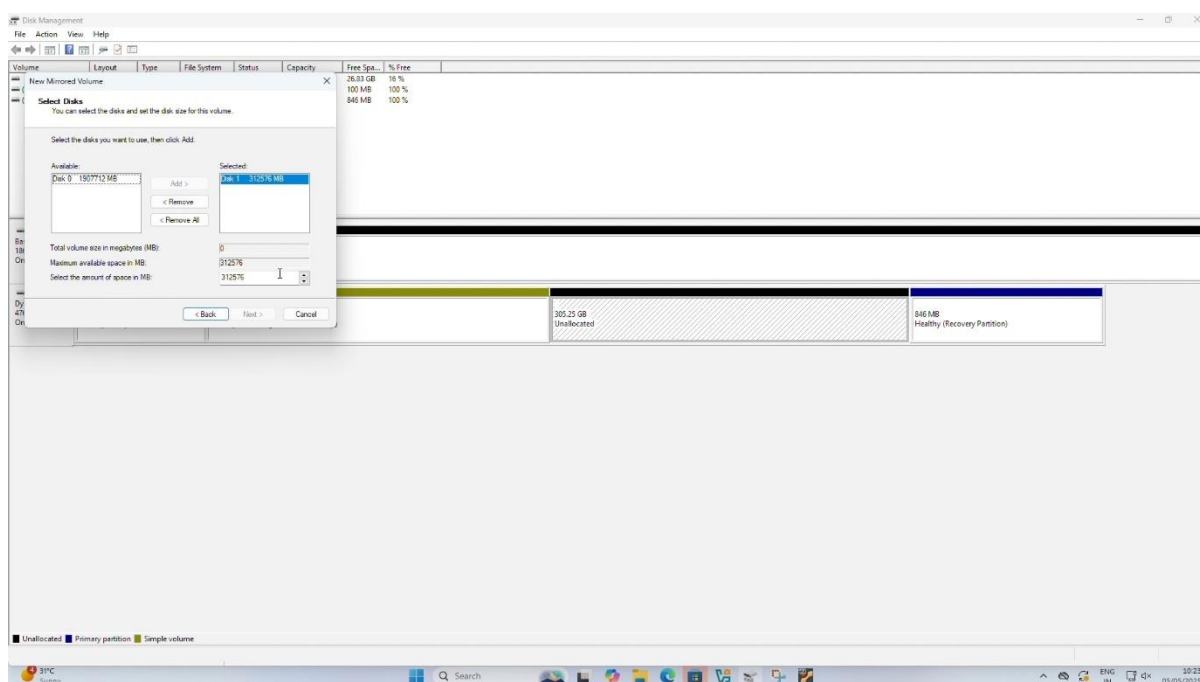
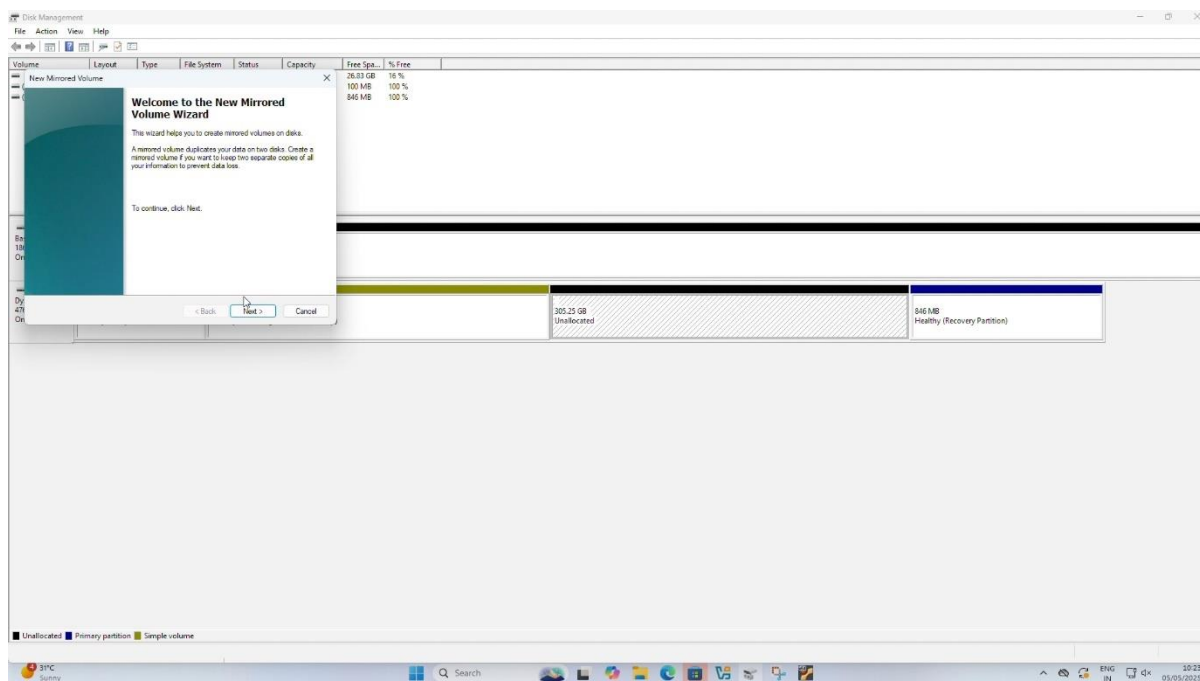
Step 1 : Press Windows + X and select Disk Management to open the disk management console.

Step 2 : Right-click on unallocated space and choose New Mirrored Volume to start the volume creation wizard.

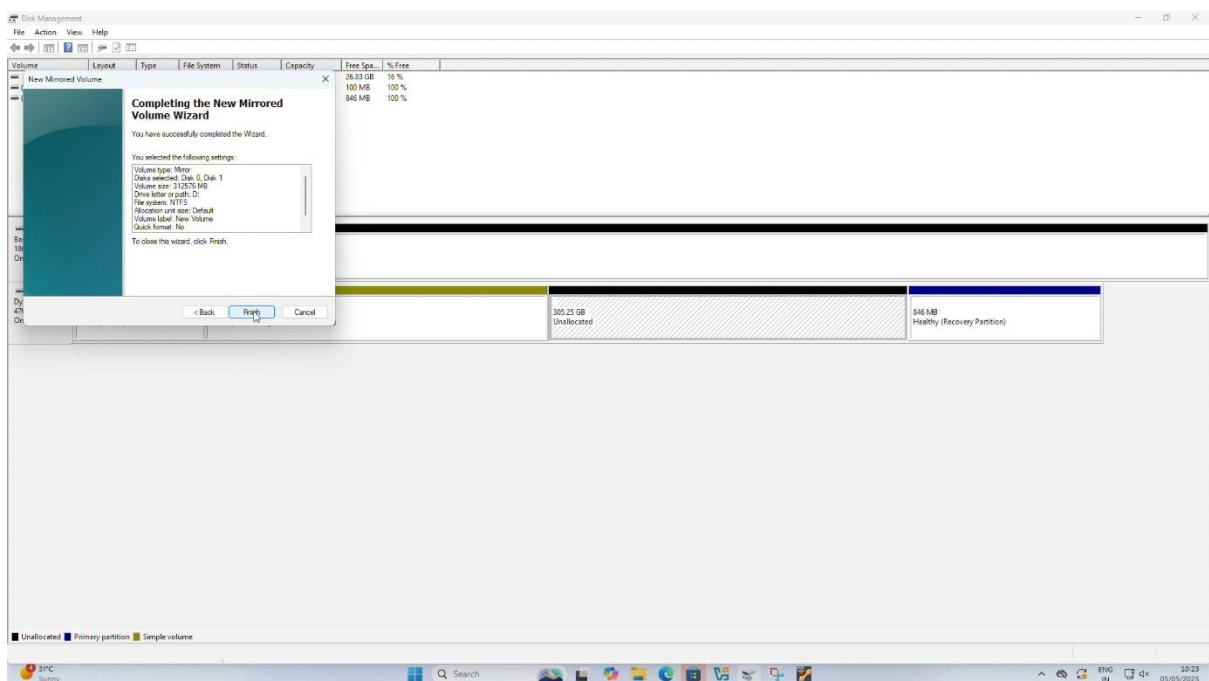
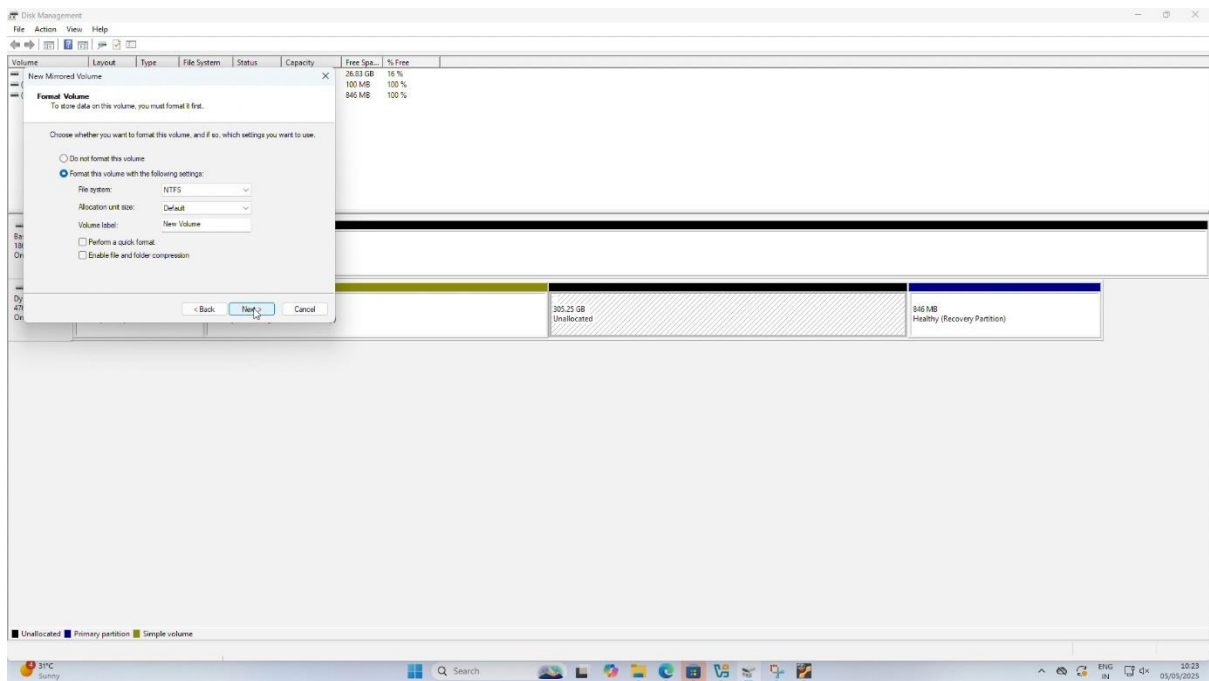


Step 3 : Select the disks you want to mirror, and click Next.

Step 4 : Assign a drive letter or mount path to the new mirrored volume and click Next.

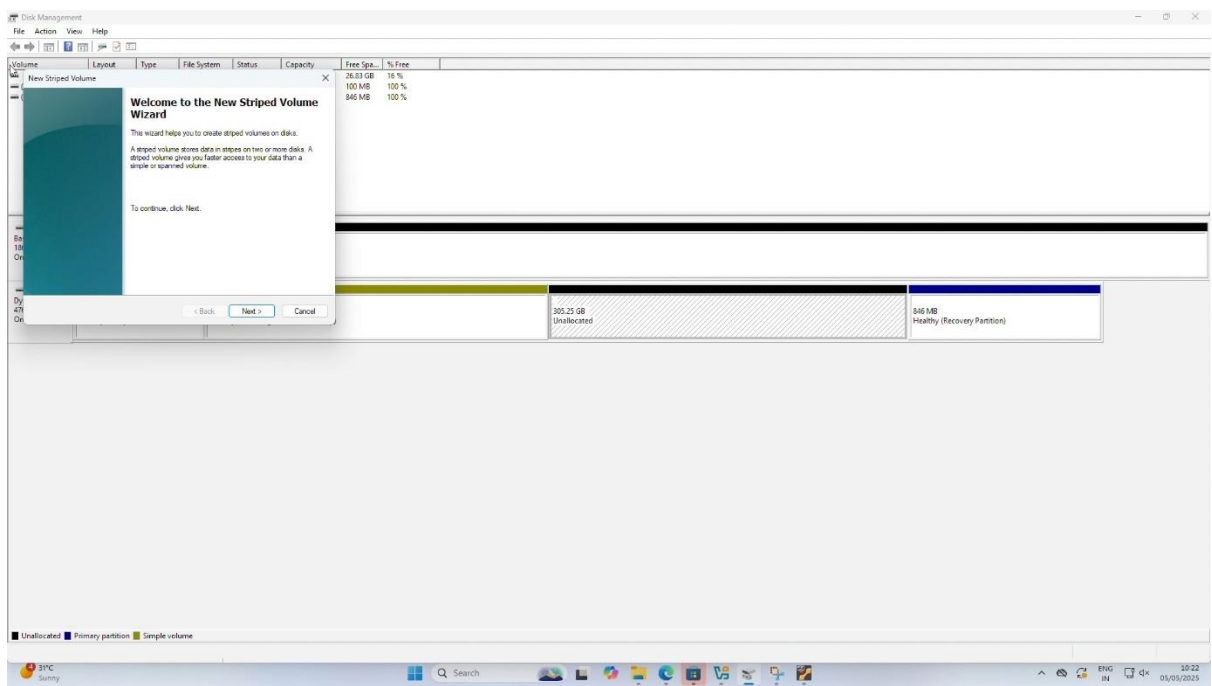
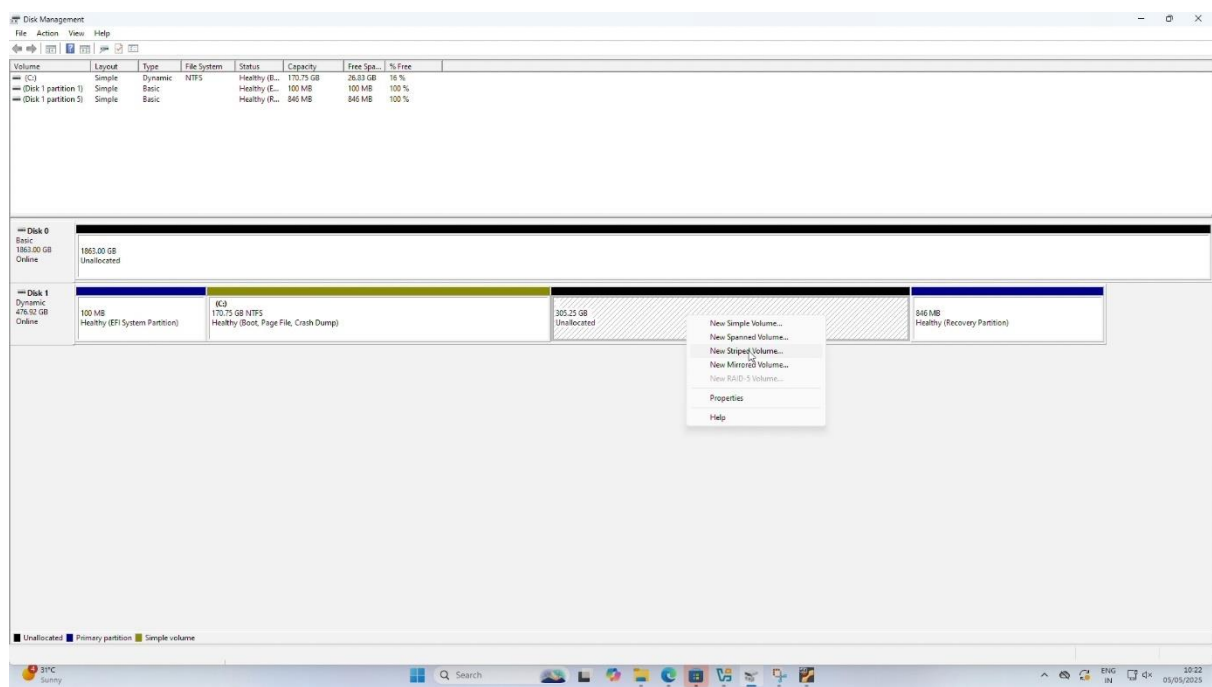


Step 5 : Choose the file system (e.g., NTFS) and perform a quick format, then click Next and Finish to create the mirrored volume.

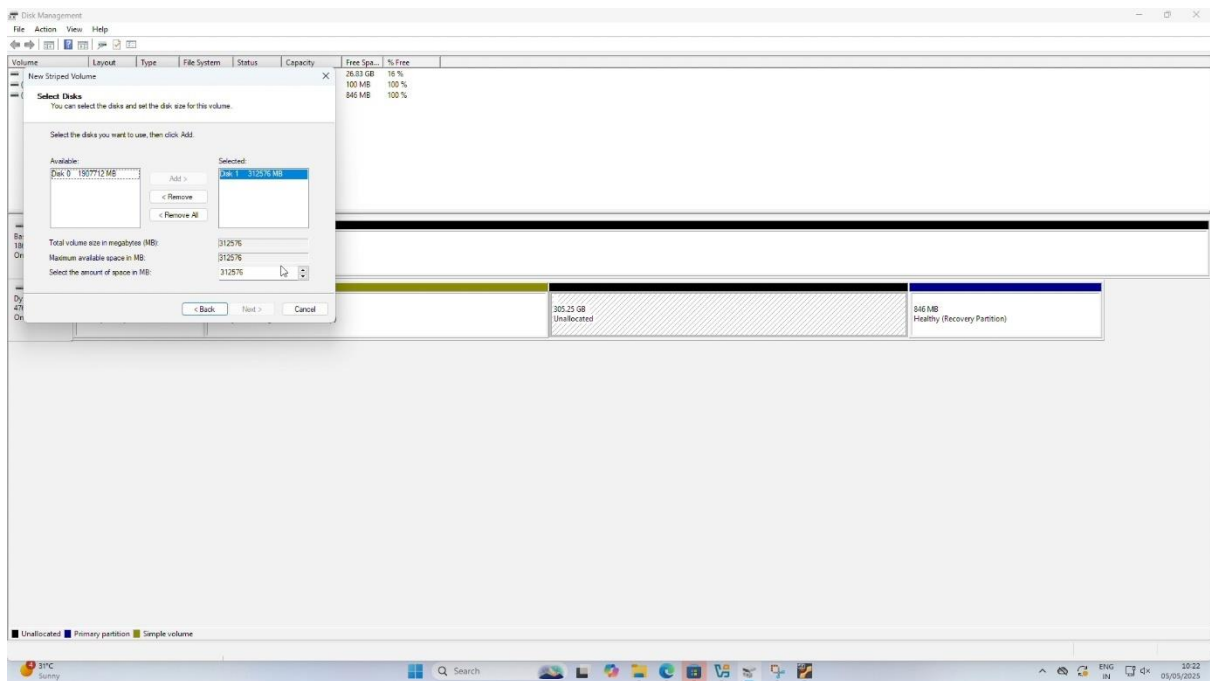


CREATE A STRIPED VOLUME

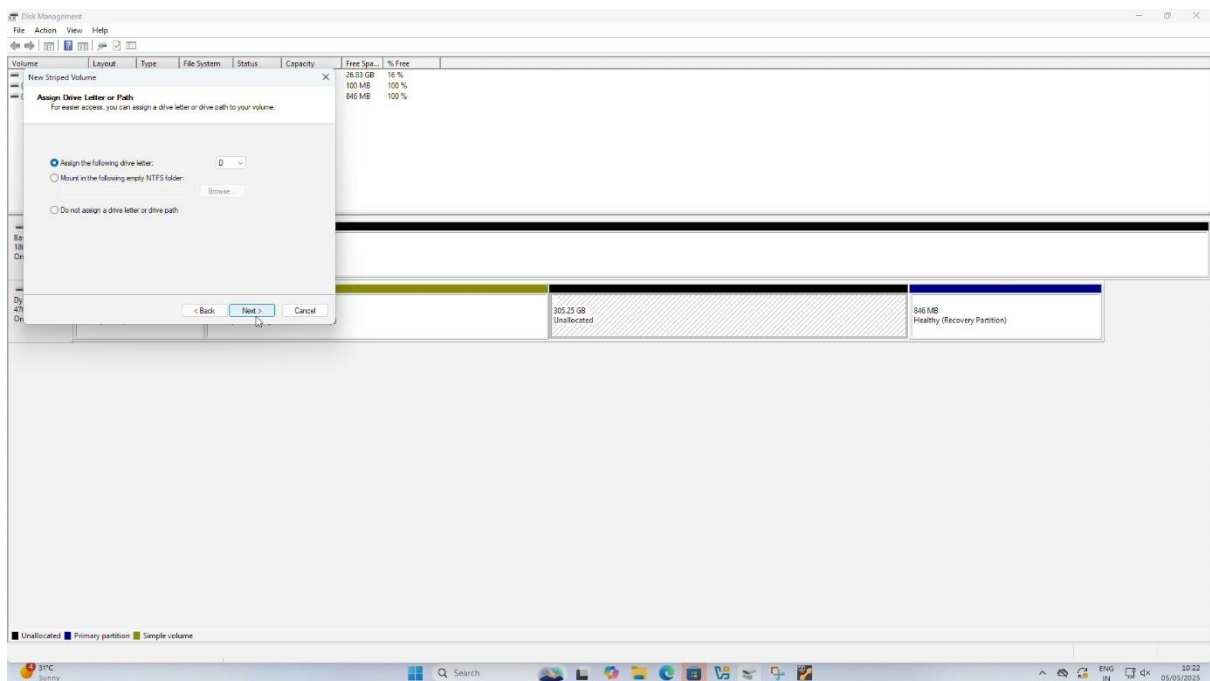
- Step 1 : Press Windows + X and select Disk Management to open the disk management console.
- Step 2 : Right-click on unallocated space and choose New Striped Volume to start the volume creation wizard.



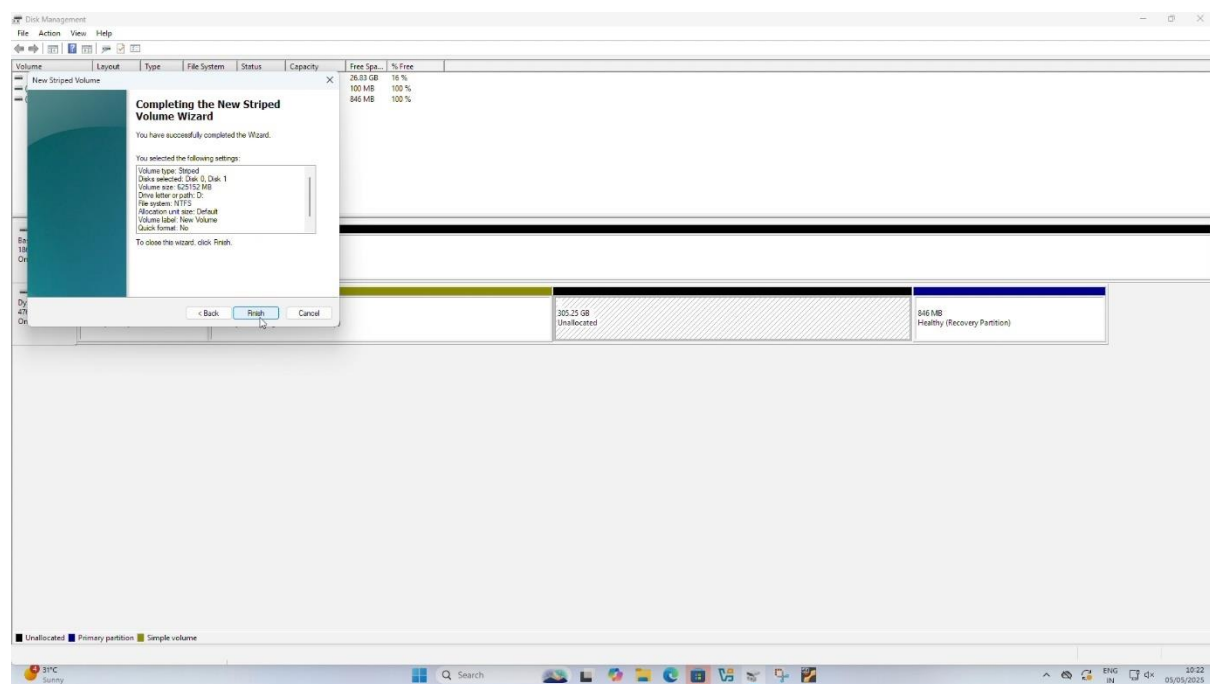
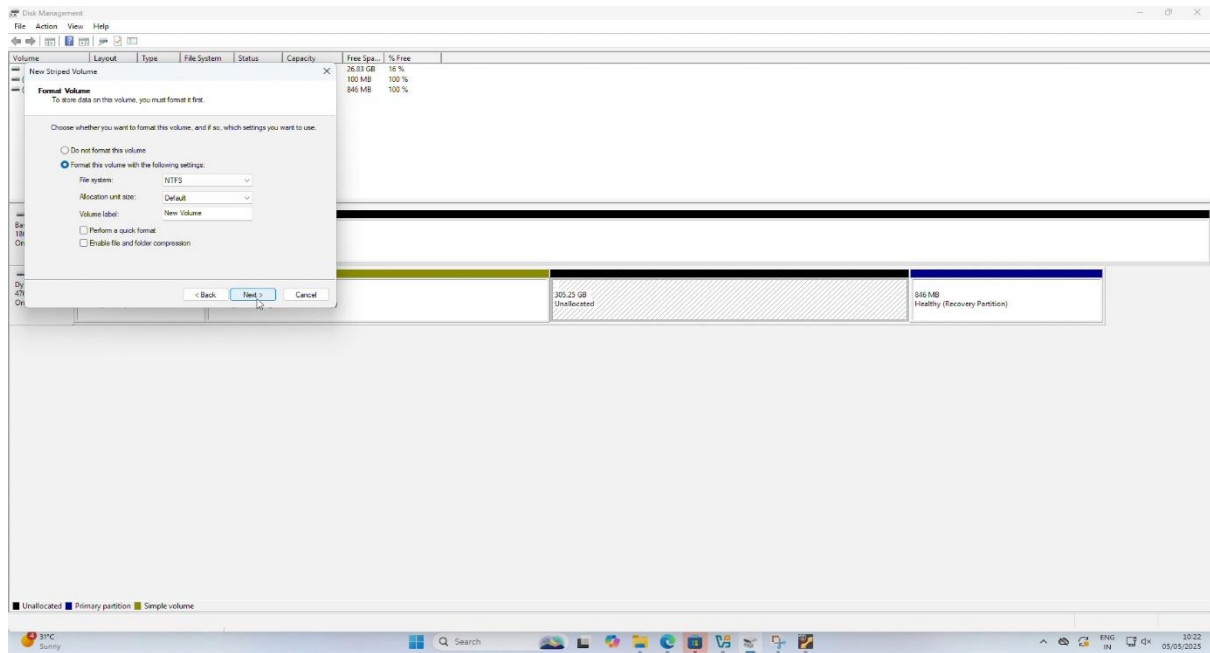
Step 3 : Select the disks you want to stripe and click Next.



Step 4 : Assign a drive letter or mount path to the new striped volume and click Next.



Step 5 : Choose the file system (e.g., NTFS) and perform a quick format, then click Next and Finish to create the striped volume.



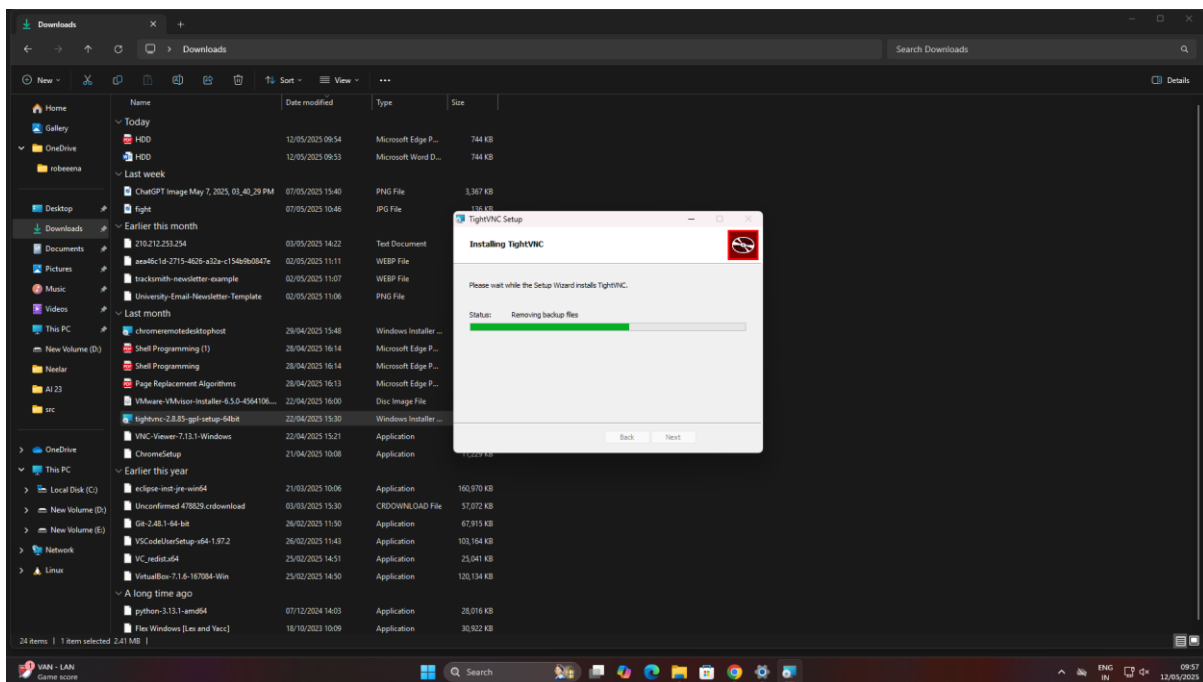
RESULT

DESKTOP VIRTUALIZATION USING VNC

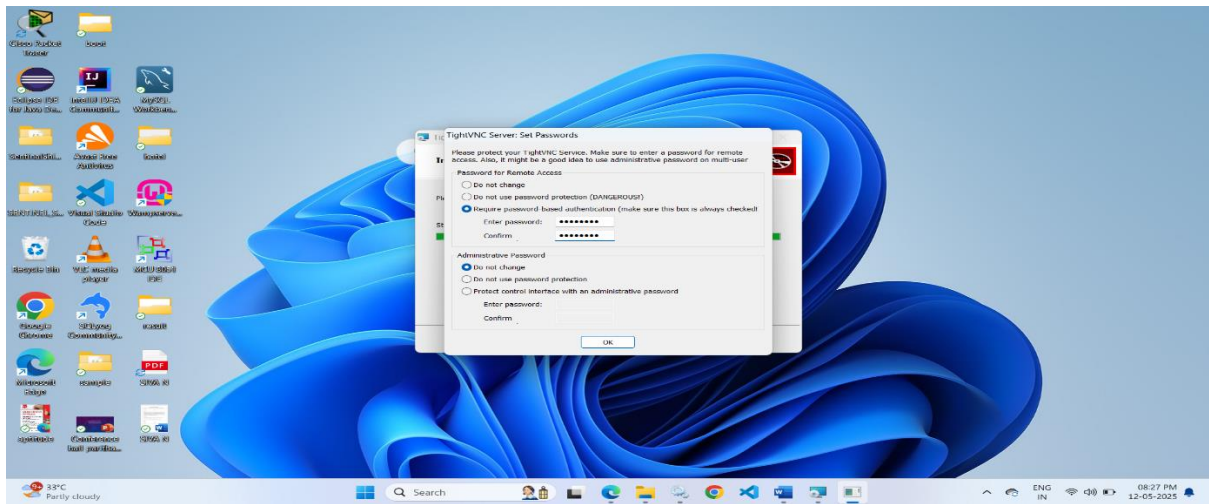
AIM

PROCEDURE

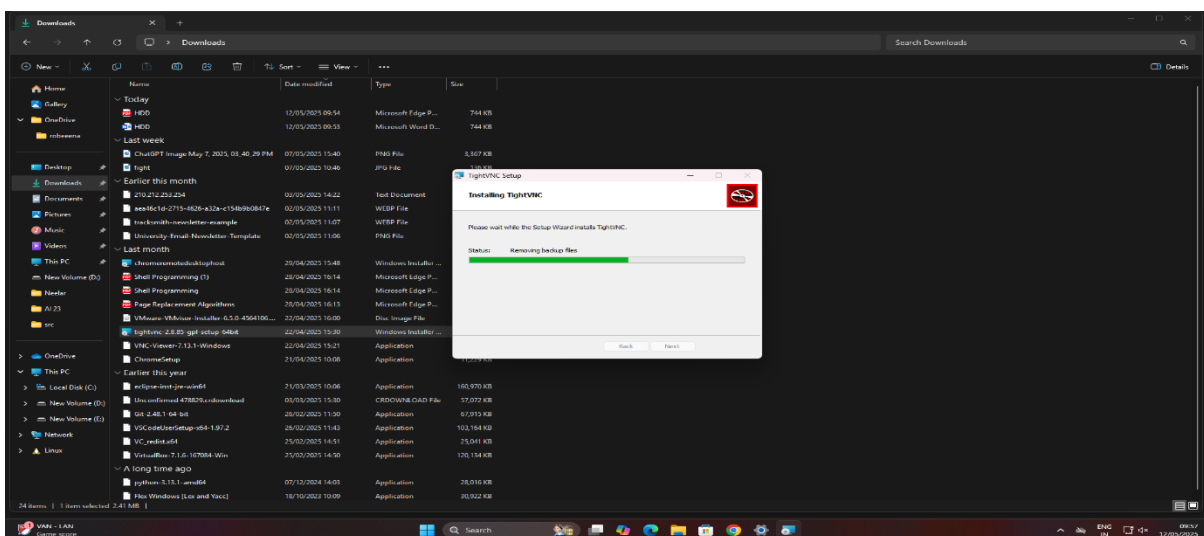
Step 1 : Go to the TightVNC website and download and Install TightVNC for Windows.



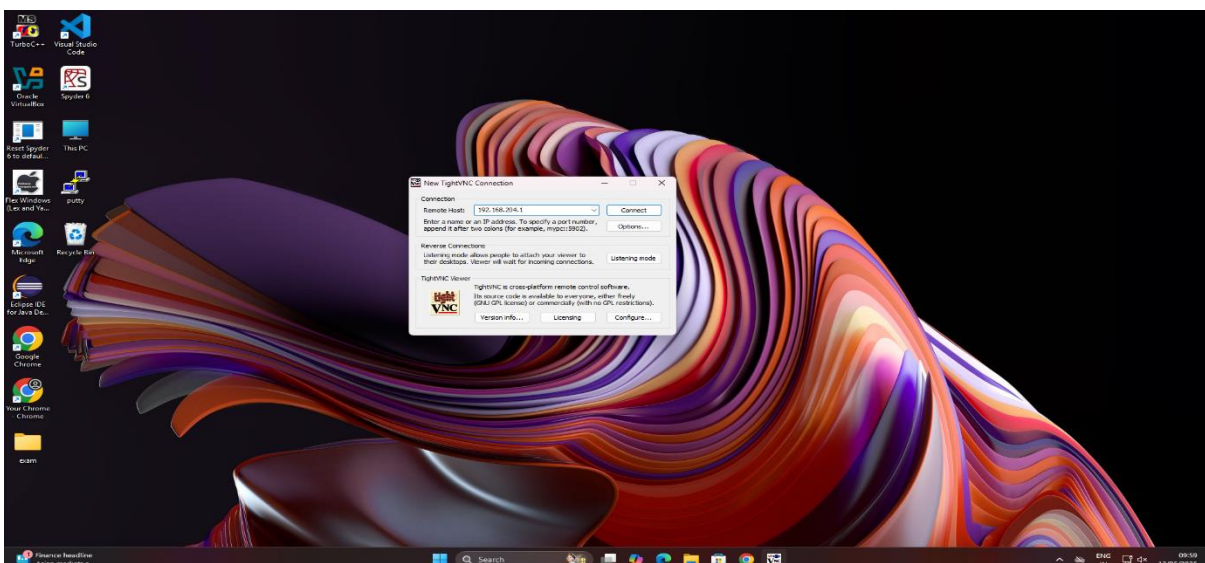
Step 2 : After installation, open TightVNC and set a VNC password for remote access to your Windows machine.



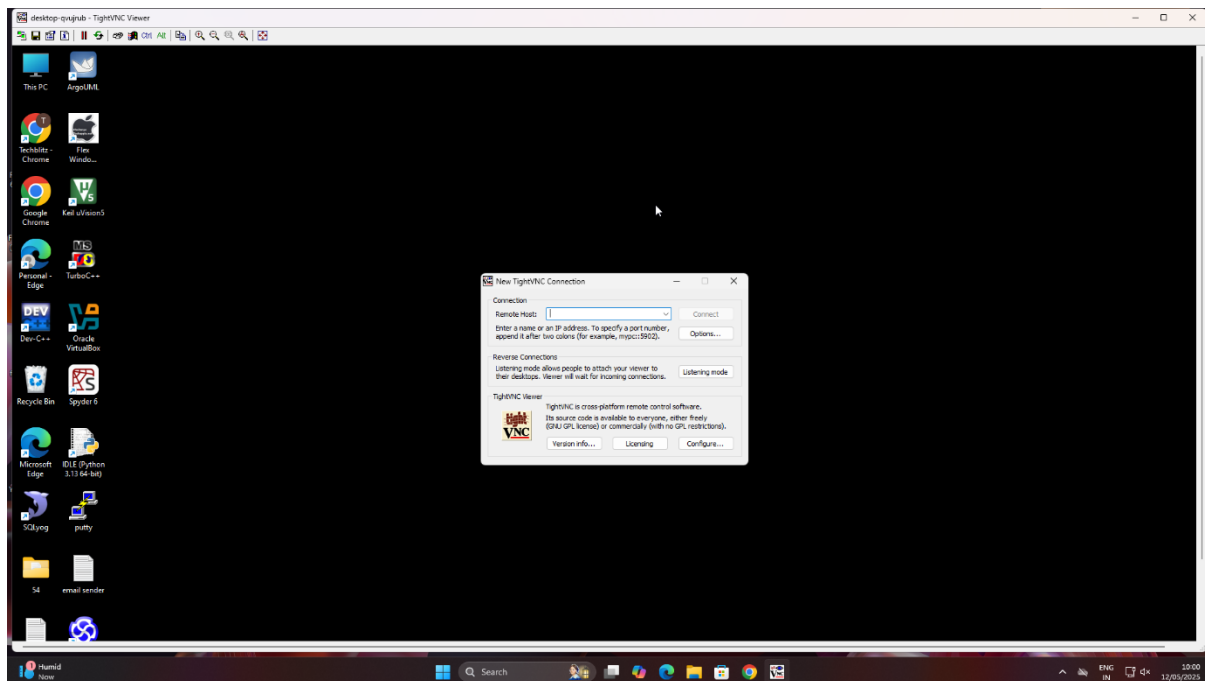
Step 3 : Go to the TightVNC website and download TightVNC for your client device.



Step 4 : Open TightVNC on client device and enter the IP Address of the Host then click connect.



Step 5 : You should now be connected to your Windows desktop remotely and able to control it.



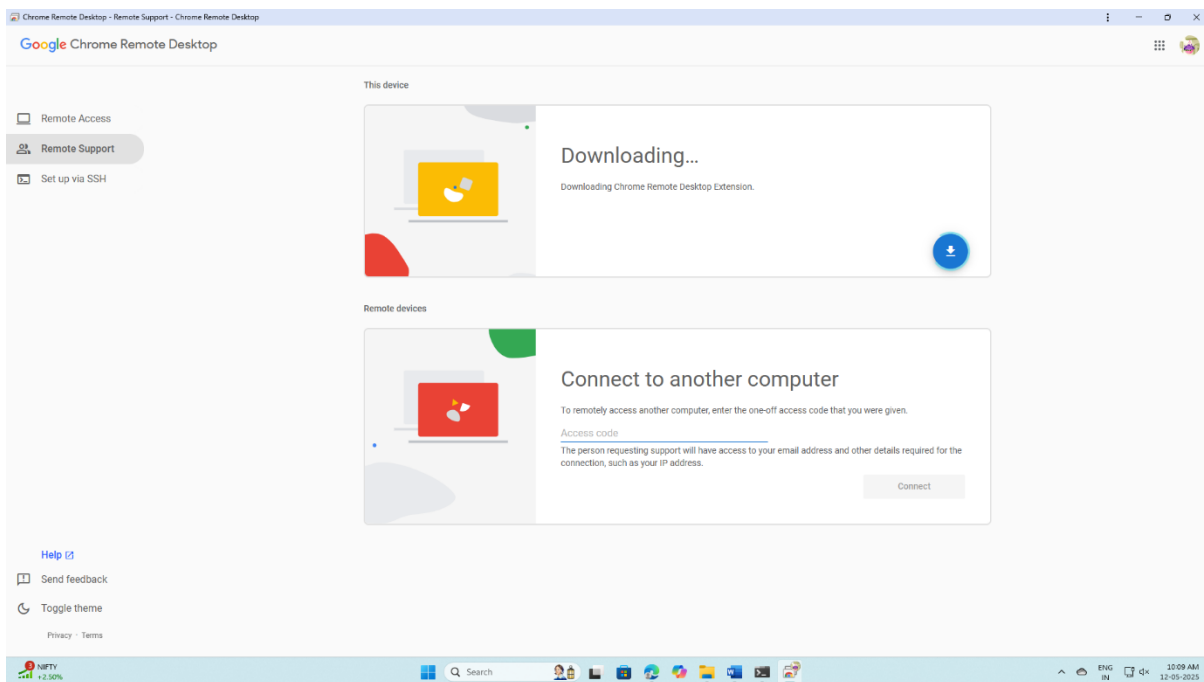
RESULT

DESKTOP VIRTUALIZATION USING CHROME REMOTE DESKTOP

AIM

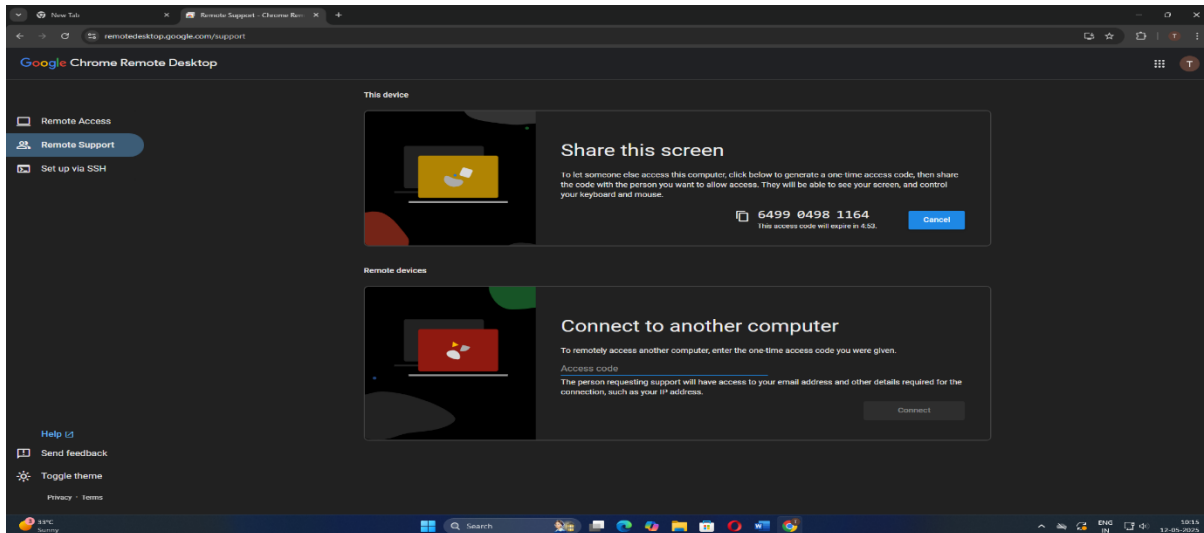
PROCEDURE

Step 1 : Go to the Chrome Web Store and install the Chrome Remote Desktop extension.



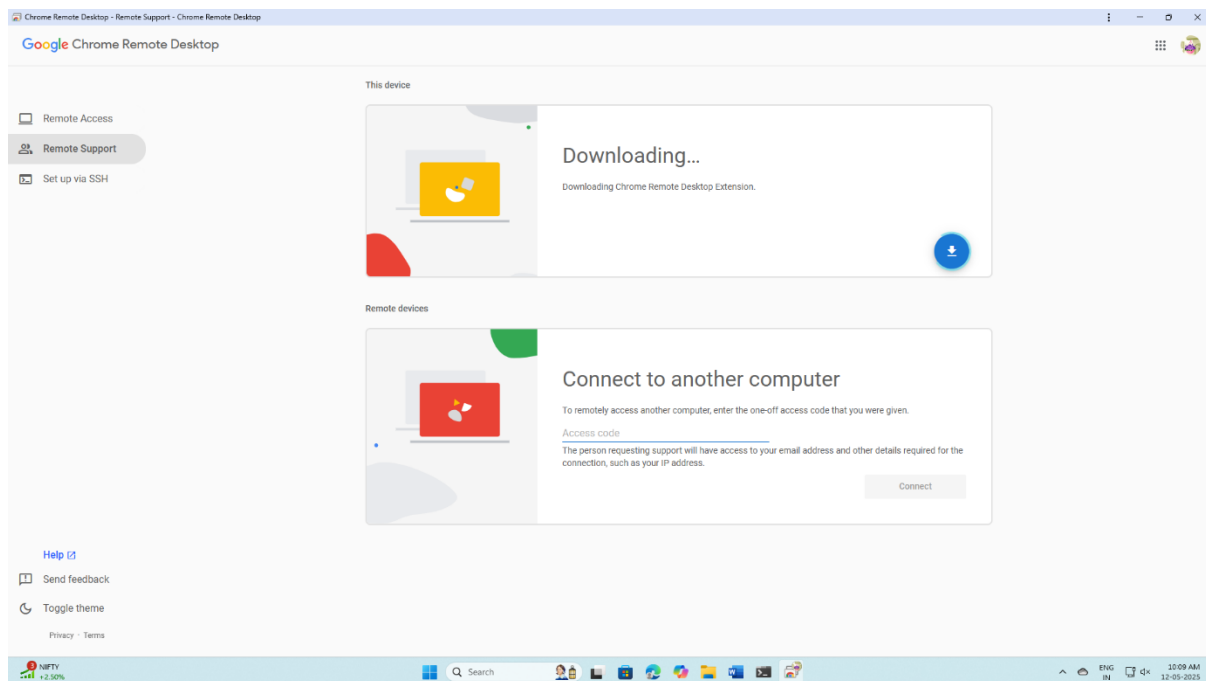
Step 2 : Open the extension, sign in with your Google account, and click Enable Remote Connections.

Step 3 : Set a PIN for secure remote access to your machine.



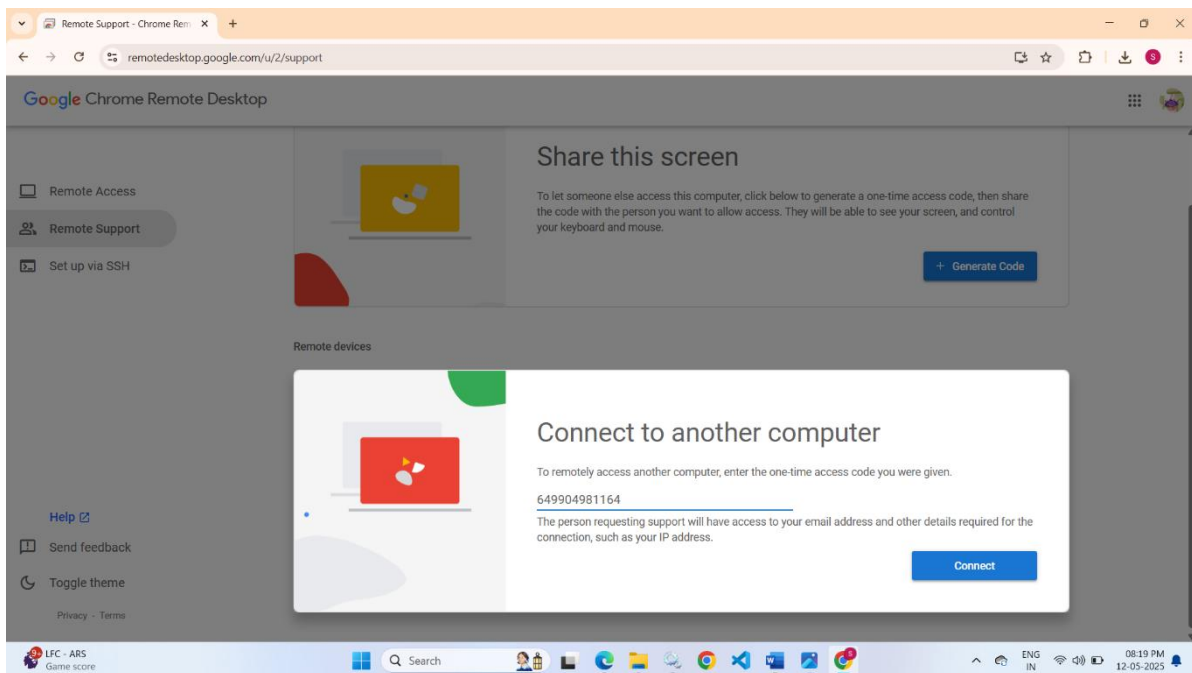
Step 4 : Click Done, and the host machine is now ready to accept remote connections.

Step 5 : In Client Device Go to the Chrome Web Store and install the Chrome Remote Desktop extension if not already installed.

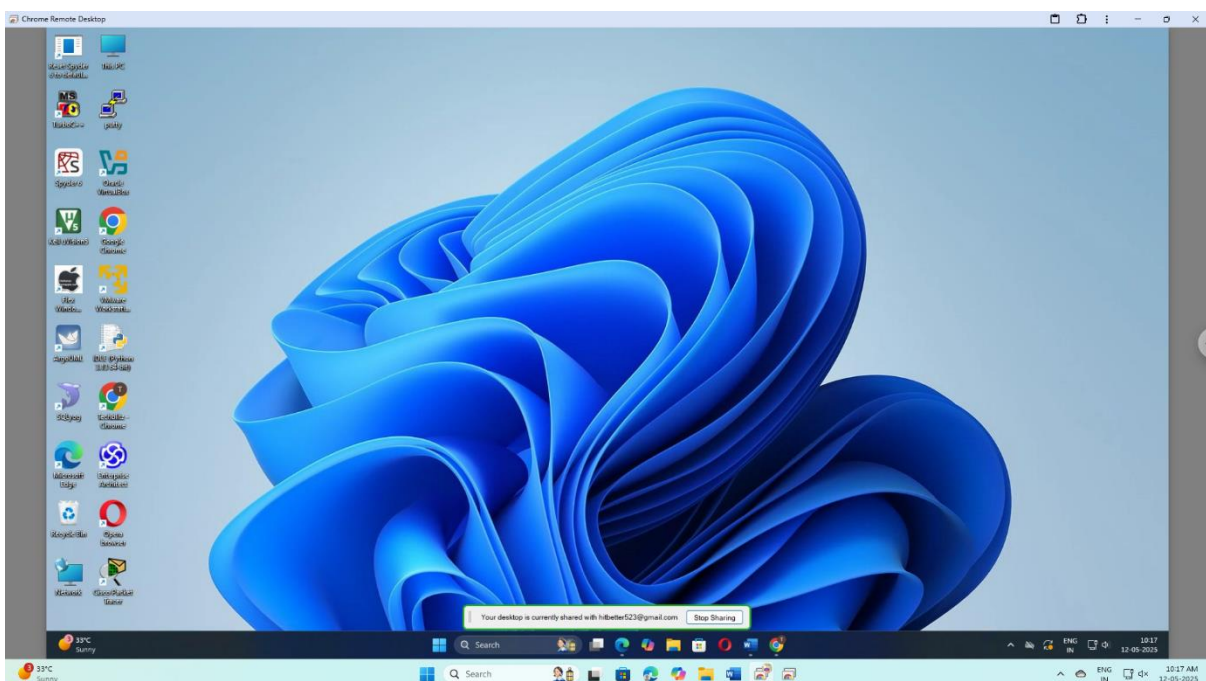


Step 6 : Sign in with the same Google account used on the host machine.

Step 7 : In the Chrome Remote Desktop Interface Click on Acces My Computer then enter the Pin.



Step 8 : You should now be connected and can remotely control the host machine's desktop.



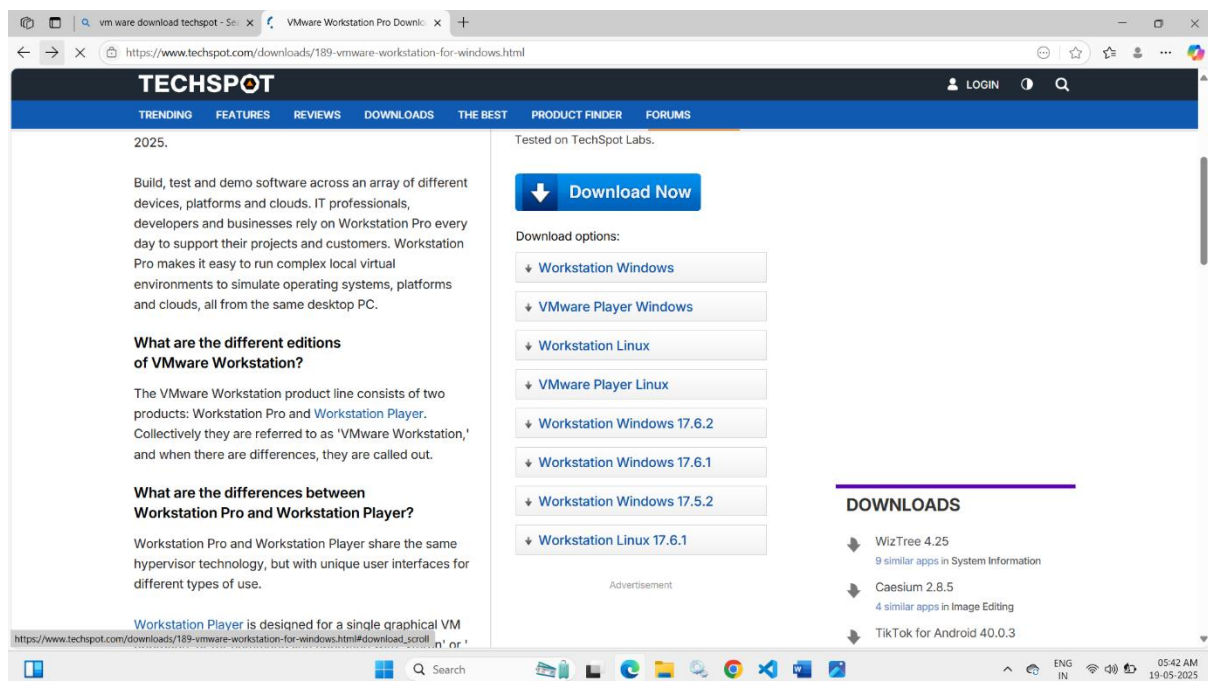
RESULT

CREATE TYPE 2 VIRTUALIZATION ON ESXI 6.5 SERVER

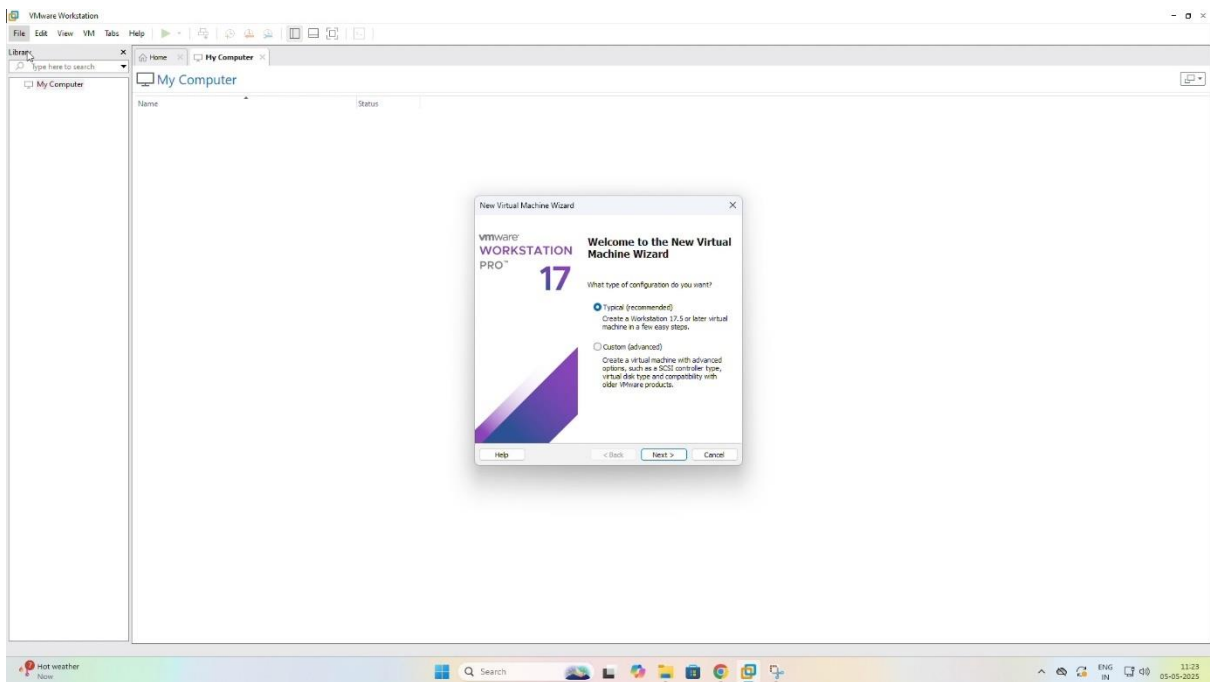
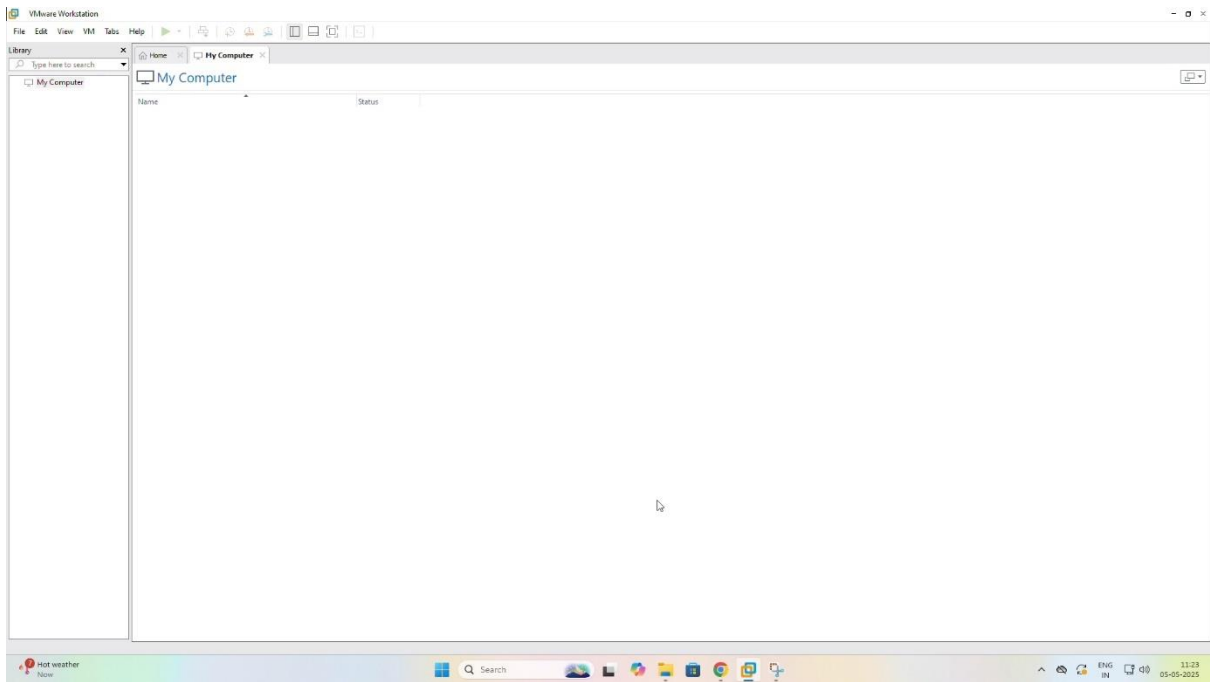
AIM

PROCEDURE

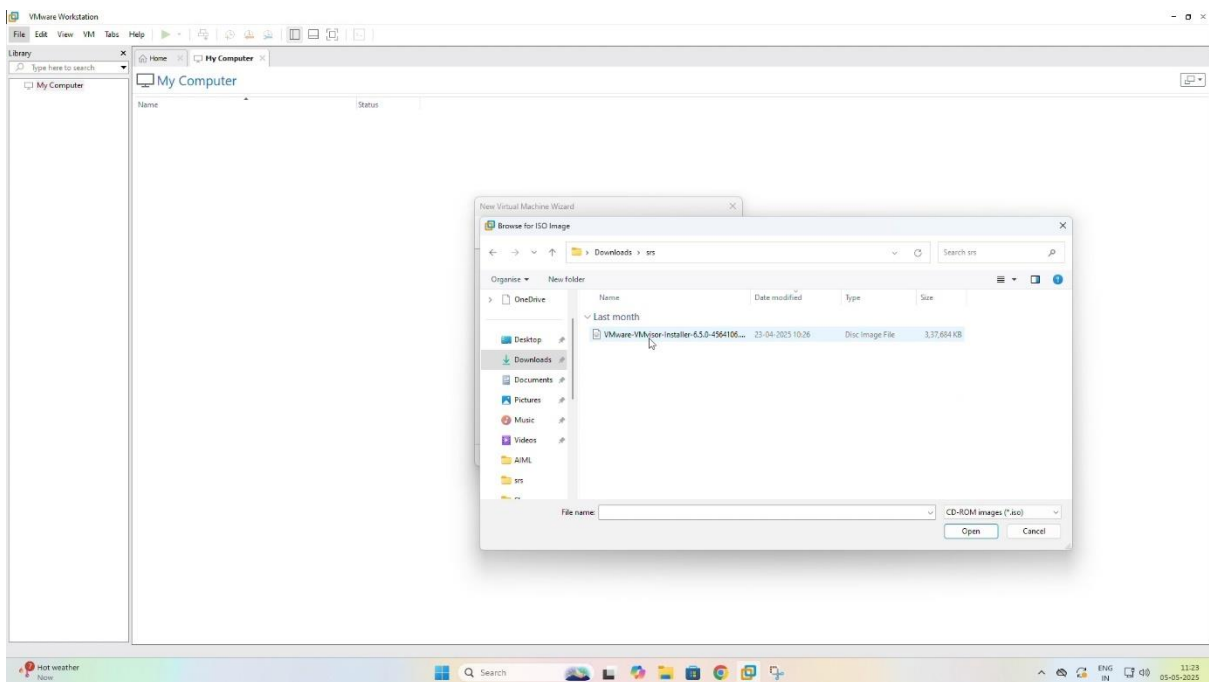
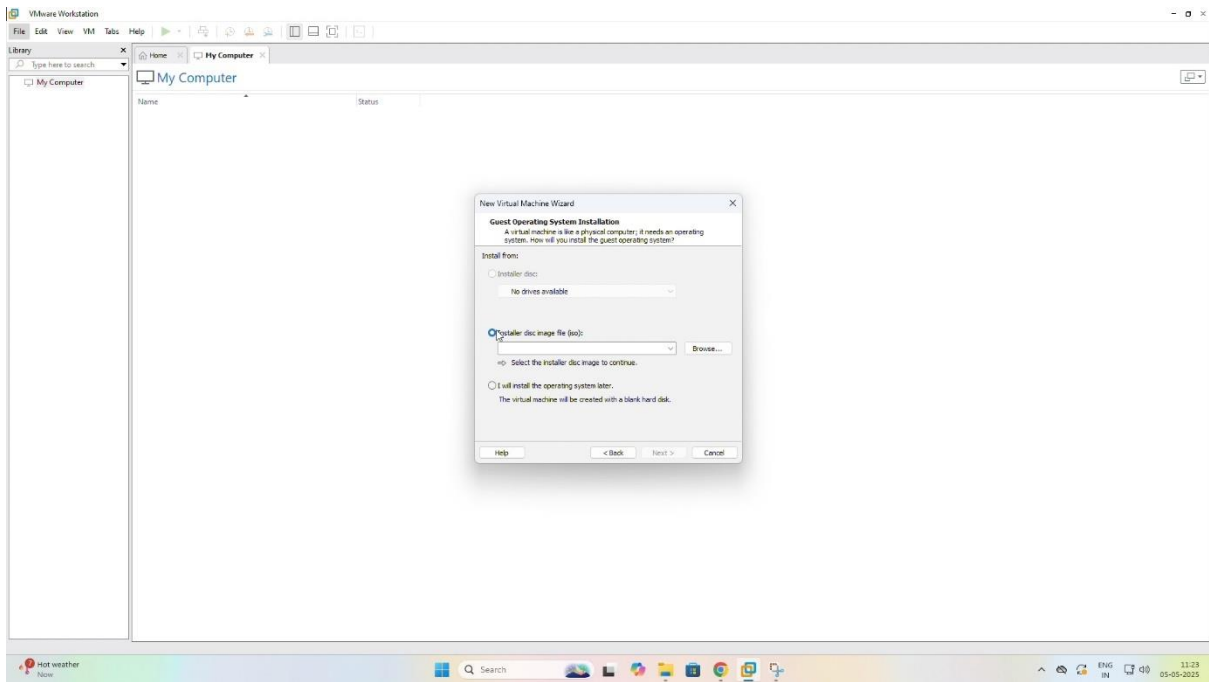
Step 1 : Download and install VMware Workstation or VMware Player on your physical machine, ensuring the host supports the required resources for virtualization.



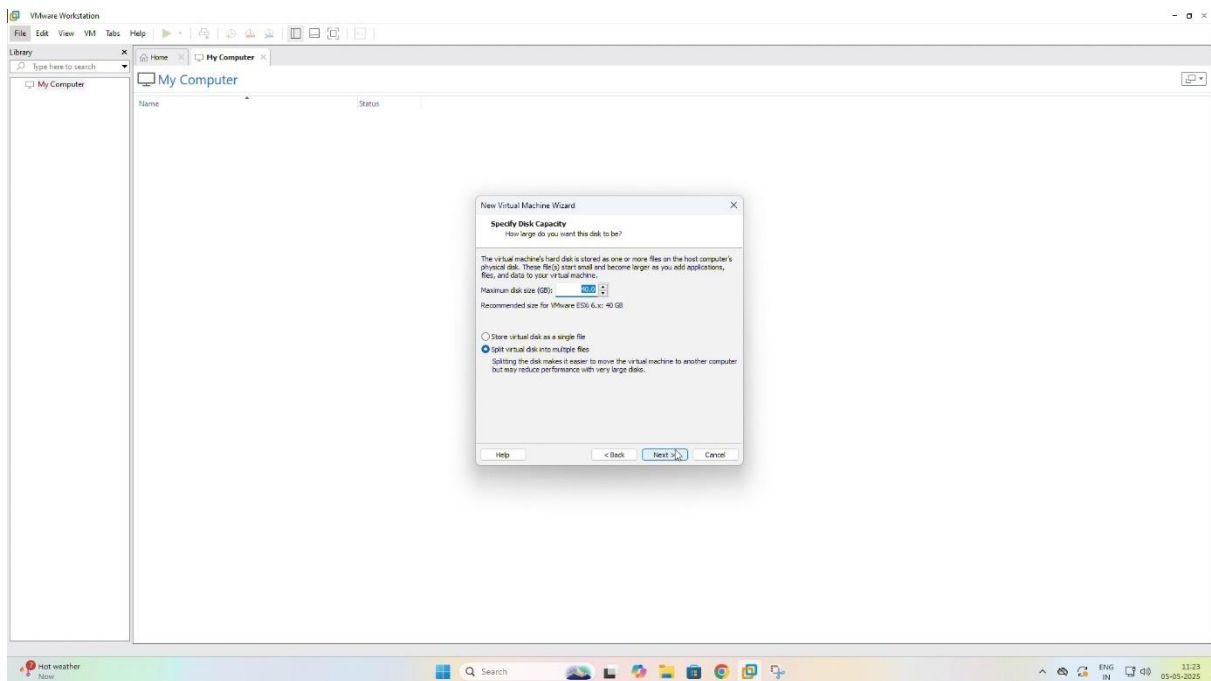
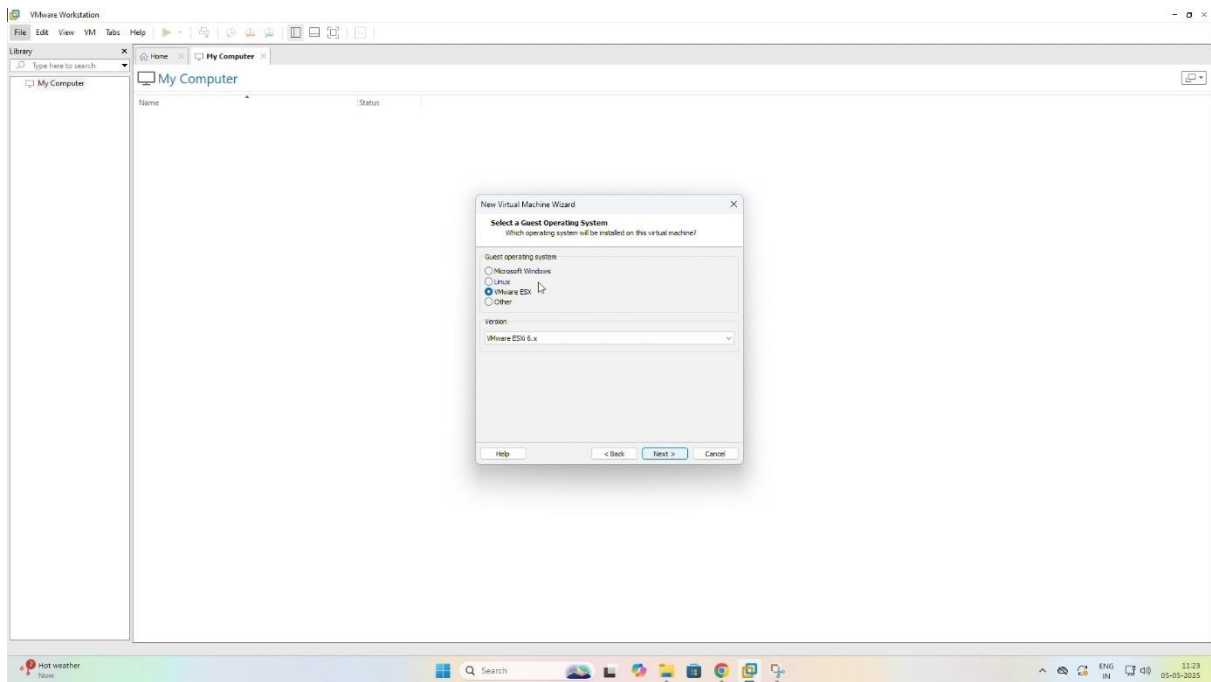
Step 2 : Open VMware Workstation, click on Create a New Virtual Machine, select Typical, and proceed with the setup process for your ESXi installation.

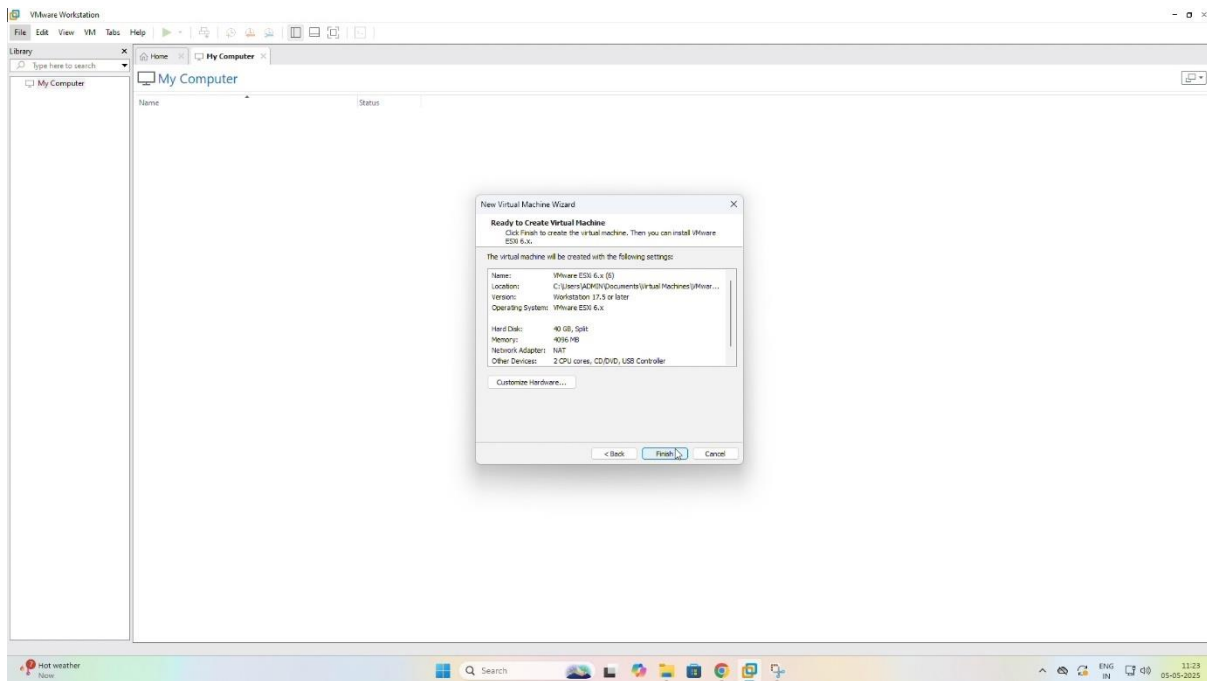


Step 3 : Select Installer disc image file (iso) and browse to the location of your ESXi ISO file to set it as the installation media

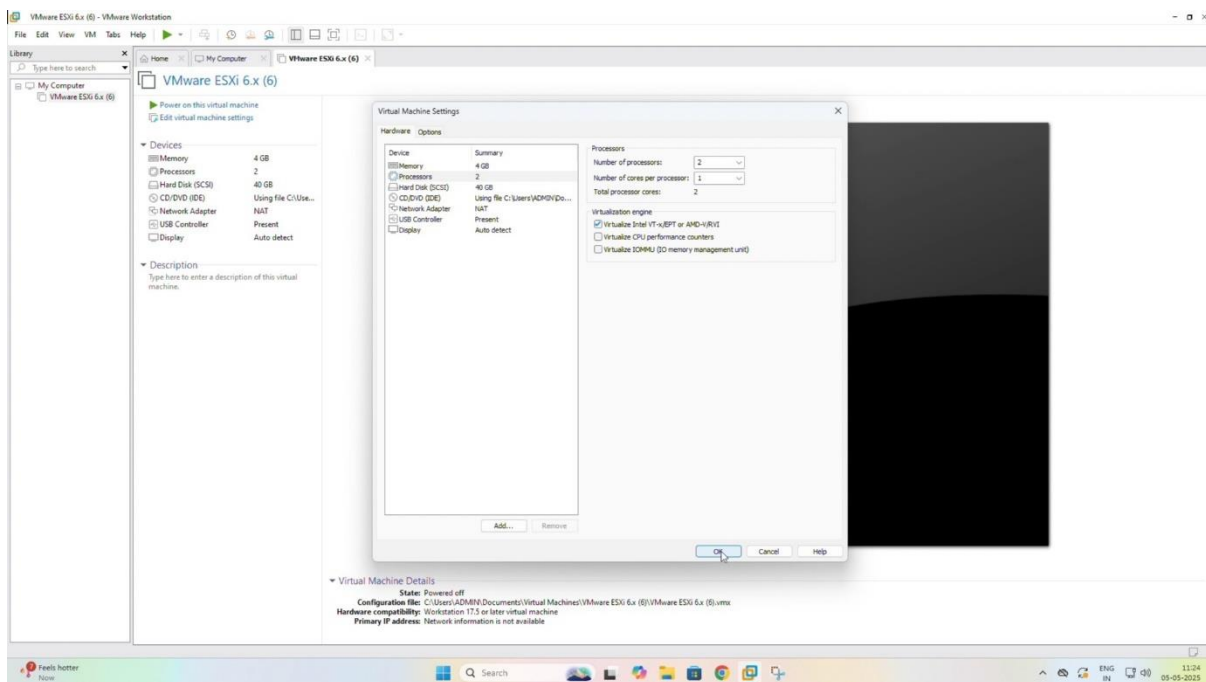


Step 4 : Allocate at least 2 CPU cores, 4GB RAM, 20GB of disk space, and configure the network adapter (either Bridged or NAT) for your virtual machine.

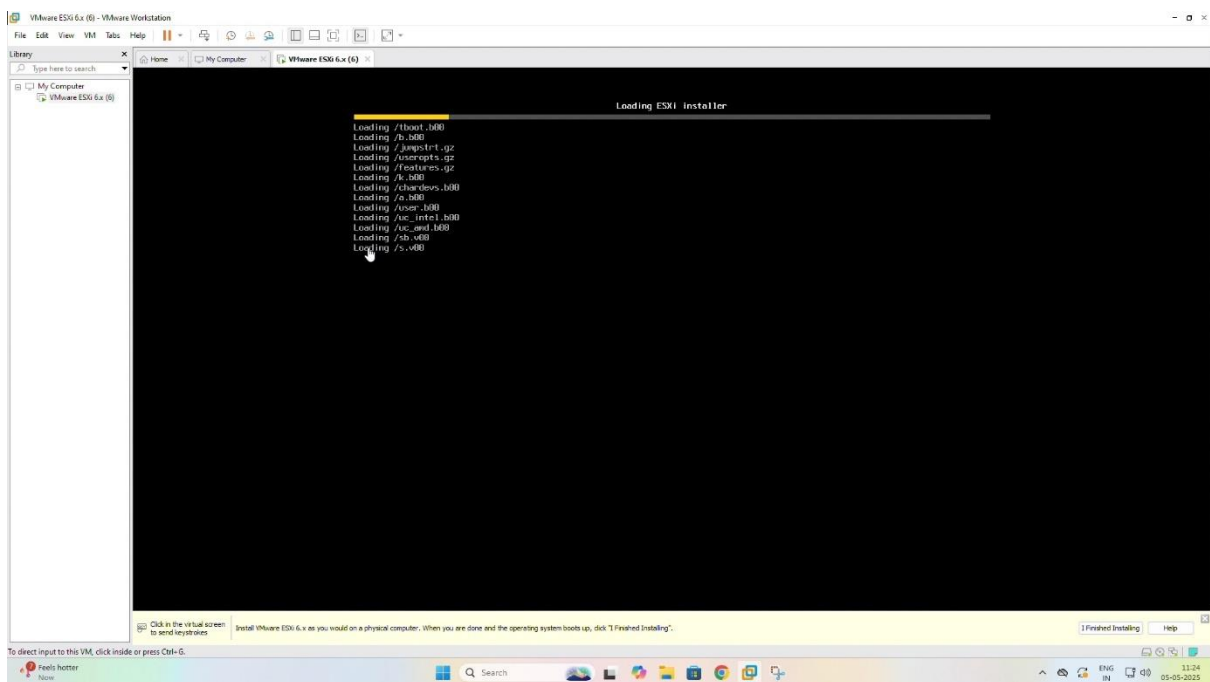
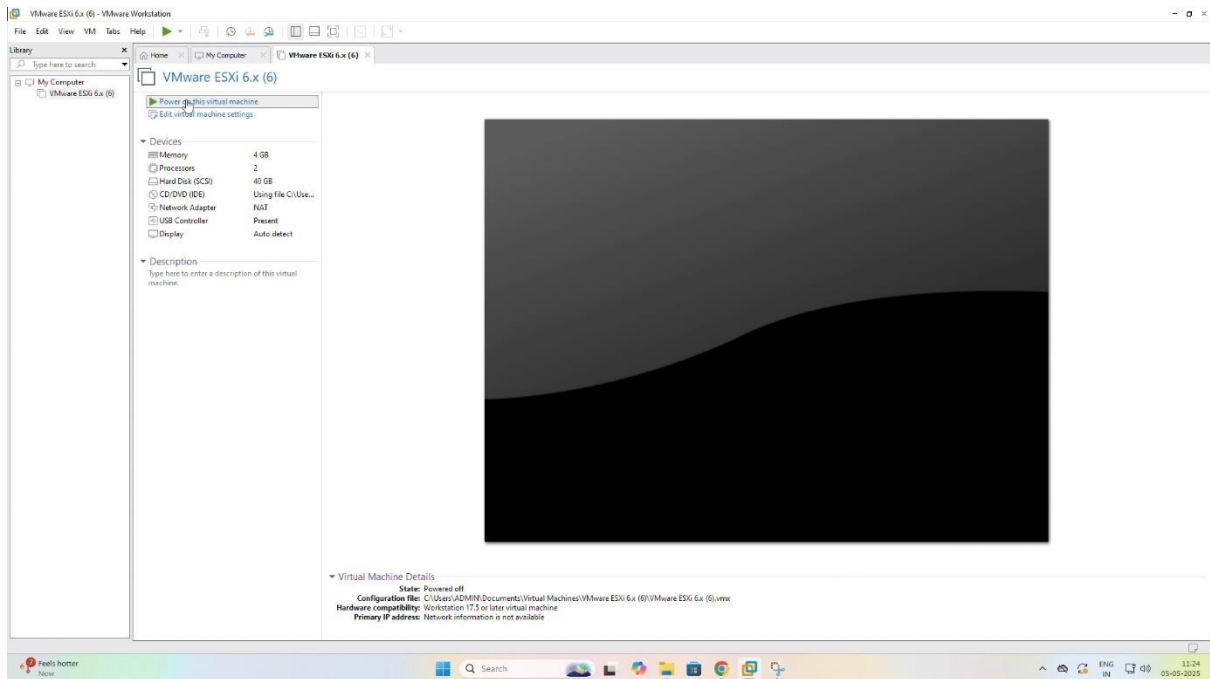




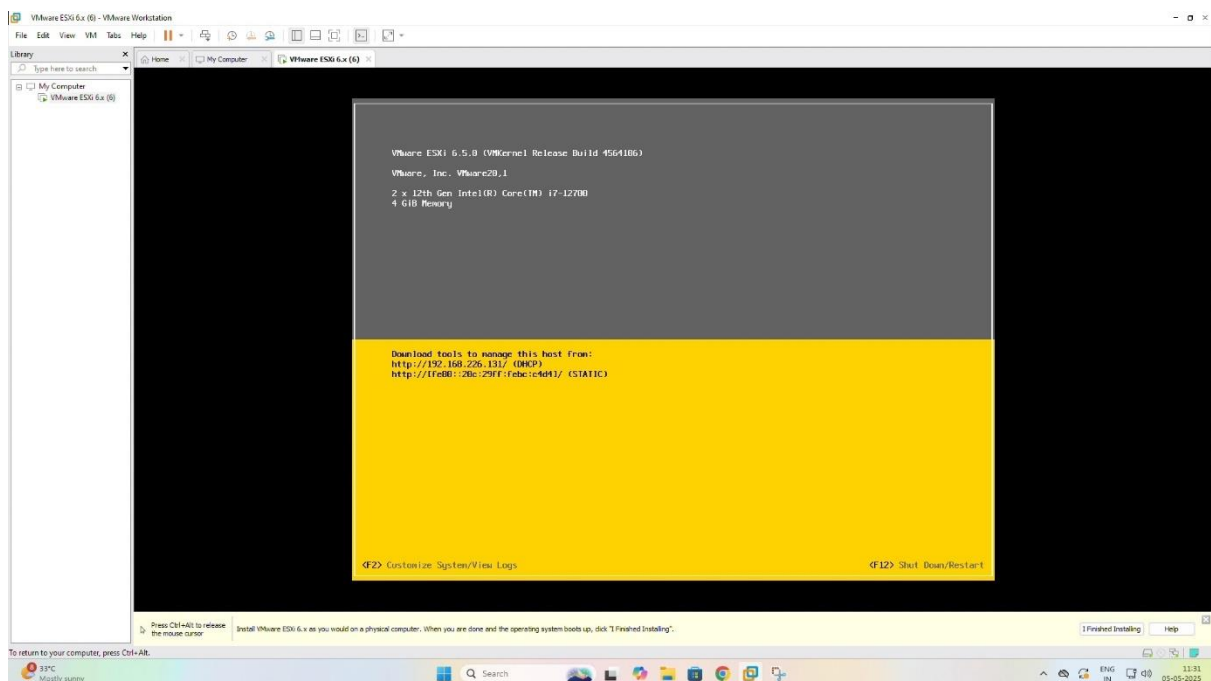
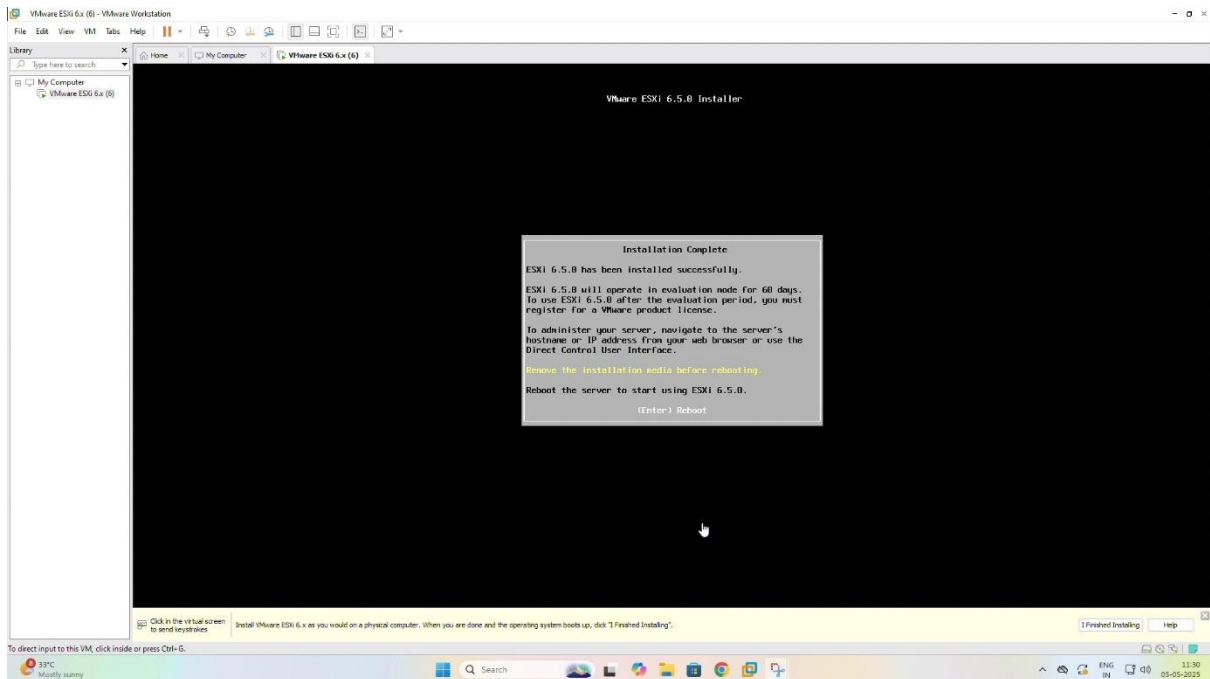
Step 5 : Enable nested virtualization by checking the "Virtualize Intel VT-x/EPT or AMD-V/RVI" box under the Processors section in VMware Workstation settings.



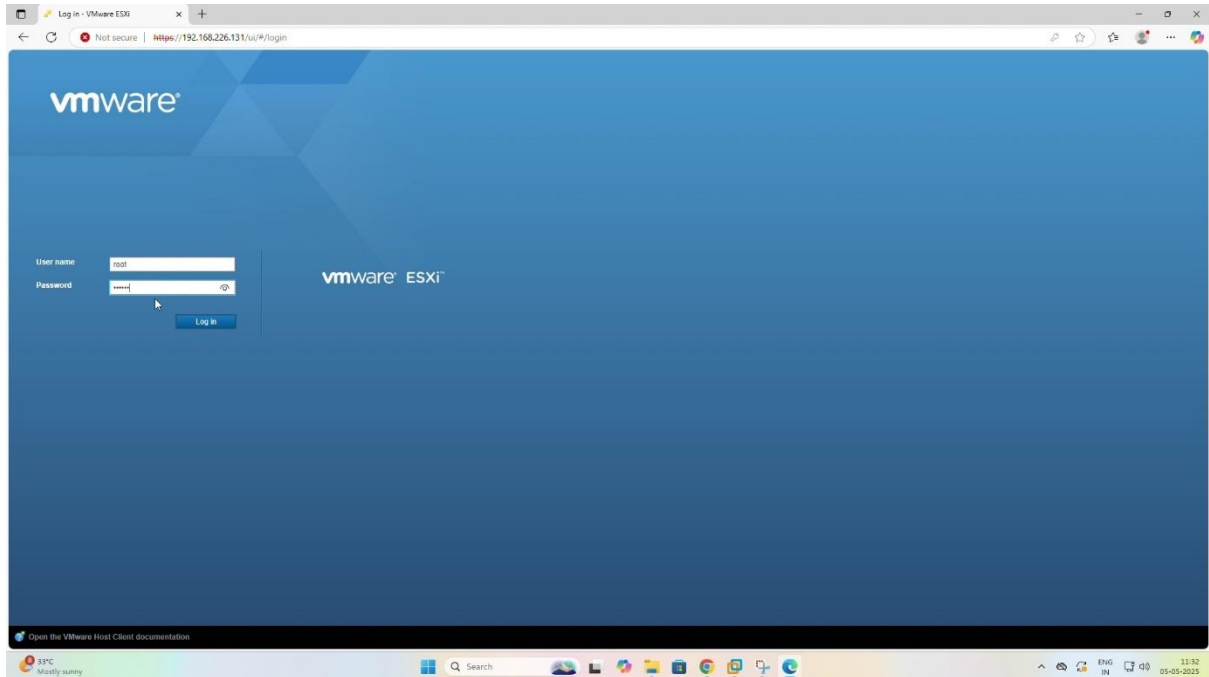
Step 6 : Power on the VM, boot from the ESXi ISO to begin the installation process, and follow the on-screen prompts to install ESXi on the virtual hard disk.



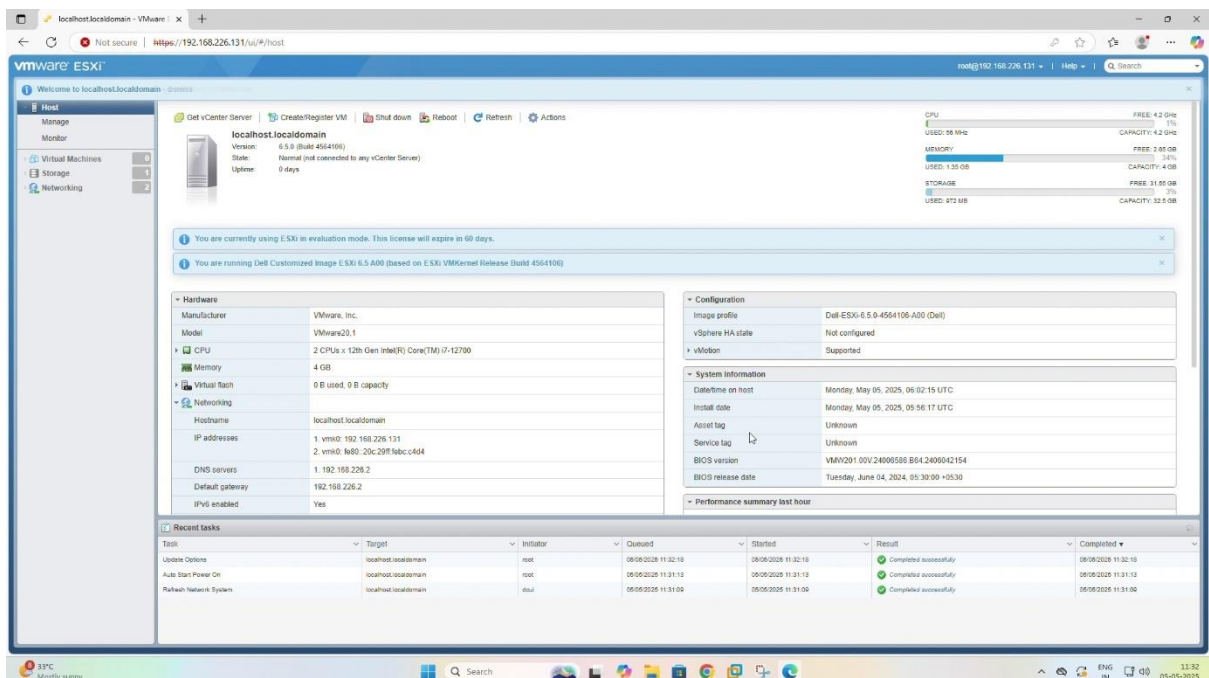
Step 7 : Once installed, the VM will reboot, and the ESXi console will show the IP address to access the ESXi server remotely via a web browser.

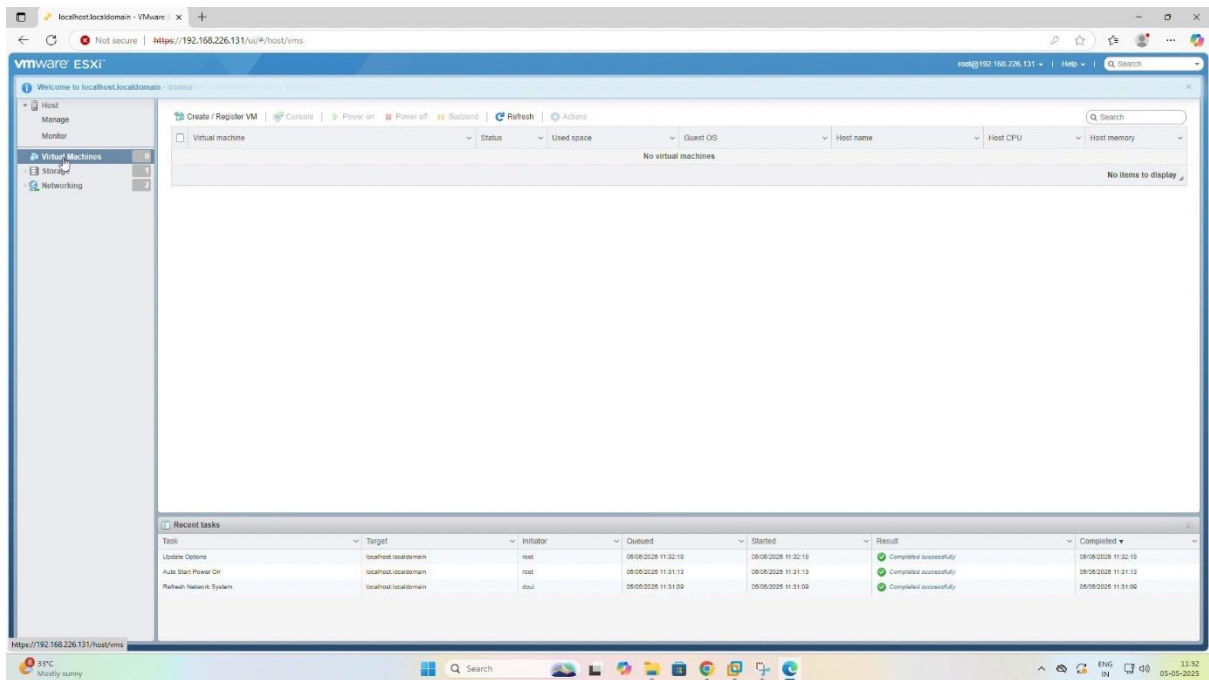


Step 8 : Open a web browser on your host machine, navigate to https://<ESXi_IP>/ui, and log in using the root password set during installation to manage ESXi.

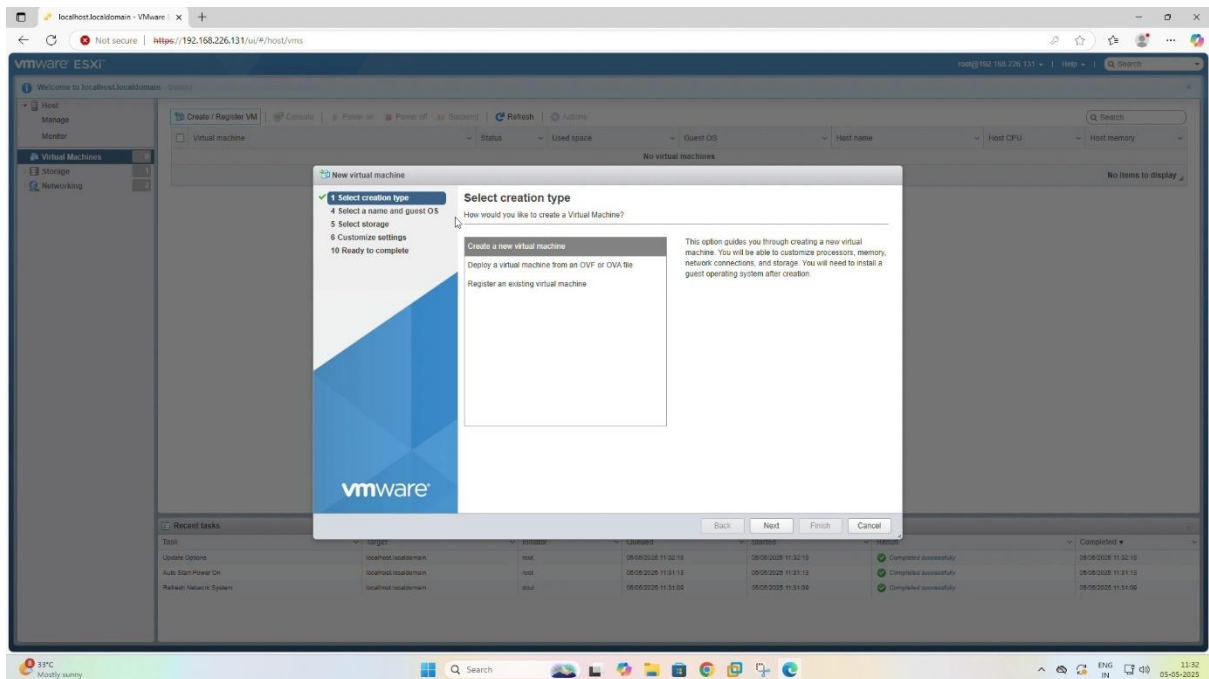


Step 9 : From the ESXi dashboard, click on Virtual Machines in the left sidebar to manage and create new VMs.

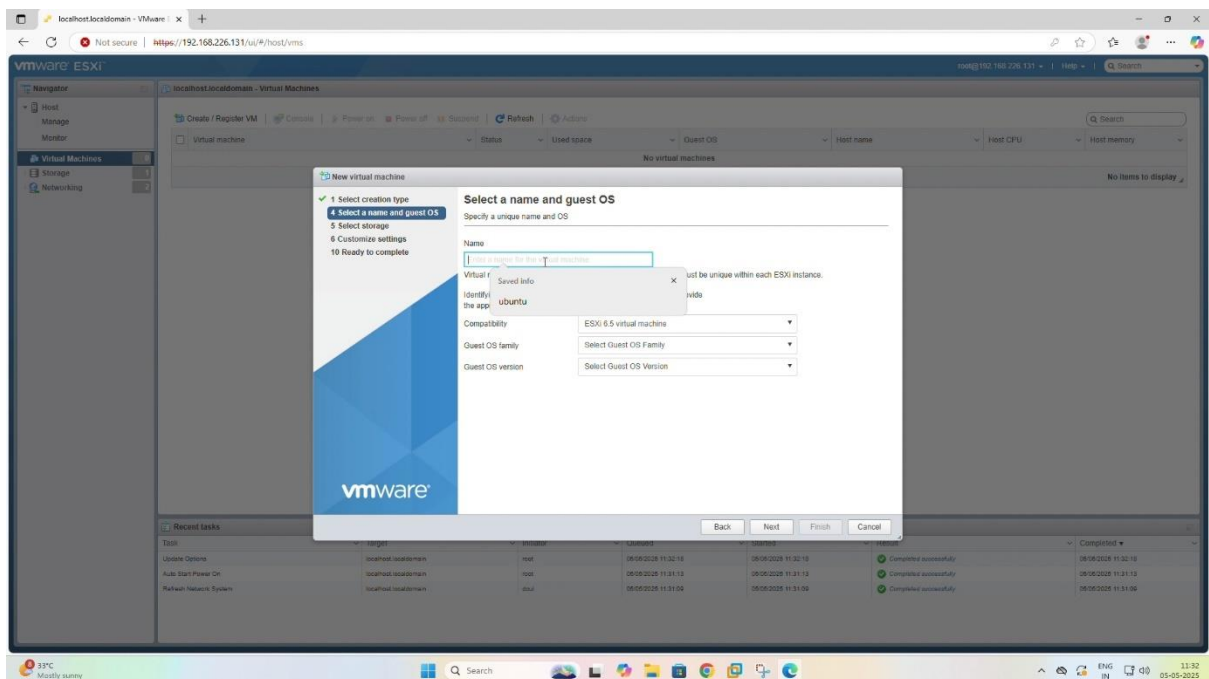




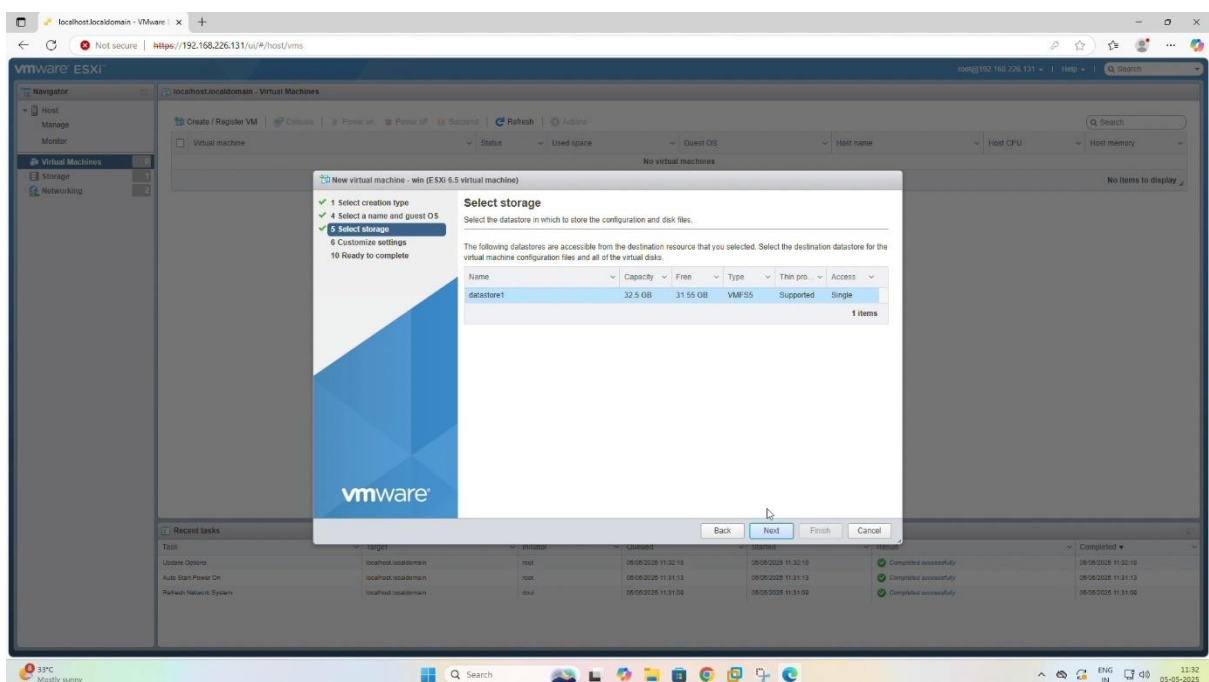
Step 10 : Click the Create/Register VM button at the top-right corner of the Virtual Machines page to start the creation process.



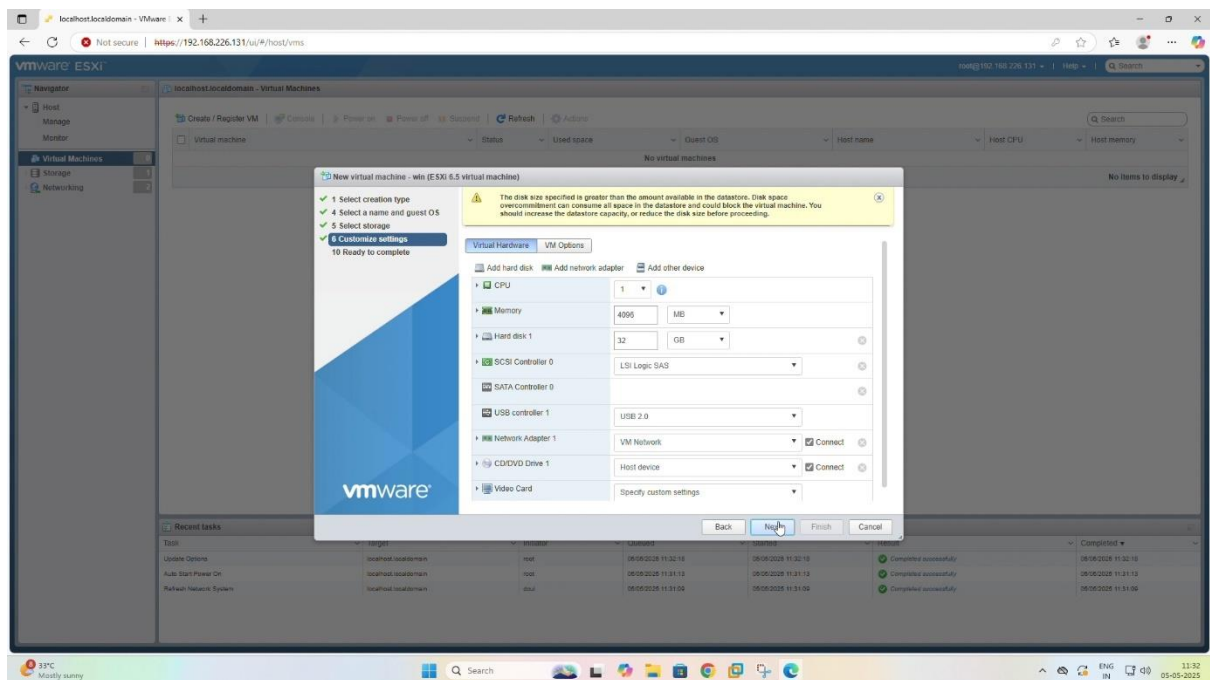
Step 11 : Enter a name for the VM (e.g., “Ubuntu_VM”), select the appropriate compatibility version, and click Next.



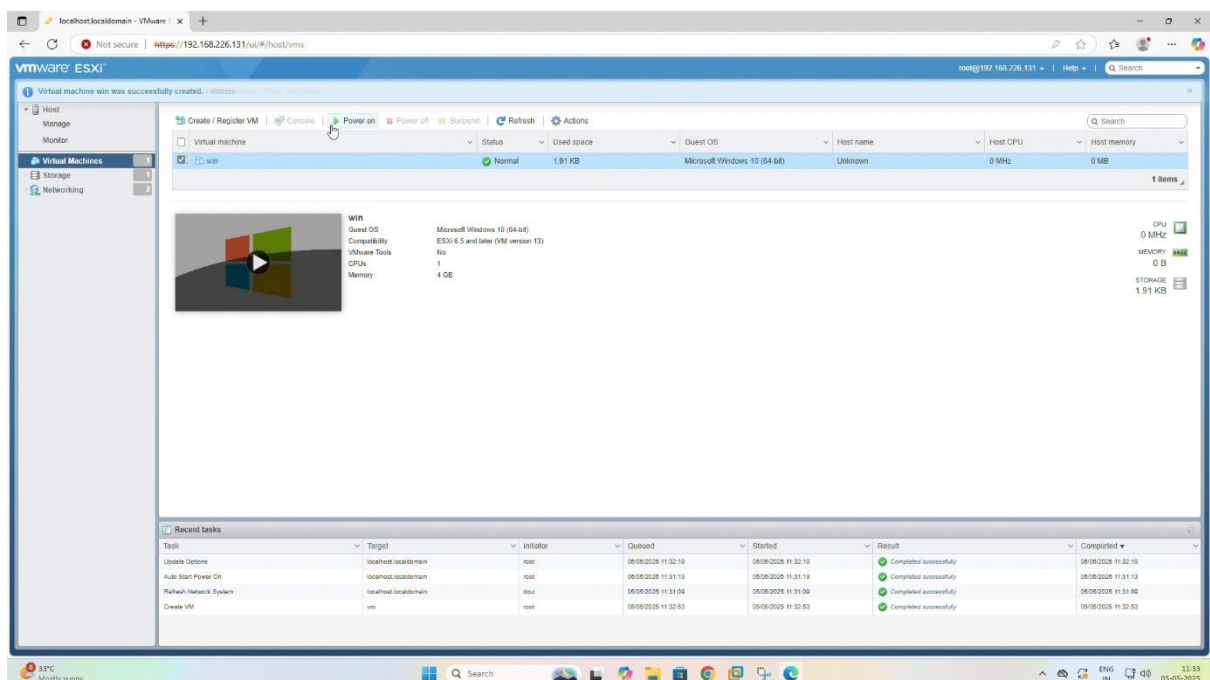
Step 12 : Choose a datastore to store the VM files, then click Next to proceed to the next configuration step.



Step 13 : Allocate the required virtual CPU, memory, disk size, and configure network adapters for the VM as per your needs.



Step 14 : Review your settings, click Finish to create the VM, and the VM will appear in the Virtual Machines list.



Step 14 : Select the newly created VM, click on Power On, to begin installing the OS on the VM.

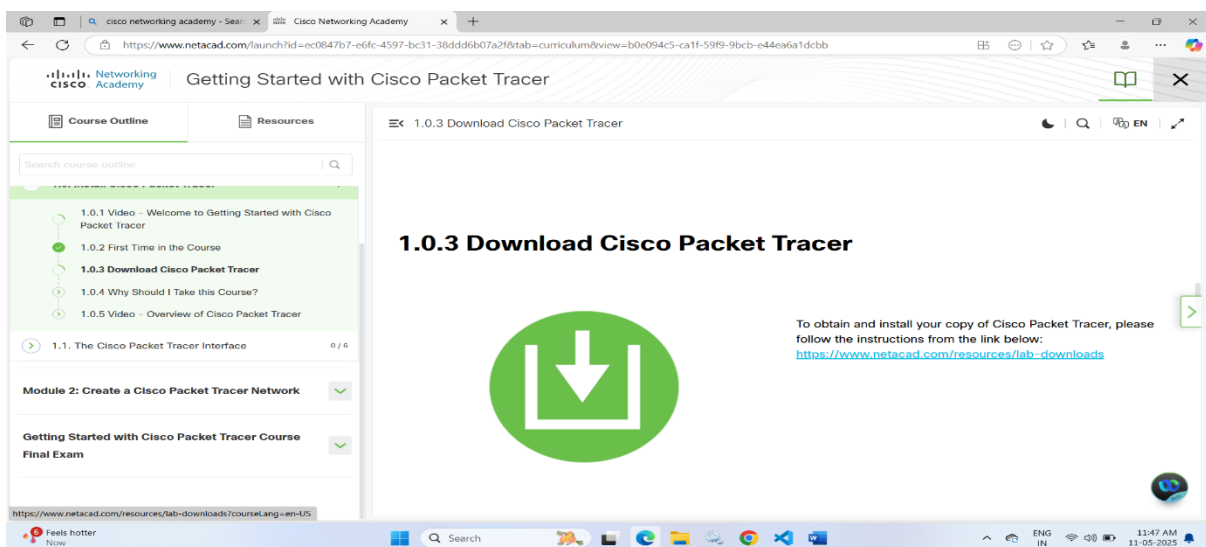
RESULT

CREATE A VLAN IN CISCO PACKET TRACER

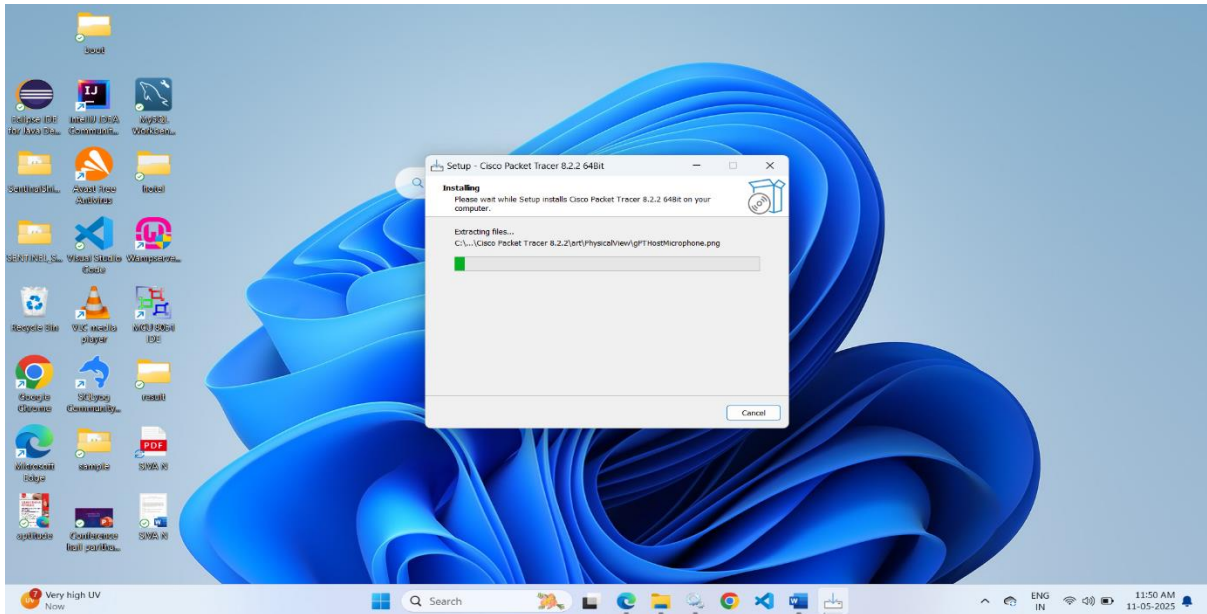
AIM

PROCEDURE

Step 1 : Go to the Cisco NetAcad website, log in or create an account, then download the latest Cisco Packet Tracer version for your OS (Windows/Linux/macOS).

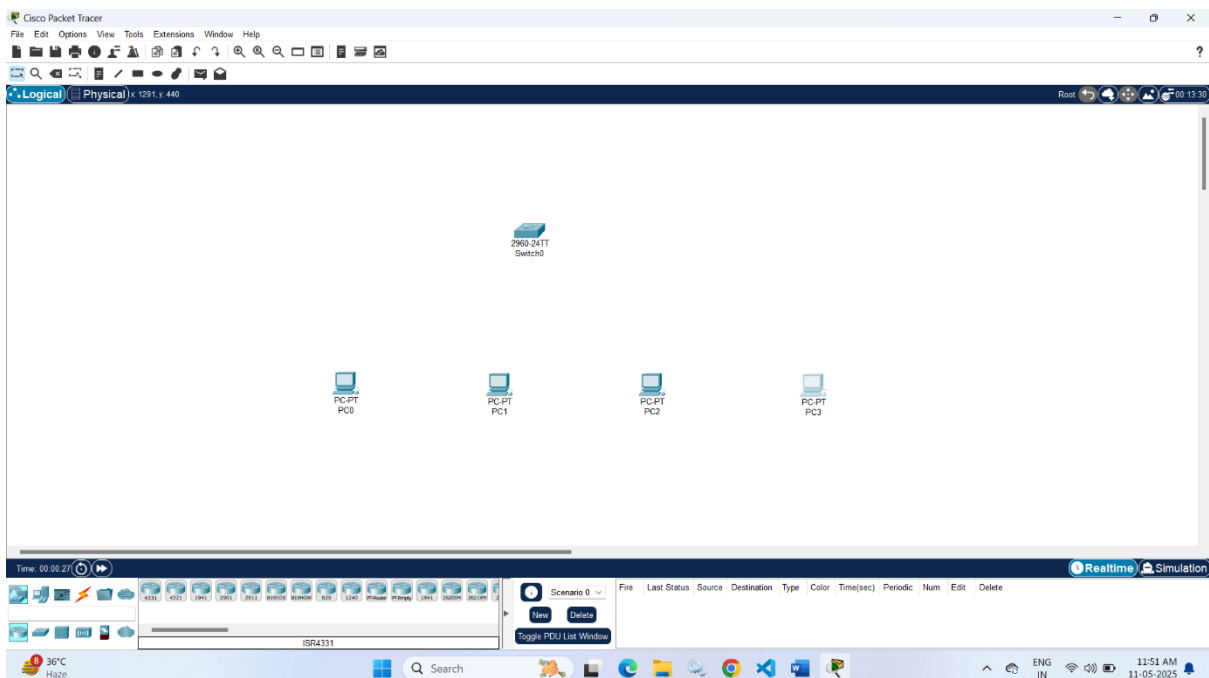


Step 2 : Run the downloaded installer, follow the installation prompts, accept the license agreement, and complete the installation.

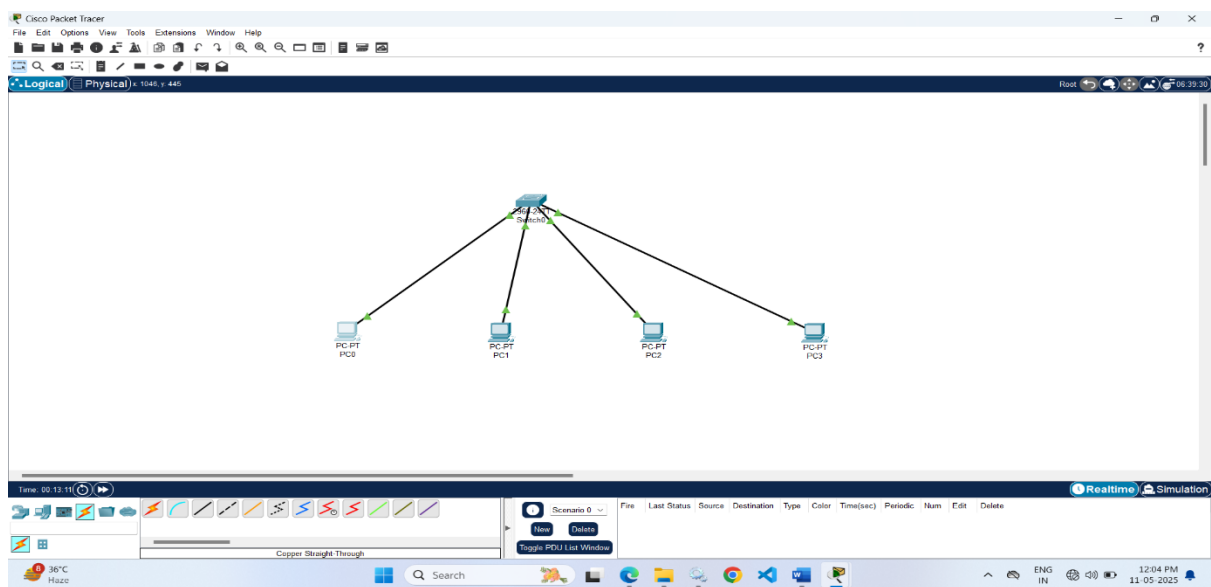
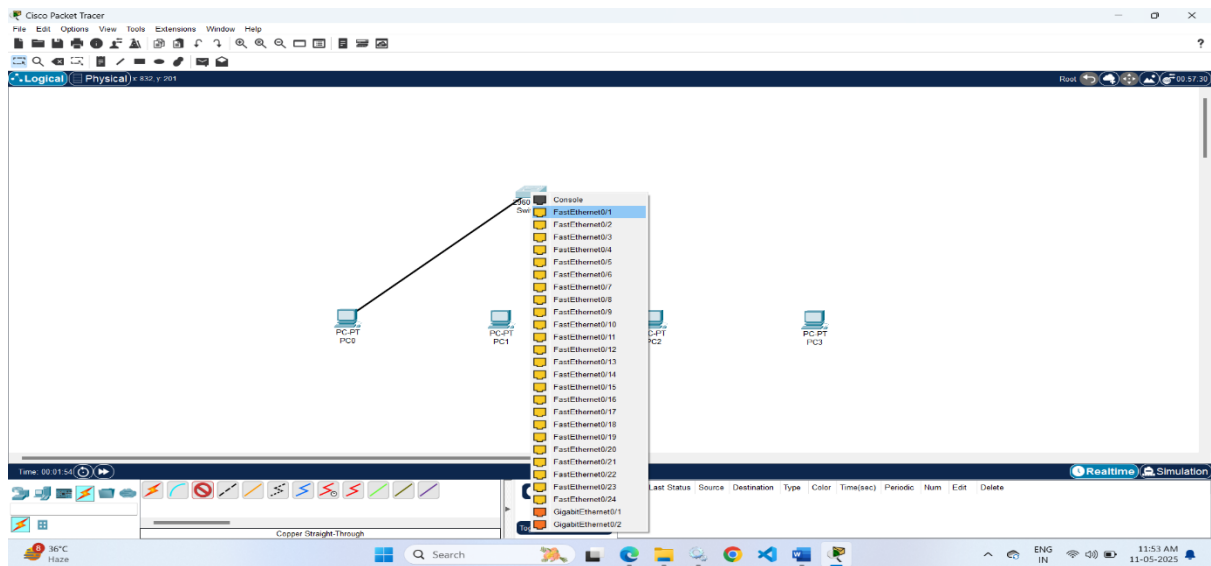
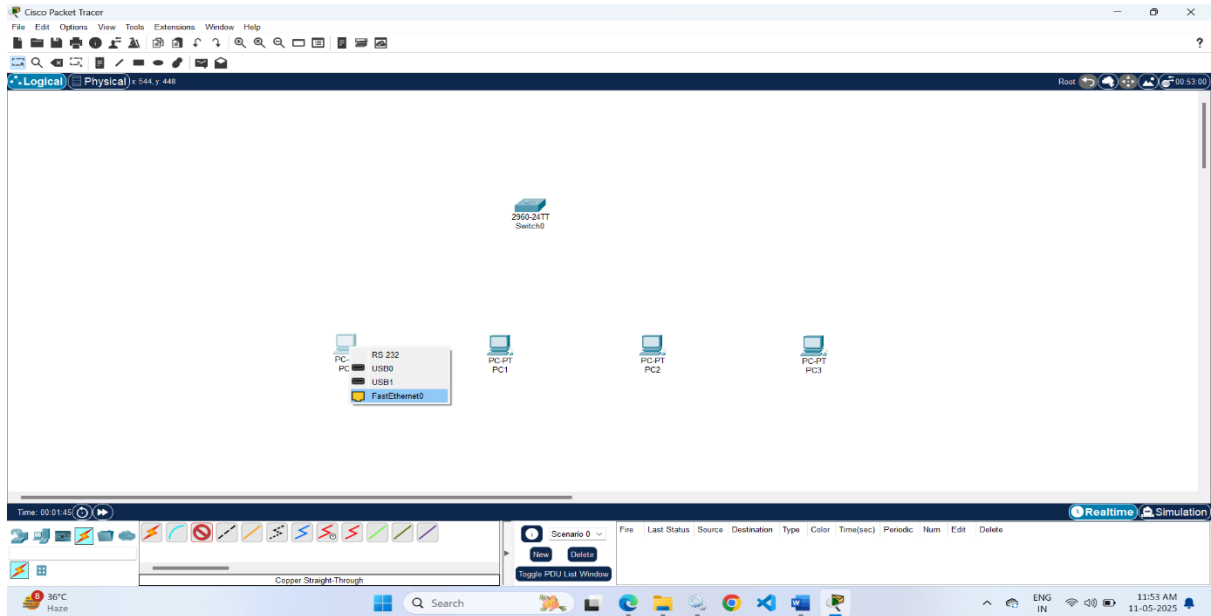


Step 3 : Open Cisco Packet Tracer, click "New" to open a blank workspace to start your simulation.

Step 4 : From the bottom left, drag 1 switch (2960) and 4 end devices (PC0 to PC3) onto the center canvas.

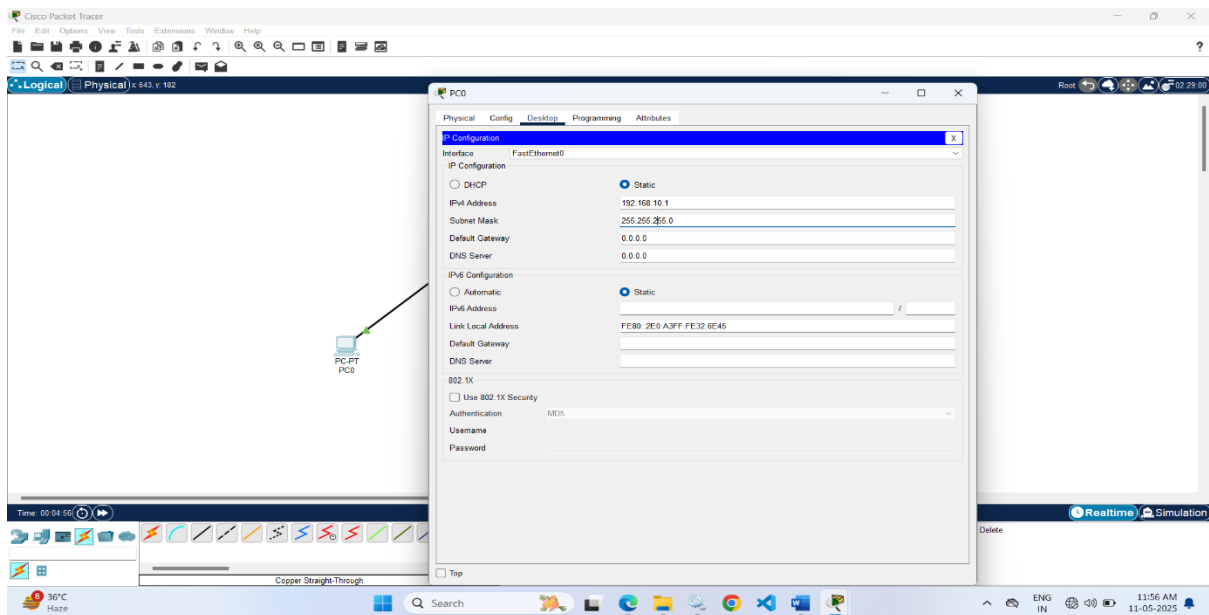


Step 5 : Use the "Connections" lightning icon, choose Copper Straight-Through cable, and connect each PC to Fa0/1–Fa0/4 ports on the switch.

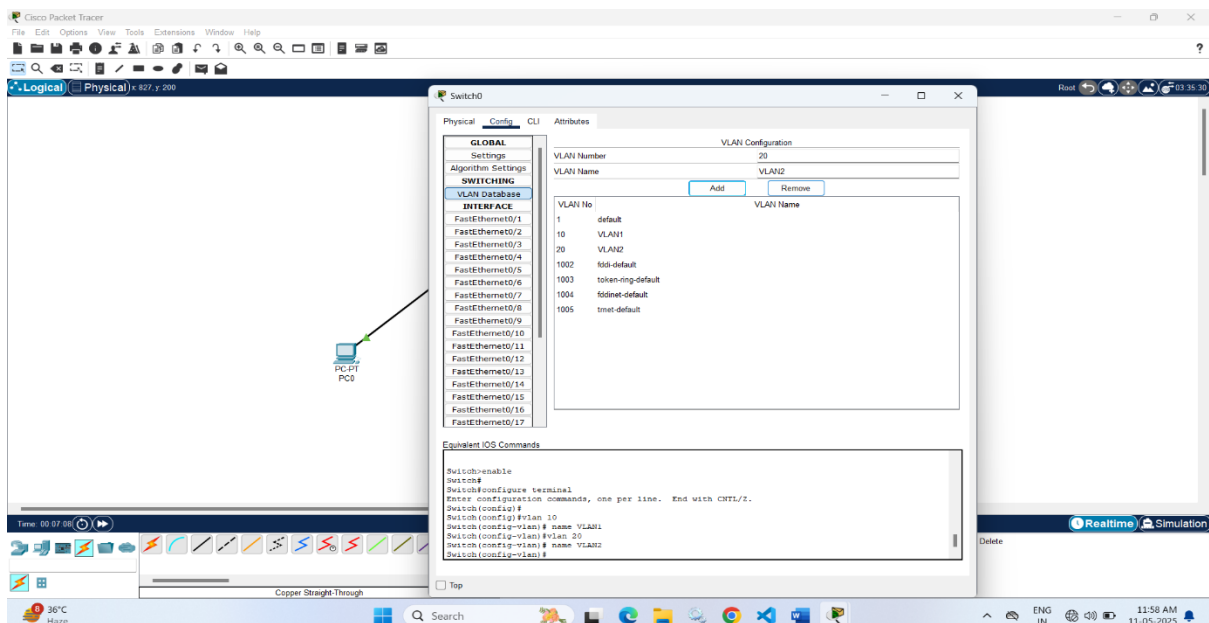


Step 6 : Click each PC → Desktop → IP Configuration → set:

- PC0: 192.168.10.1 / 255.255.255.0
- PC1: 192.168.10.2 / 255.255.255.0
- PC2: 192.168.20.1 / 255.255.255.0
- PC3: 192.168.20.2 / 255.255.255.0

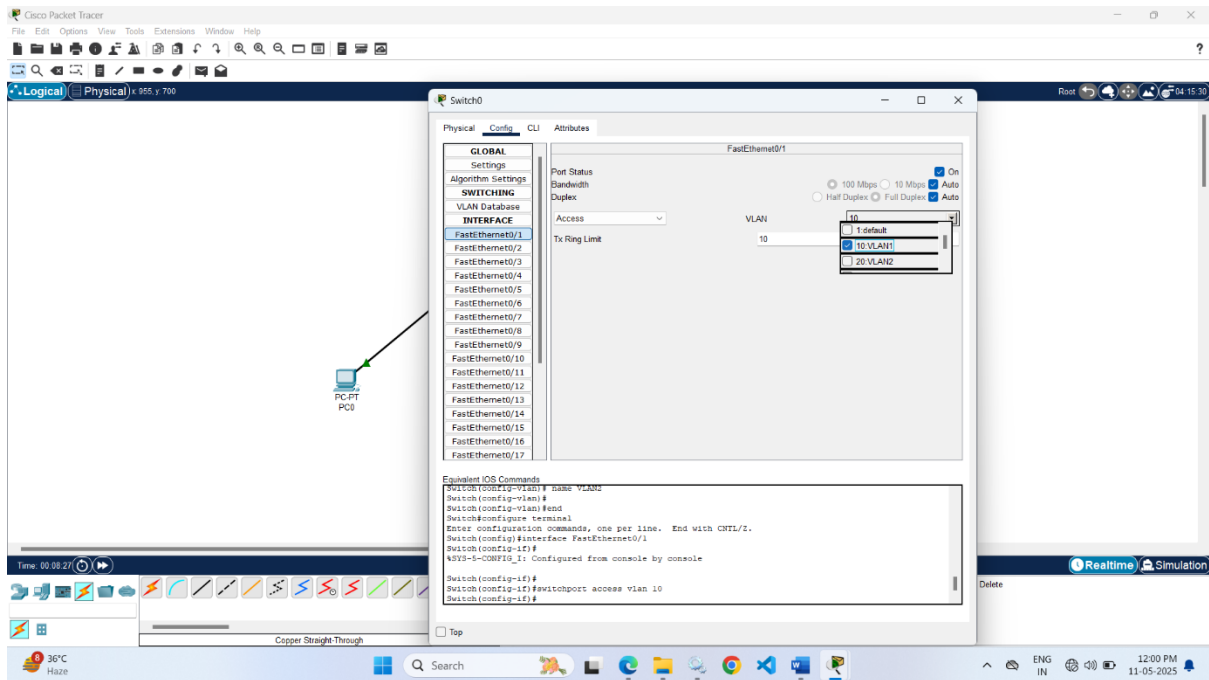


Step 7 : Click the switch → Config tab → VLAN Database → add VLAN 10 (name: VLAN10) and VLAN 20 (name: VLAN20).

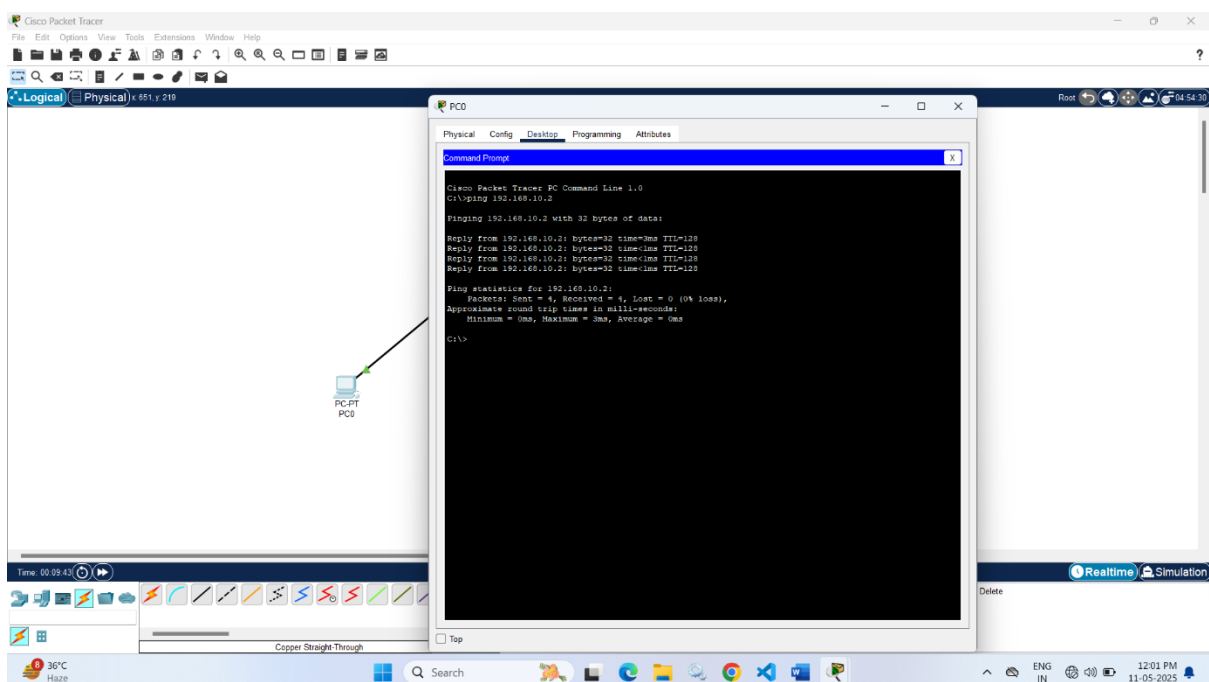


Step 8 : In the switch Config tab:

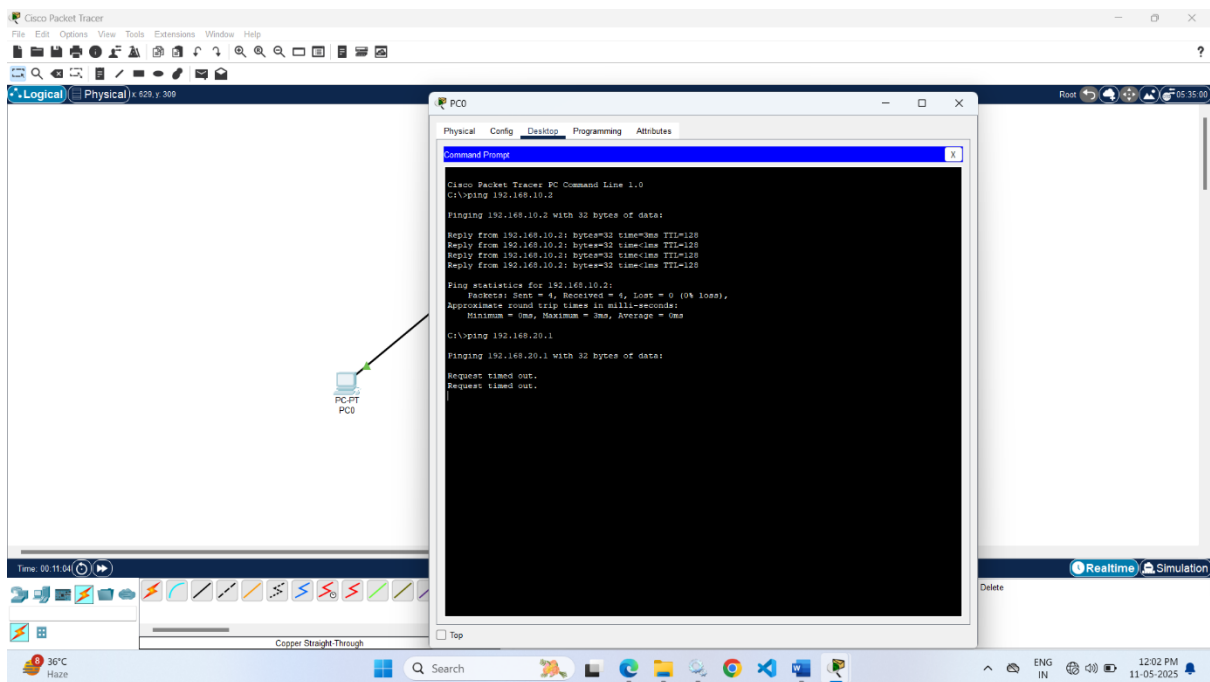
- Click FastEthernet0/1 and 0/2, set Port Mode = Access, VLAN = 10
- Click FastEthernet0/3 and 0/4, set Port Mode = Access, VLAN = 20



Step 9 : Click PC0 → Desktop → Command Prompt → ping 192.168.10.2 should succeed, and PC2 → ping 192.168.20.2 should also succeed.



Step 10 : From PC0 or PC1, ping any IP in VLAN 20 (e.g., ping 192.168.20.1) → it should fail because VLANs are isolated without a router.



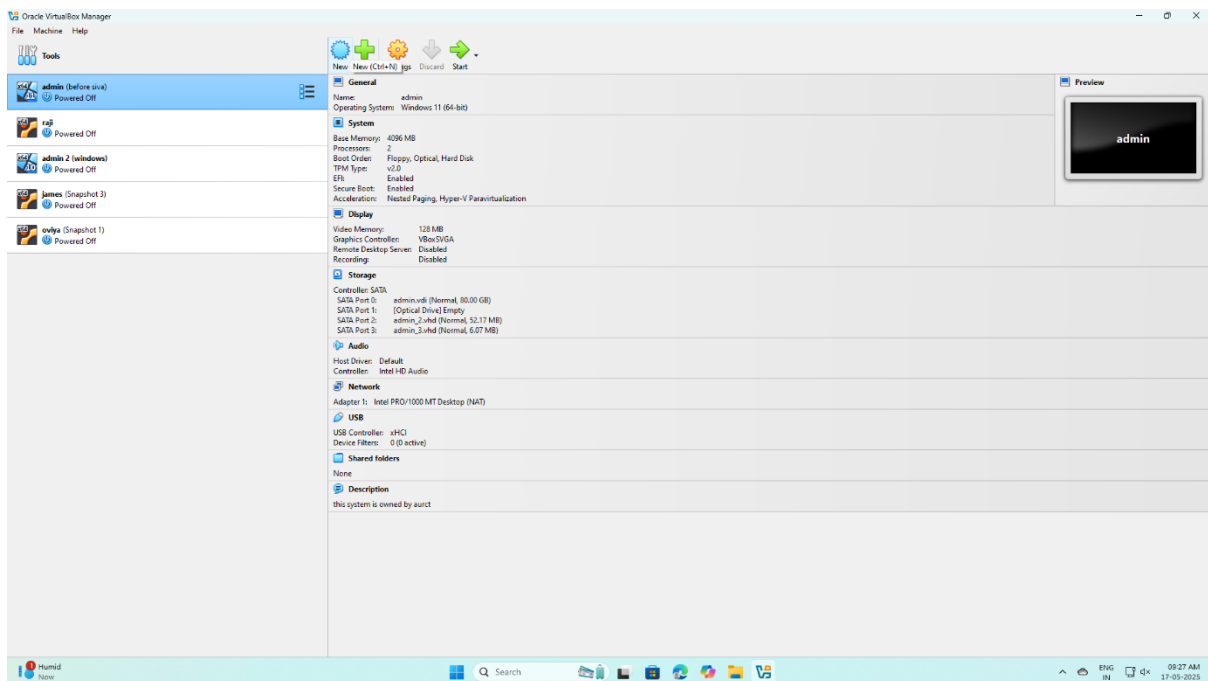
RESULT

INSTALL KVM IN LINUX

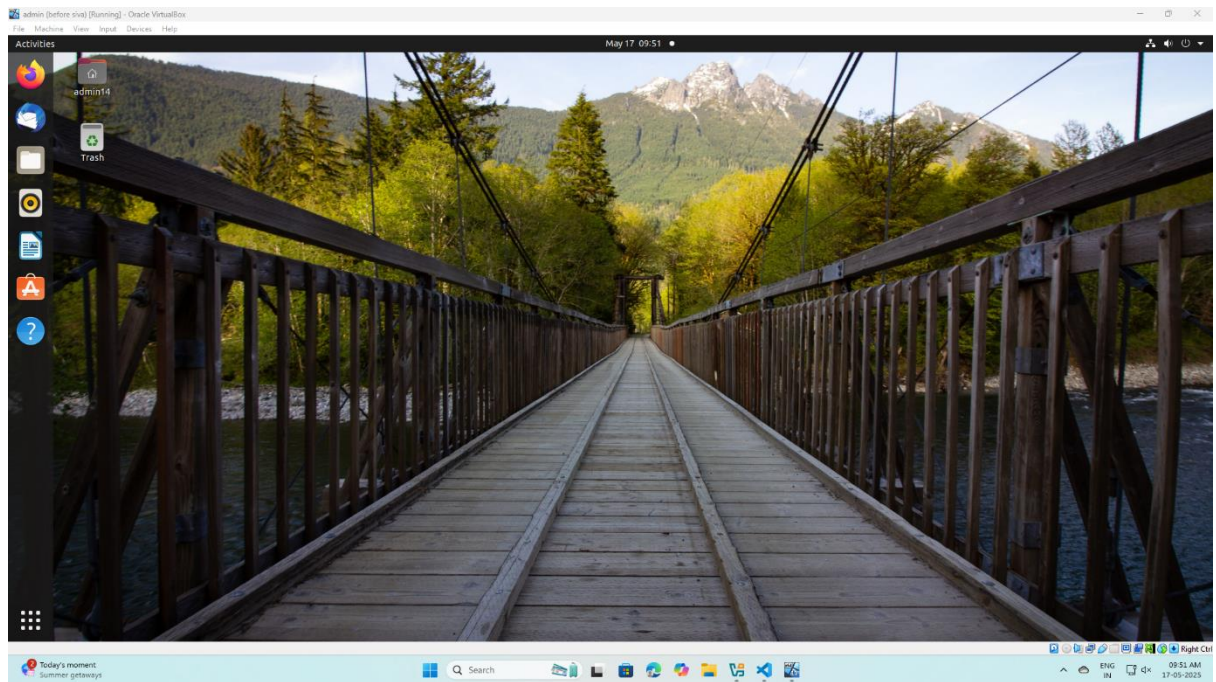
AIM

PROCEDURE

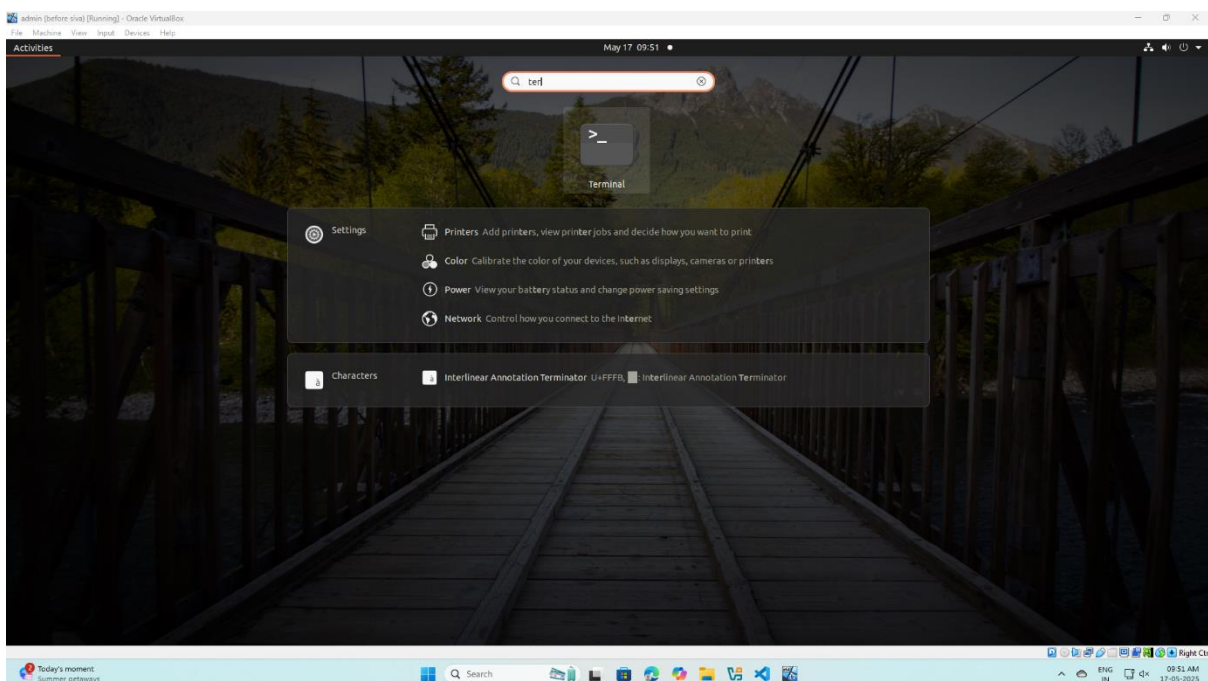
Step 1 : Launch Oracle VirtualBox, select your Ubuntu VM from the list, and click the "Start" button (green arrow) to boot the guest OS.



Step 2 : Once Ubuntu finishes booting, enter your username and password, then press Enter to access the desktop environment.

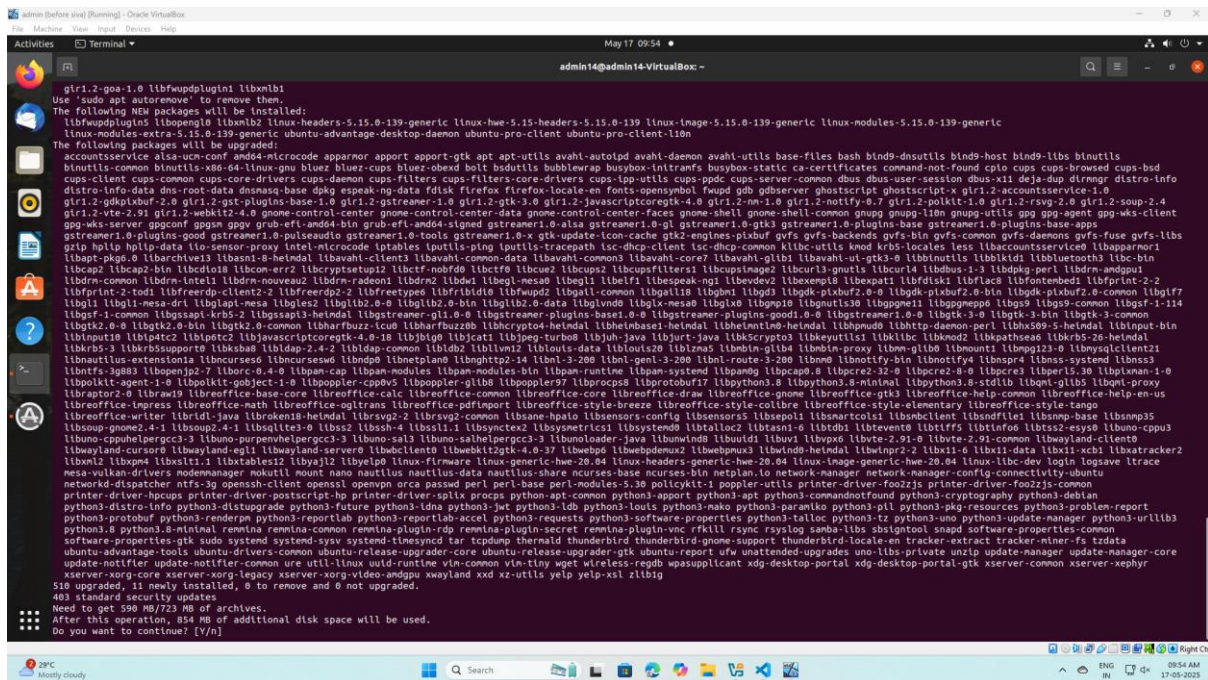


Step 3 : Click the "Show Applications" (grid icon in the bottom-left), type "Terminal" in the search bar, and click the Terminal icon to open it.

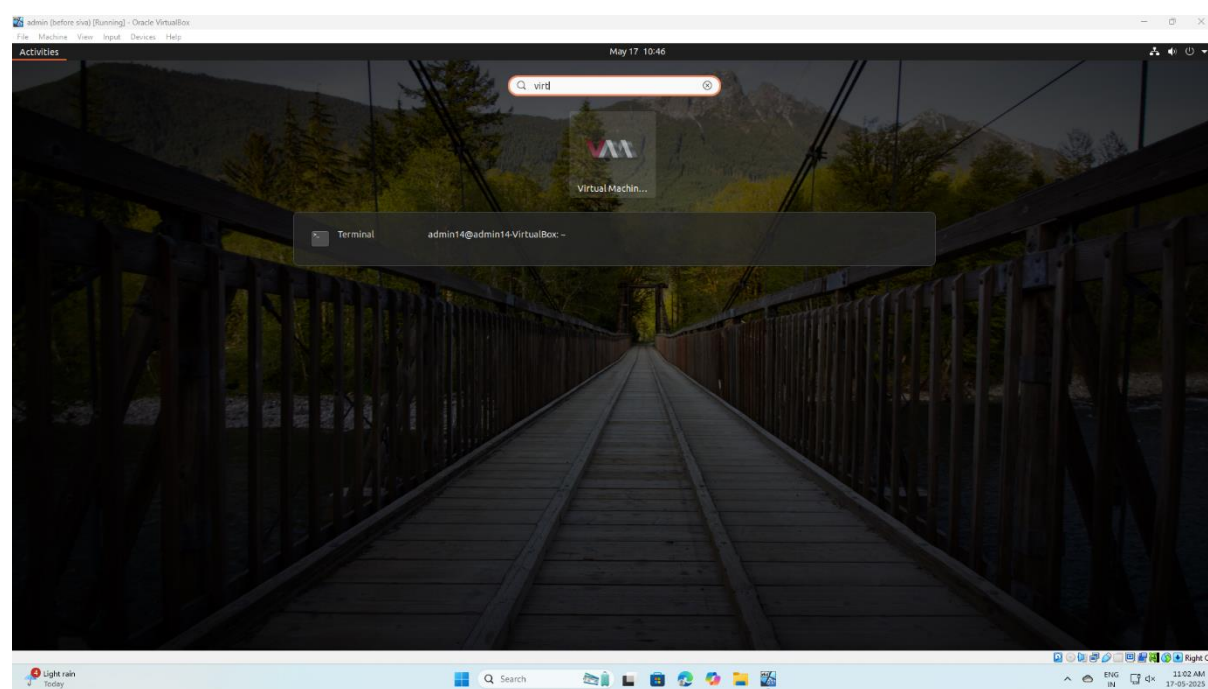


Step 4 : Run `sudo apt update && sudo apt upgrade -y` to make sure your system is fully updated with the latest packages and security patches.

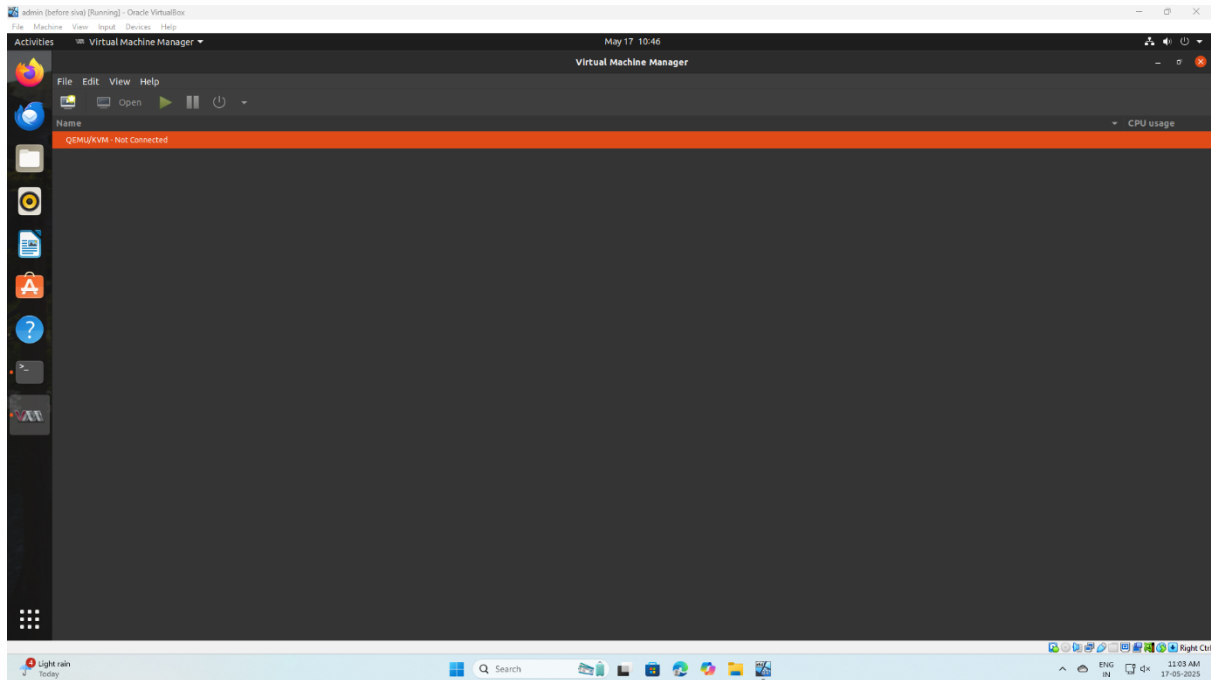
Step 6 : Run `sudo usermod -aG libvirt $(whoami)` to allow your user to manage KVM virtual machines.



Step 8 : Launch Virtual Machine Manager By Running the Command virt-manager in the Terminal.



Step 9 : Once Virt-Manager is open, it should automatically connect to the local virtualization environment (if installed correctly). You should see your local machine listed under "Local" or "QEMU/KVM."



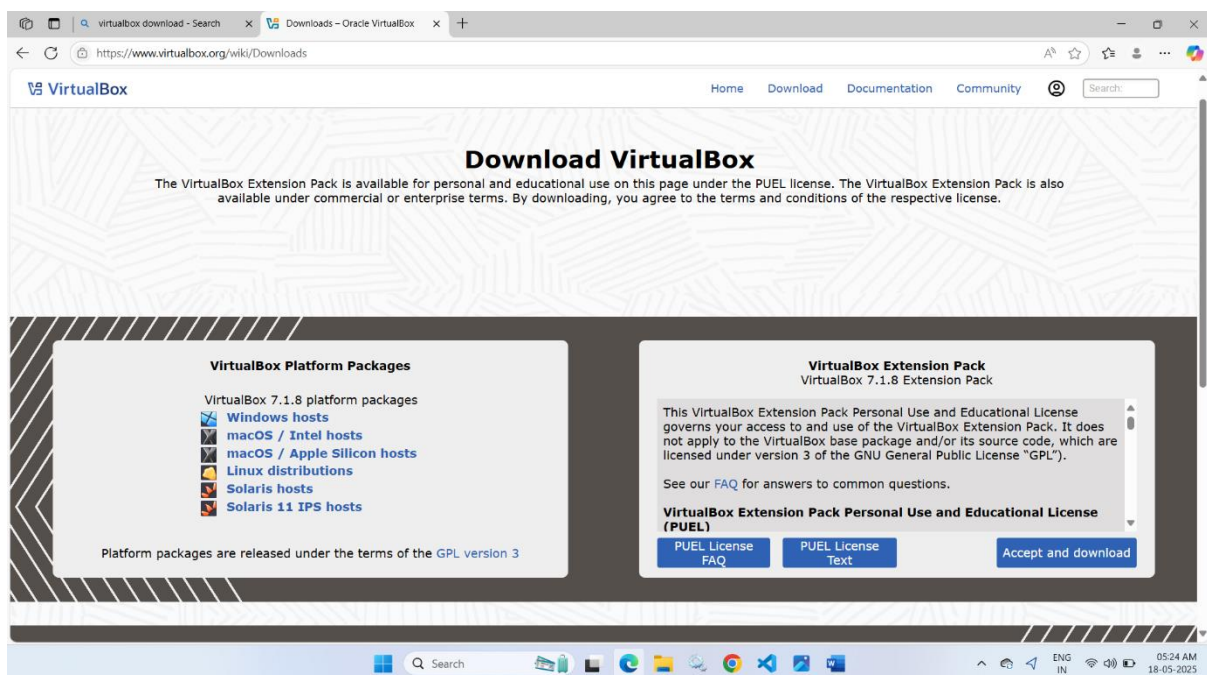
RESULT

CREATE NESTED VIRTUAL MACHINE(VM UNDER ANOTHER VM)

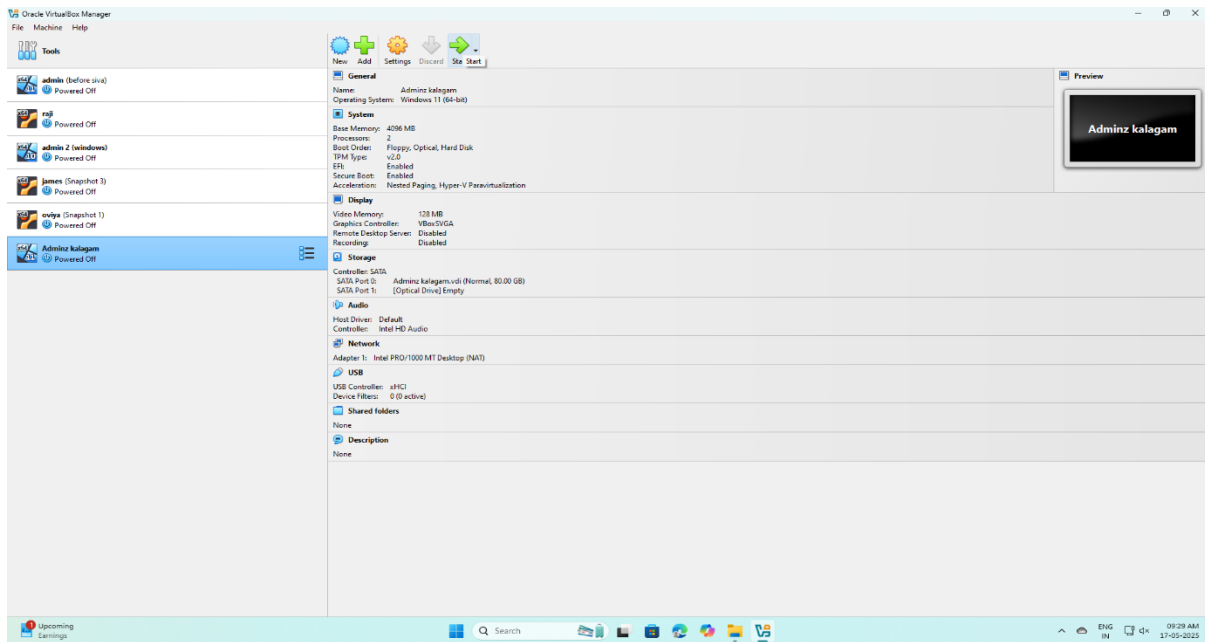
AIM

PROCEDURE

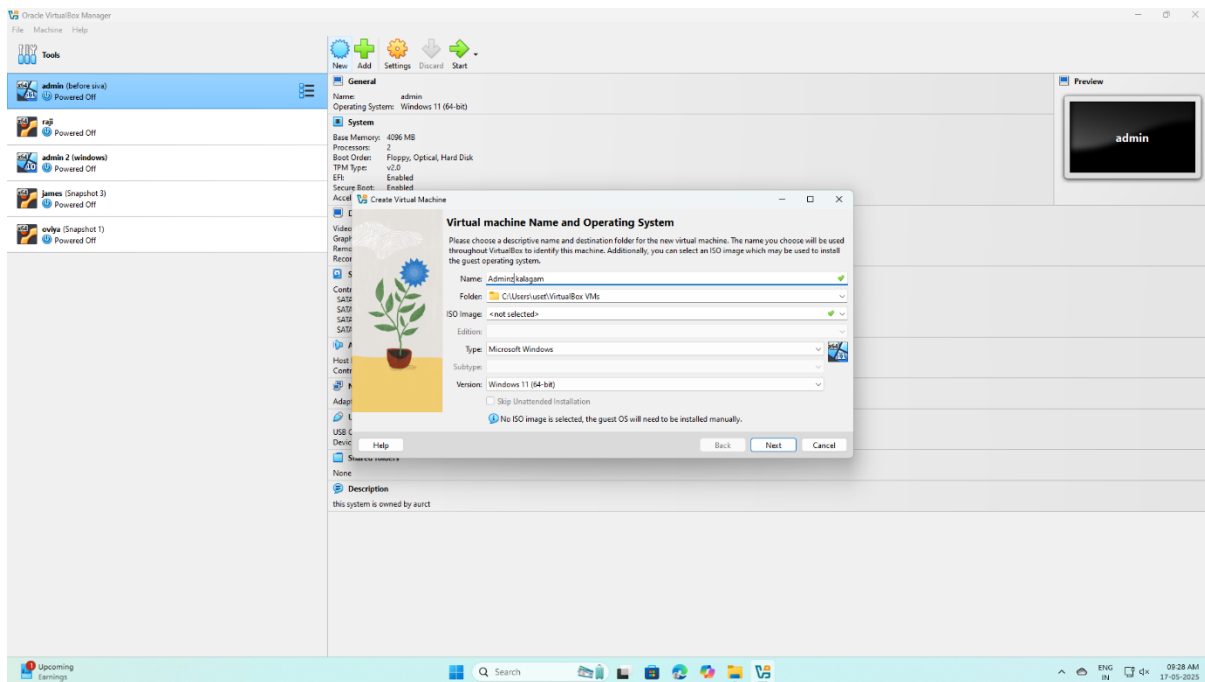
Step 1 : Download VirtualBox and Ubuntu ISO from their official websites to get the latest versions needed for virtualization and OS installation.



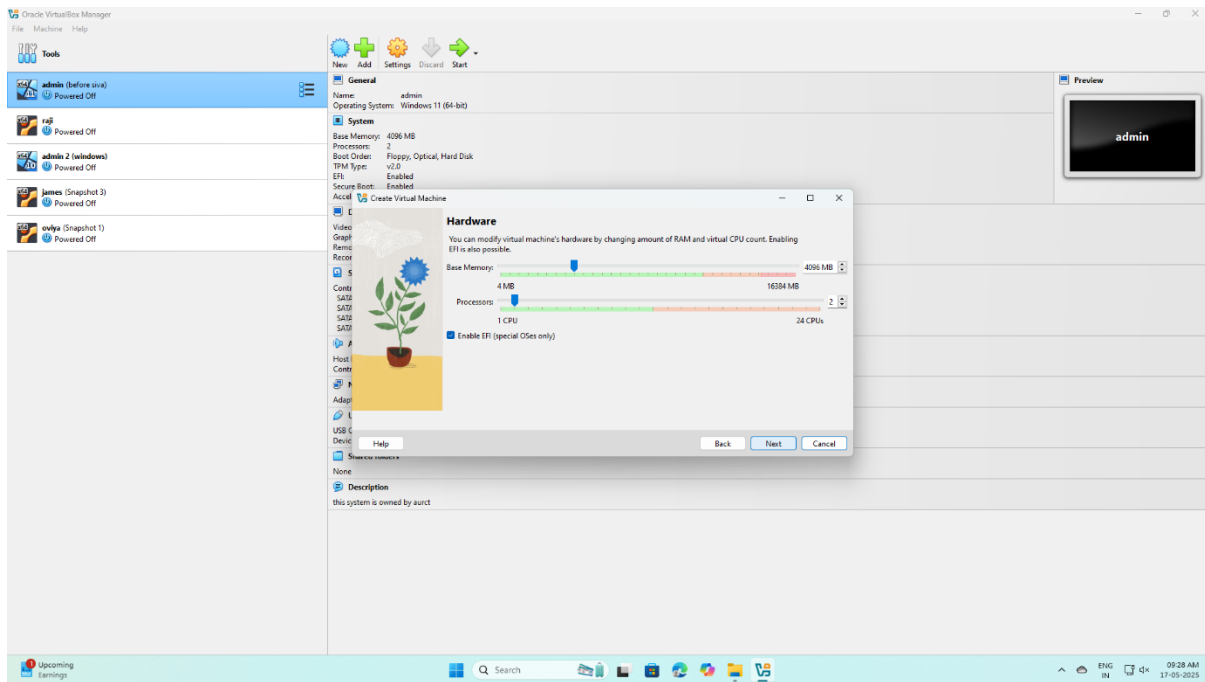
Step 2 : Open VirtualBox and click “New” to begin creating a new virtual machine for Ubuntu.



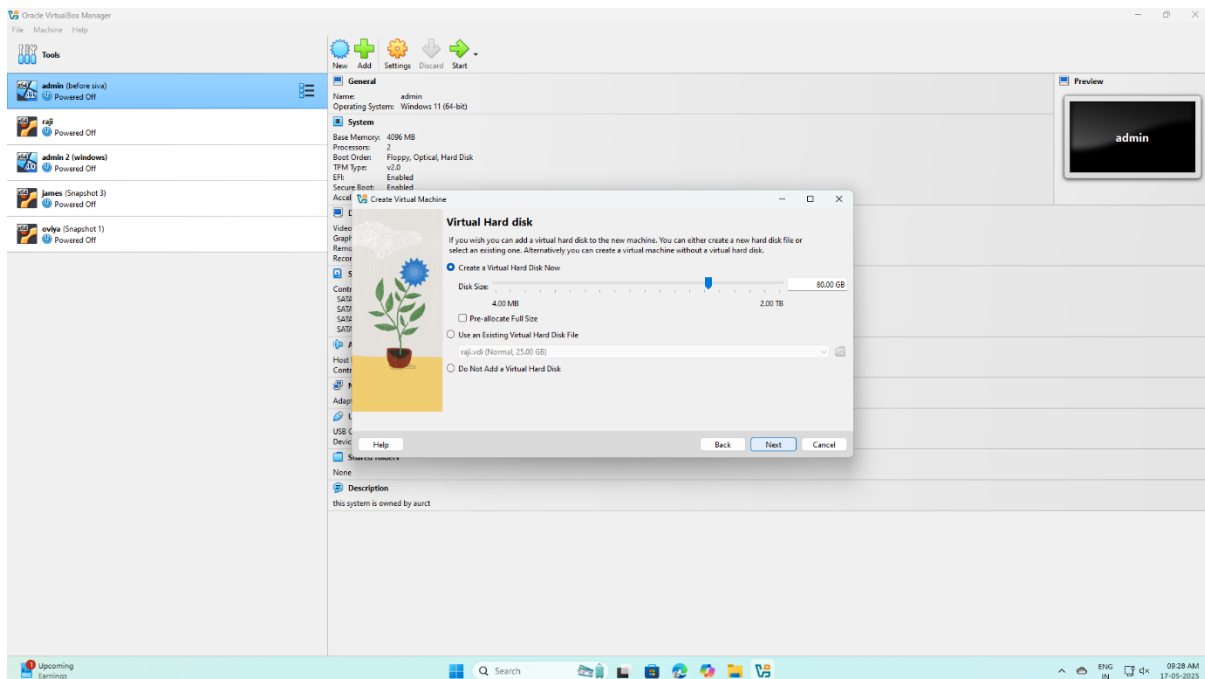
Step 3 : Enter VM name and OS type, like “Ubuntu 22.04”, set Type to “Linux” and Version to “Ubuntu (64-bit)”, and check “Skip Unattended Installation”.



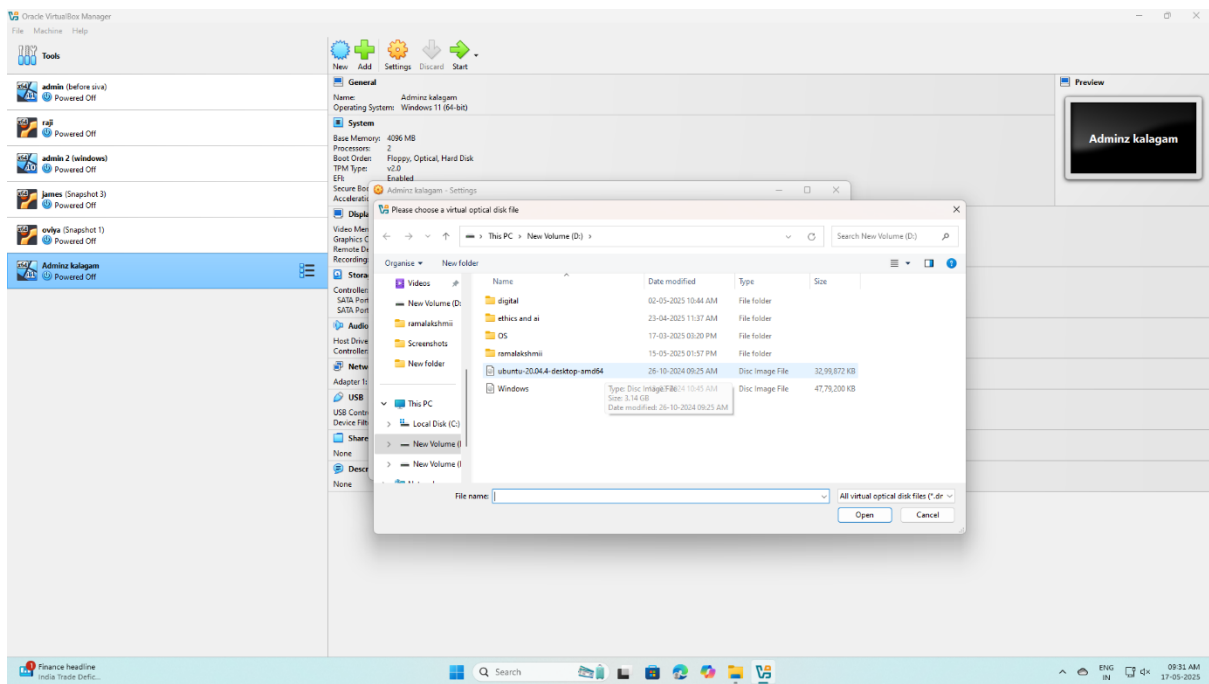
Step 4 : Allocate memory (RAM) by choosing at least 2048 MB (2 GB) to ensure Ubuntu runs smoothly.



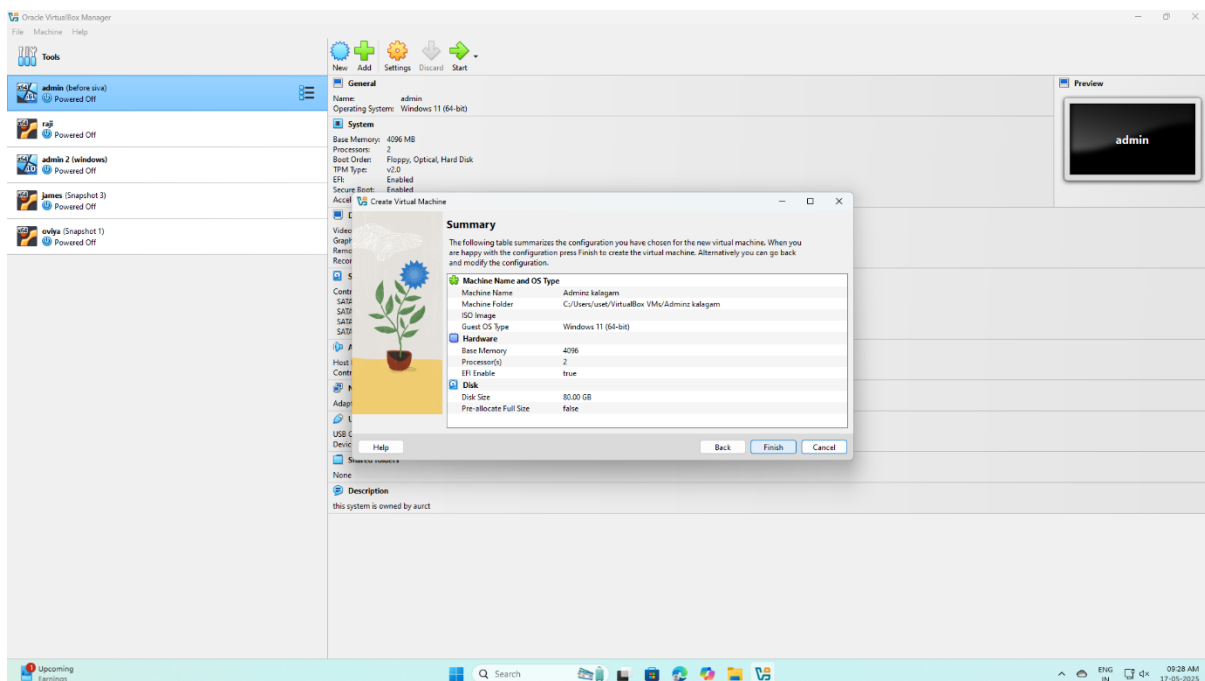
Step 5 : Create a virtual hard disk, choose VDI, select “Dynamically allocated”, and set disk size to at least 25 GB for system and app files.



Step 6 : Upload Ubuntu ISO before finishing VM creation by clicking the folder icon next to “ISO Image”, selecting your downloaded Ubuntu ISO, and clicking “Choose”.

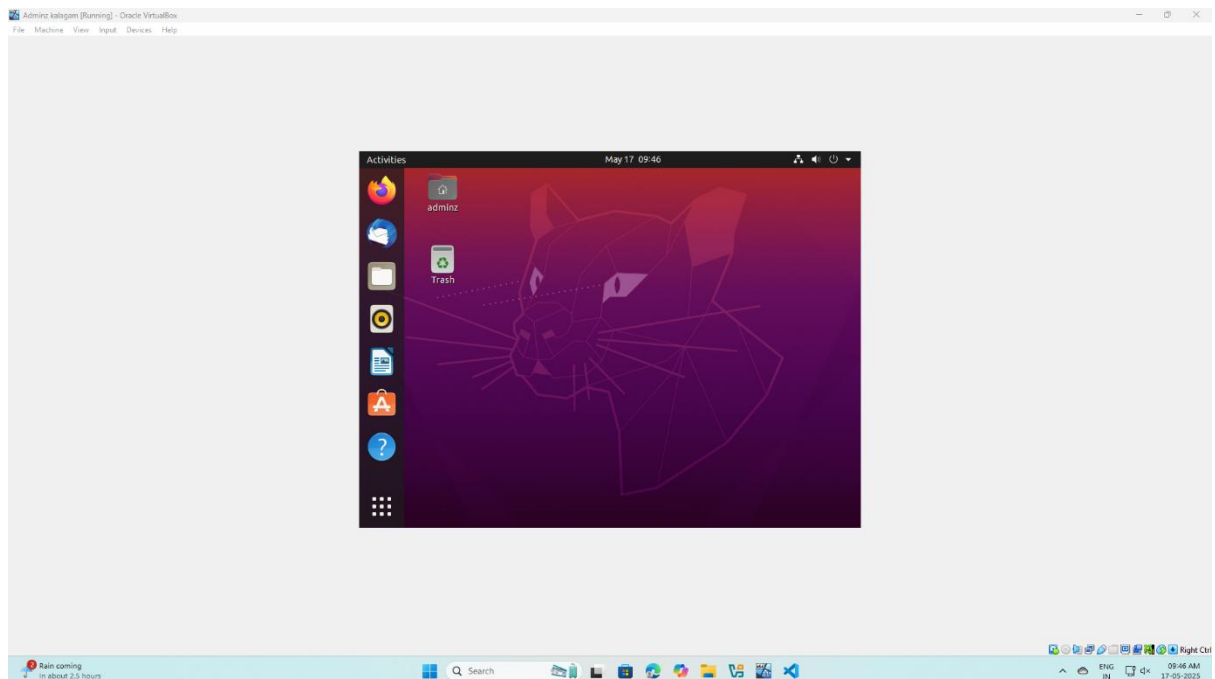
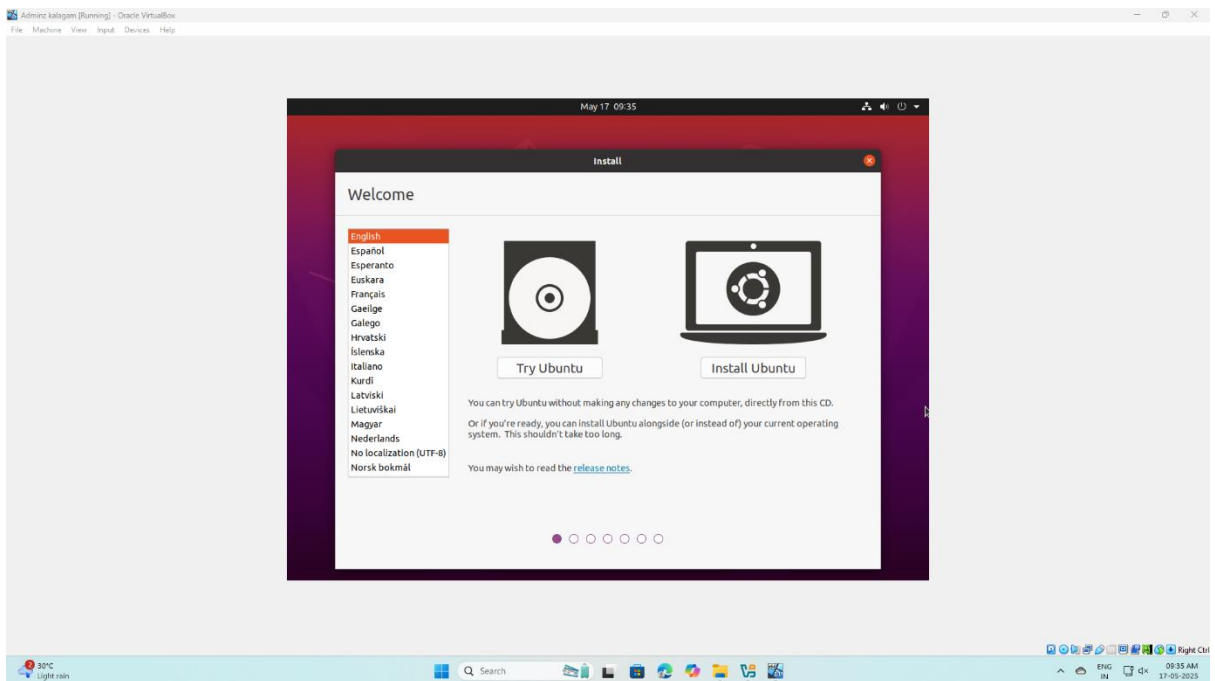


Step 7 : Click Finish to create the VM, which now has the Ubuntu ISO mounted and ready to boot.

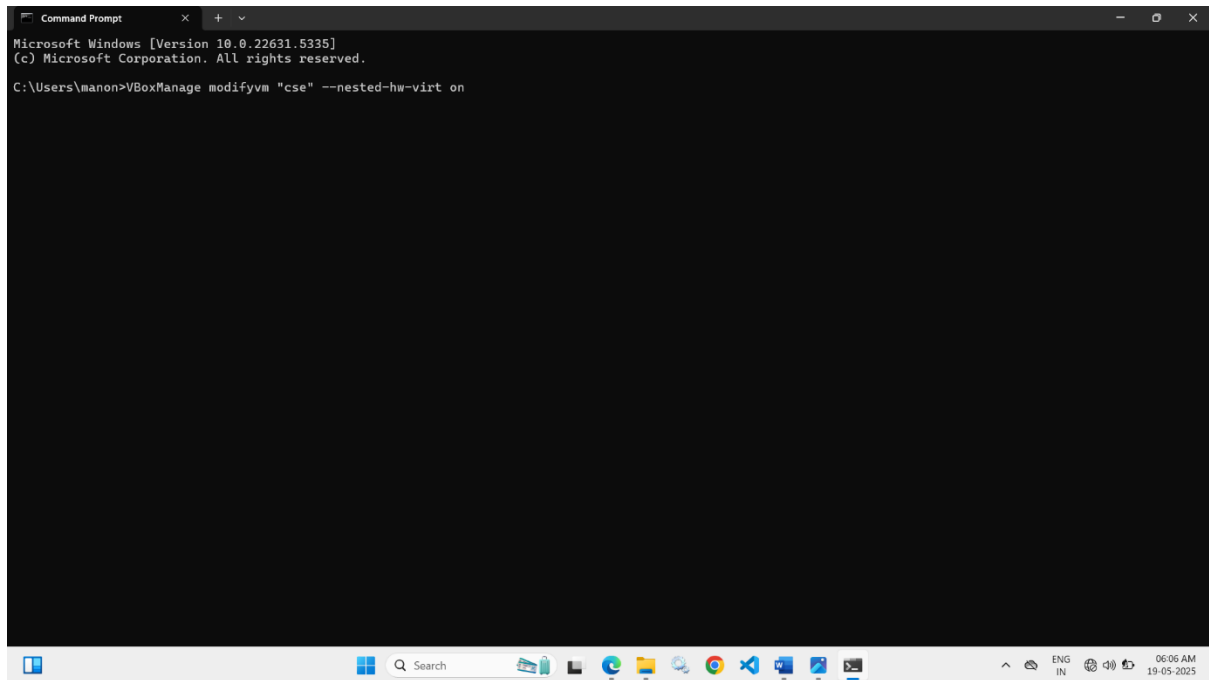


Step 8 : Start the VM, which will automatically boot into the Ubuntu installer from the ISO.

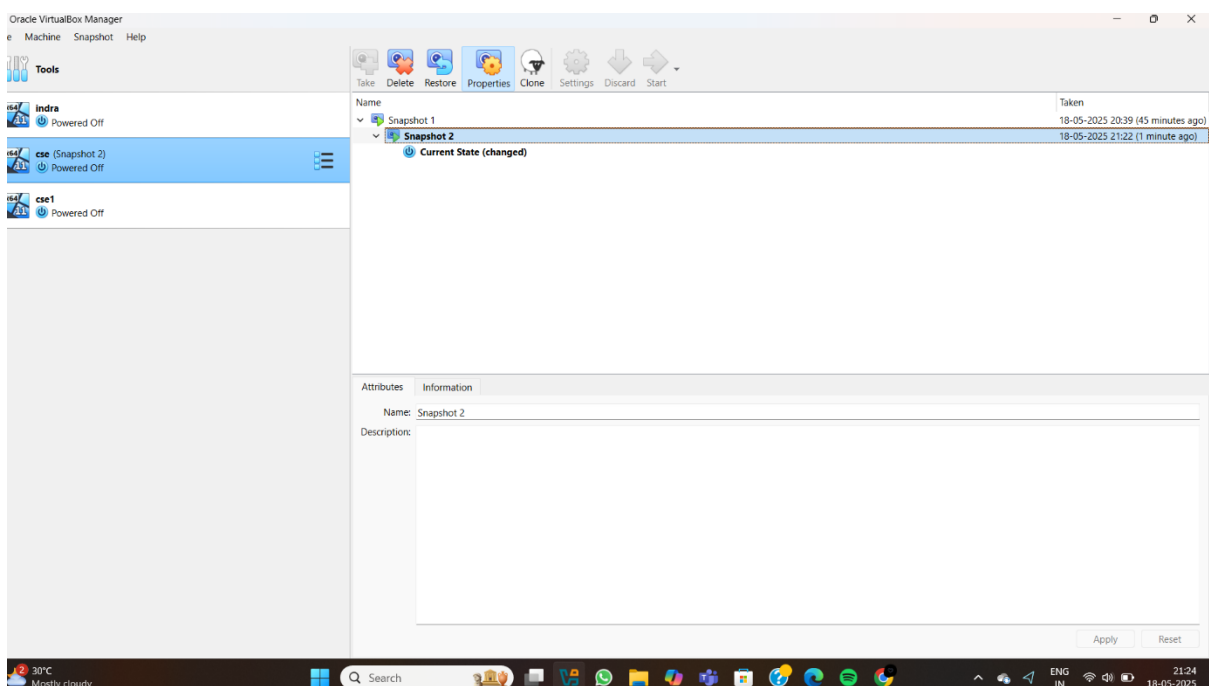
Step 9 : Choose “Install Ubuntu” and proceed with the normal installation options like keyboard layout and software selection.



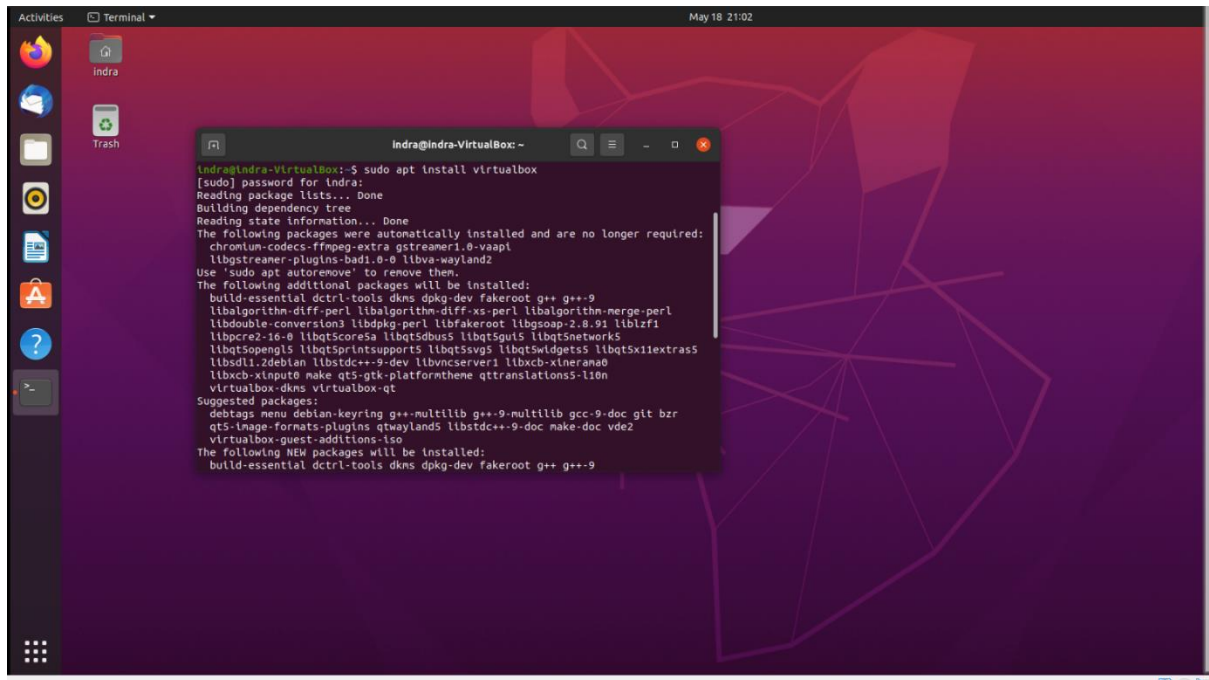
Step 10 : Open PowerShell as Administrator on the Windows host, and execute the command `VBoxManage modifyvm "Ubuntu_VM_Name" --nested-hw-virt on` which will enable nested virtualization for that specific virtual machine and require you to restart the Ubuntu guest for the changes to take effect.



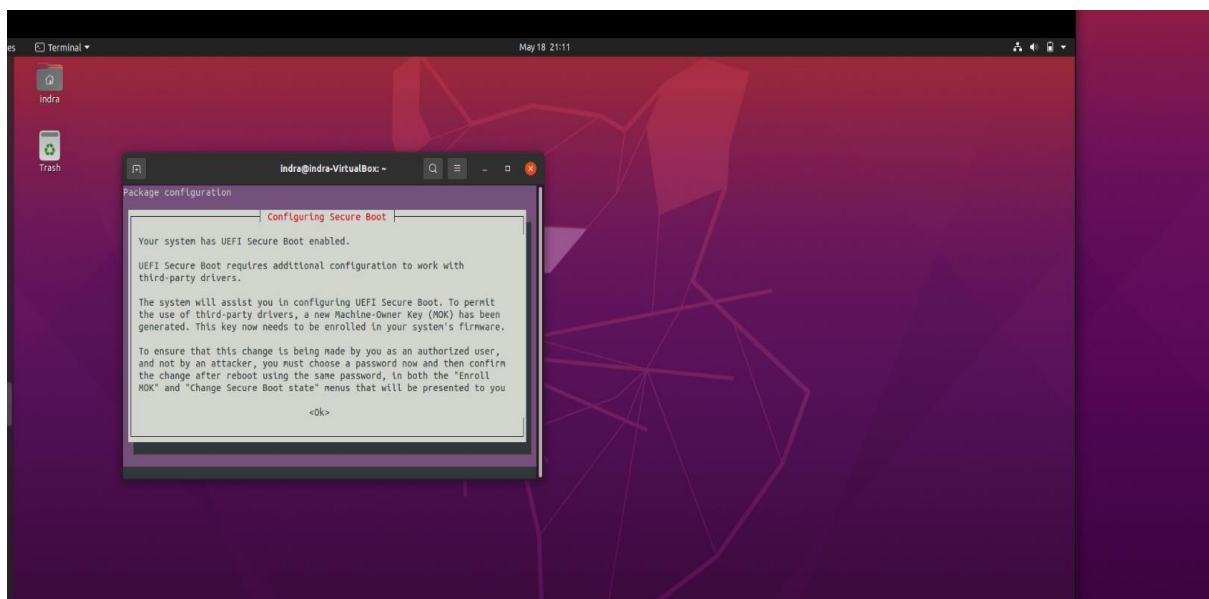
Step 11 : Launch VirtualBox on your Windows host, select the Ubuntu guest VM from the list of VMs, and click the Start button to power on the Ubuntu guest operating system.



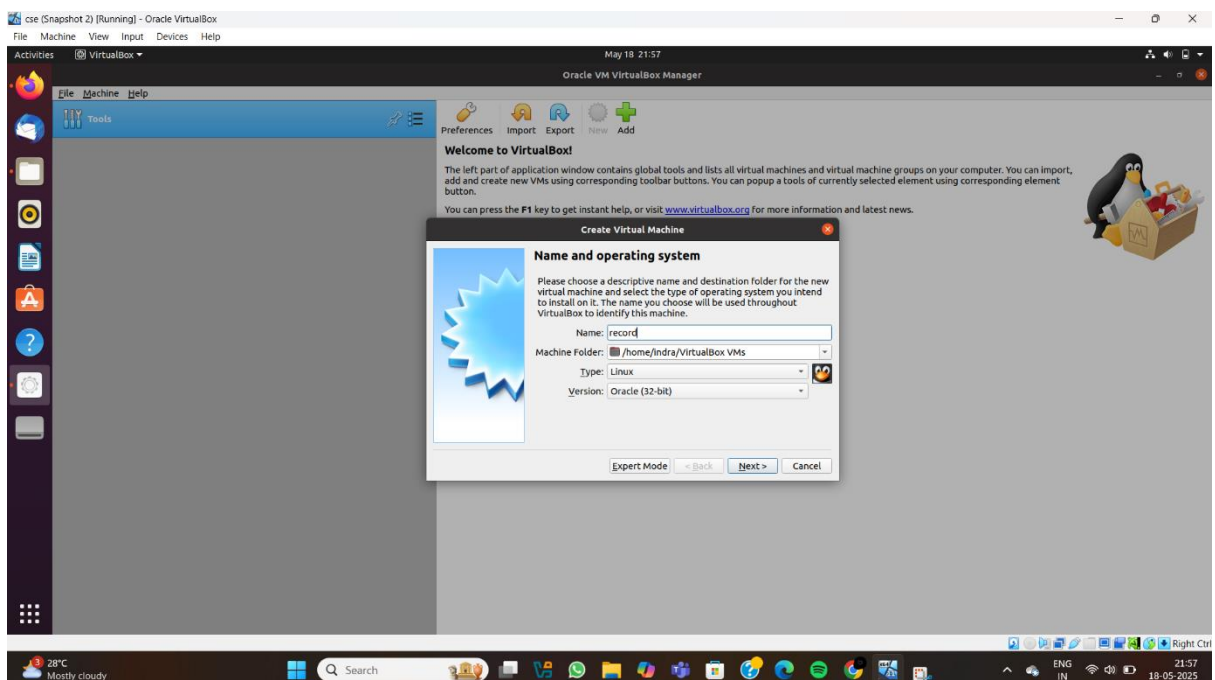
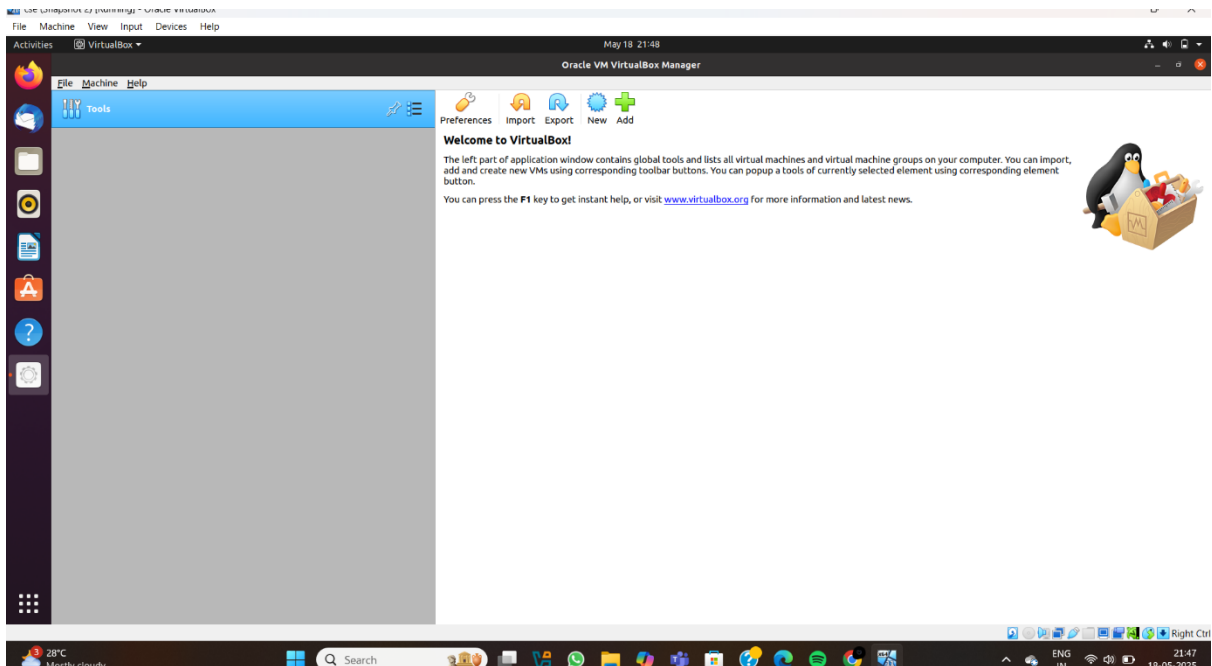
Step 12 : Once the Ubuntu guest VM is running, open the Terminal within Ubuntu and run the commands `sudo apt update` to update the package repository, followed by `sudo apt install virtualbox` to install VirtualBox on the Ubuntu guest system, allowing you to create and manage virtual machines within Ubuntu.

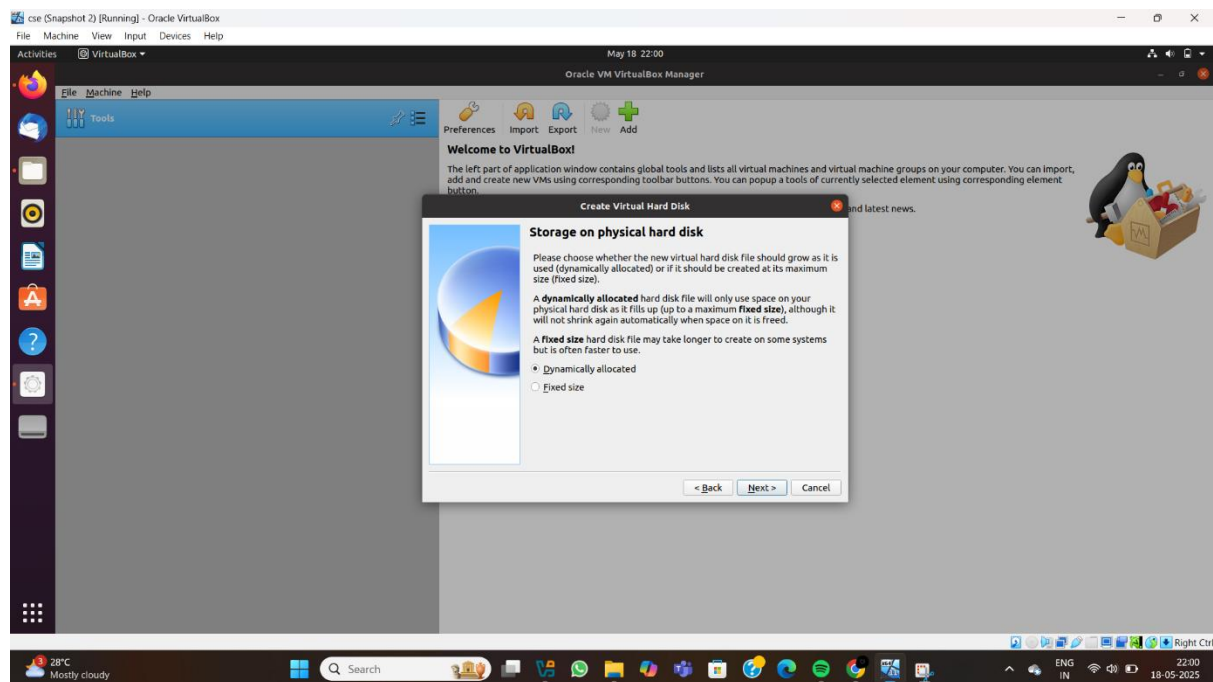
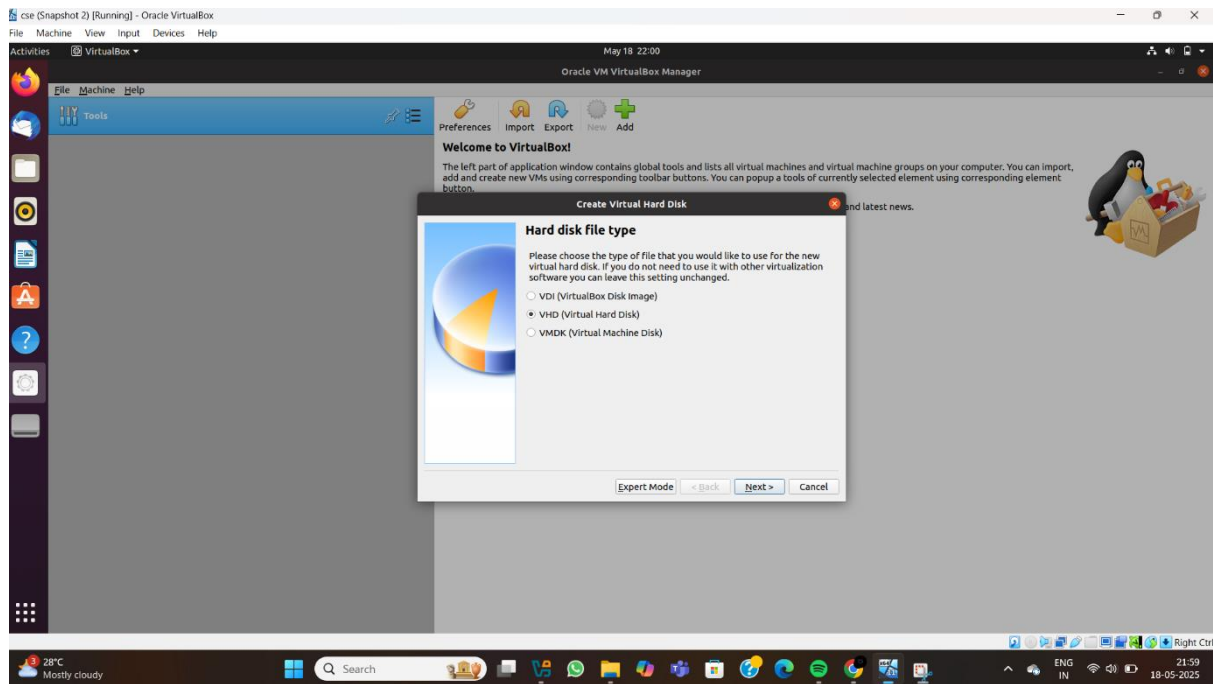


```
Indra@Indra-VirtualBox: ~  
[sudo] password for Indra:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  chromium-codecs-ffmpeg-extra gstreamer1.0-vaapi  
  libgstreamer-plugins-bad1.0-0 libva-wayland2  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  build-essential dctrl-tools dkms dpkg-dev fakeroot g++ g++-9  
  libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl  
  libdouble-conversion3 libdpkg-perl libfakeroot libgsoap-2.8.91 liblzfi  
  libpcre2-16-0 libqt5core5a libqt5dbus5 libqt5gui5 libqt5network5  
  libqt5opengl5 libqt5printsupport5 libqt5svg5 libqt5widgets5 libqt5xml5 libqt5xmlpatterns5  
  libstdc++12 libstdc++9-dev libvncserver1 libxcb-xinerama0  
  libxcb-xinput0 make qt5-gtk-platformtheme qttranslations5-l10n  
  virtualbox-dkms virtualbox-qt  
Suggested packages:  
  debtags menu debian-keyring g++-multilib g++-9-multilib gcc-9-doc git bzr  
  qt5-image-formats-plugins qtwayland5 libstdc++9-doc make-doc vde2  
  virtualbox-guest-additions-iso  
The following NEW packages will be installed:  
  build-essential dctrl-tools dkms dpkg-dev fakeroot g++ g++-9
```

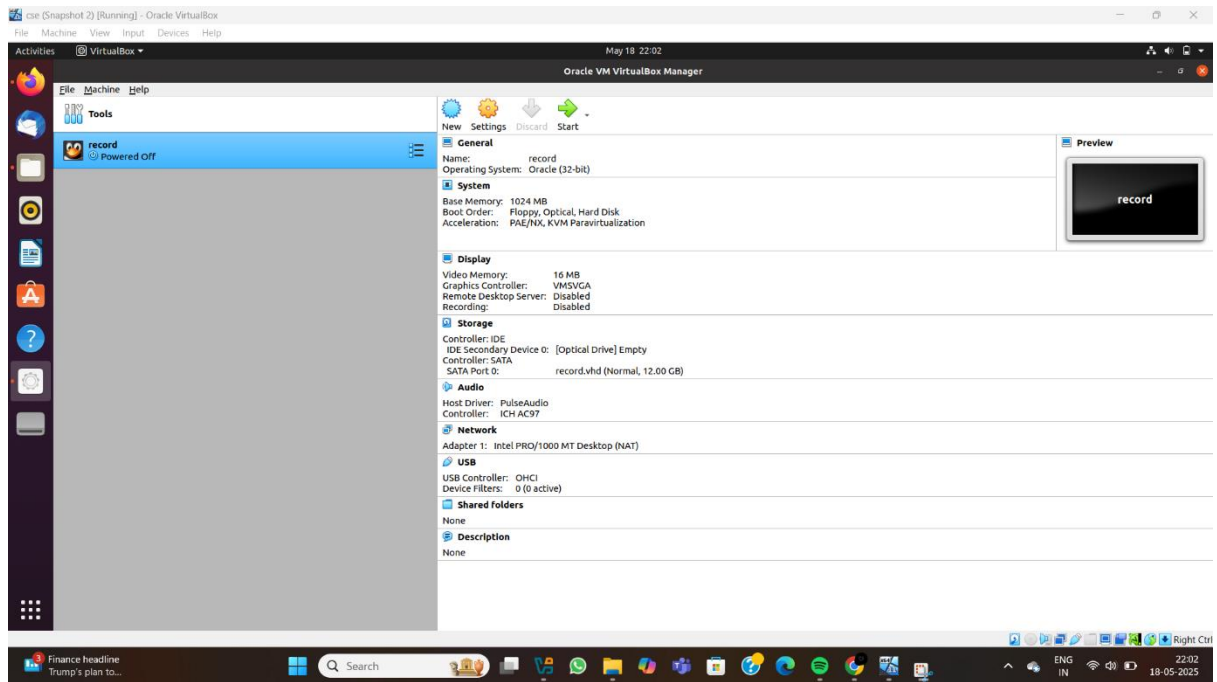


Step 13 : Open VirtualBox inside the Ubuntu guest VM, click the New button to create a new virtual machine, choose the appropriate operating system type and version, allocate sufficient memory (RAM) and virtual hard disk space, and follow the on-screen prompts to complete the VM creation process.





Step 14 : After creating the nested virtual machine, select it from the VirtualBox Manager in Ubuntu, click Start to launch the nested VM, and the virtual machine should boot up and allow you to install an operating system or perform other tasks just as you would with any regular virtual machine.



RESULT